

Floating population: migration with(out) family and Chinese economic development*

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Abstract

Migrants around the world leave dependent family members behind to cope with high living costs and restrictive migration policies at destination. Yet these features are absent from standard equilibrium models of migration. Using Chinese data, we develop and estimate a quantitative spatial model in which rural households choose whether to migrate, which members to send, where to go, and how to allocate their consumption across members and locations. We show that the ability to migrate without family is essential to reap the aggregate productivity gains from rural-urban migration. We also evaluate the effect of actual and potential housing and migration policies on migration patterns, productivity, and welfare.

Keywords: family migration, migration policy, housing policy, spatial equilibrium.

JEL Classification: R12, J61, O15.

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Standard economic models of migration, following [Harris and Todaro \(1970\)](#) and [Borjas \(1987\)](#), present it as an individual worker decision based on expected wages at origin and destination. Yet many migrants around the world have a family, which they might bring with them or leave behind: 39% of Ecuadorian emigrants in 2005 had left at least one child behind ([Herrera Mosquera 2006](#)), as did about 55% of Filipino migrants in 2007 ([Cortes 2015](#)). Important factors in this decision are higher living costs at destination as well as migration policies, which often impose additional costs on migrant families.¹ These critical aspects of migration decisions are less studied by the economic literature, and lacking especially in models that quantify migration frictions and the aggregate productivity gains from reducing them ([Bryan and Morten 2019](#), [Tombe and Zhu 2019](#)).

Taking into account the family dimension of migration decisions is challenging for at least three reasons. First, we need a model that allows households to decide not only whether and where to migrate, but also who in the family migrates and who stays behind. Second, data on migrants’ family arrangements, earnings, living costs and migration policies is rarely available for all origins and destinations, especially for international migrants, but also for internal migrants. Third, wages, living costs and migration policies at destination respond endogenously to immigration, which makes it harder to identify their effect on migration decisions.

In this paper, we study China between 2000 and 2005, where a so-called “floating population” of over 30 million rural workers migrated despite high living costs and restrictive migration policies (*hukou* system), leaving millions of children behind ([UNICEF 2018](#)). We develop a new model in which households decide whether and where to migrate, and also *how* to migrate, i.e., as a father or a mother alone, as a couple leaving children behind or with the entire family. We estimate the model using rich census data on migrants’ family arrangements, earnings and living costs across all origins and destinations, and exploiting exogenous variation in wages, living costs and migration policies at destination. The model allows us to quantify the effect of policies such as housing subsidies and *hukou* reform on the magnitude, the direction, and the type of migration flows, as well as on aggregate productivity and welfare.

In more detail, in the first part of the paper we document a number of empirical facts on rural-urban migration in China in the early 2000s. For this, we use detailed census data on bilateral migration flows, income, rents and family arrangements, as well as data on expenditures and remittances from the Longitudinal Survey on Rural Urban Migra-

¹Migration policies are not only relevant for international migrants. A survey of 185 countries shows that 80% of them had policies to reduce rural-urban migration—a fast-growing trend (from 38% in 1996), especially in developing countries ([United Nations 2013](#)). They range from land tenure insecurity to registration booklets limiting access to public services and under-servicing of migrant settlements ([Feler and Henderson 2011](#), [Song 2014](#), [Selod and Shilpi 2021](#)), which impose larger costs on families.

tion in China (RUMiC). Using these data, we show that rural migrants can be broadly classified into five categories: individual migrants without dependent family members (63%); couples migrating with their children (12%), couples leaving their children behind (12%), fathers migrating alone (9%), and mothers migrating alone (4%). We show that rural migrants disproportionately concentrate in high-wage destinations, despite their high housing costs, typically live in poor housing conditions, and remit a substantial part of their income. These patterns are stronger for couples leaving children behind and for mothers or fathers migrating alone. These stylized facts suggest that migrant households seek higher wages, as in standard migration models, but also respond to higher living costs at destination by leaving children behind.

The second part of our analysis introduces a quantitative spatial model of location choice based on these empirical facts. We assume that each rural household is either an individual—for example a youngster with parents that care for themselves—or a family composed of a father, a mother, and children, who are exogenously born in a given origin. Individual and family households can decide to stay or move across a set of urban locations. Our model introduces two novel features in an otherwise standard spatial model: (i) rural family households decide whether to migrate, who migrates (father alone, mother alone, both parents or the whole family), and which destination to migrate to within a nested structure; and (ii) households optimize their consumption allocation across household members and across space. In contrast with standard models where migrants are individuals who move to the city and consume entirely at destination, migrant family households might decide to leave some household members behind (children, sometimes with a parent) and allocate a fraction of their consumption at origin—which increases their incentives to locate in high-wage, high-price destinations. This interaction between the allocation of consumption and the location choice is the key novelty of our theory and a significant quantitative force.

In the third step of our analysis, we estimate the main parameters of the model, both for individual and family migrants, using plausibly exogenous variation in local prices, wages, and relative migration costs. For family households, we first use detailed consumption data to parameterize the consumption allocation across household members, goods, and space. This allows us to derive the price indices and indirect utilities that are relevant to the different options of the location choice program (family stays, father migrates, both parents migrate, whole family migrates).² Second, we use variation in exposure to international trade (Facchini et al. 2019) as an instrument for destination

²Since we estimate an elasticity of substitution between consuming non-tradables at destination and remitting close to 1, we consider an “overlapping” Cobb-Douglas formulation for preferences. We identify the consumption weights of the Cobb-Douglas utility function using the average consumption allocated to non-tradables and to remittances for households migrating with and without family dependents.

wages to estimate the elasticity of substitution between different locations *for a given family arrangement*. Third, we estimate the elasticity of substitution between the different migration types (father alone, both parents, whole family) using the inclusive value of each migration mode for each origin from the previous step (as in [Bugge et al. 2023](#)). We instrument the inclusive value with a predictor of living costs at destination based on geographic constraints to urban sprawl at the periphery of cities, in the spirit of [Saiz \(2010\)](#) and [Harari \(2020\)](#), combined with a gravity structure to capture the “family-friendliness” of migration options across origins. Fourth, we estimate the elasticity of substitution between staying and moving away from origin locations, using exogenous variation in the price of agricultural commodities combined with cropping patterns across origins, following [Imbert et al. \(2022\)](#).³ We use a similar procedure, but abstracting from family considerations and to estimate individual migrants’ decisions.

We find (i) a very high elasticity of substitution between destinations, especially when parents migrate without children, (ii) a large elasticity of substitution between family arrangements, and (iii) a substantial elasticity of emigration. In other words, the patterns of substitutability across destinations from a given origin are strongly influenced by the type of migration, something that standard migration models would not be able to capture. Overall, this procedure allows us to estimate the three nests—whether, who, and where—of the migration model, an innovation that allows us to recover more realistic bilateral migration costs.⁴ We show that these recovered migration costs are well explained by existing Hukou migration restrictions, when instrumented using grain reserves as a source of exogenous variation.

In the fourth step of our analysis, we use our framework to perform four counterfactual analyses. The first two focus on the value of migrant family strategies. First we quantify the value of the option of leaving family behind, by forcing migrants to fully consume at destination instead of remitting part of their income to family dependents. Without the possibility to remit, many rural workers would choose to stay: rural-urban migration would decrease by 67%, and up to 83% in mega cities (cities with more than 5 million inhabitants). As a result, aggregate productivity would decline by 4.65%. Second, we quantify the value of the option for mothers to leave their children behind, an option that is often absent in traditional societies: without this option, more fathers would

³To complete the estimation of the model we also estimate labor demand and land supply elasticity at destination using a shift-share design close to the one developed in [Imbert et al. \(2022\)](#).

⁴We estimate bilateral migration frictions by migration mode—including non-monetary welfare costs, or incompressible expenses induced by a certain living arrangement (e.g., travel or commuting costs) and the associated policy constraints (e.g., access to healthcare)—as the residual that separates our model from the data. This approach follows methods developed in the spatial equilibrium literature ([Redding and Rossi-Hansberg 2018](#)), which we adapt to a multiple-nest structure ([Bugge et al. 2023](#)). We show in [Appendix E.3](#) that estimating different bilateral frictions for migration with and without family is crucial in our setting.

migrate alone, but overall there would be 2.54% less migration, and aggregate productivity would decrease by 0.39%. These two counterfactuals highlights the crucial role of migrants' ability to leave family behind and remit income back home for the boom of Chinese aggregate productivity in the early 2000s.

Our third counterfactual investigates the effect of housing subsidies. In 2000 many resident tenants live in subsidized housing, up to 80% in the largest cities, while migrants' access is limited, especially in the largest cities (about 20%). By 2010 most housing subsidies have been cut, especially in smaller cities. We find that this policy decreased migration slightly, with negative effects on aggregate productivity (-0.5%). By contrast, we simulate an expansionary housing policy that would have provided the same access to subsidized housing to migrants as to residents, with the same generosity as in 2000. We show that this policy would have increased migration by almost 50%, almost doubled the fraction of migrants with families, and more than doubled migration to the largest cities, with gains in aggregate productivity of about close to 4%. Since additional subsidies impose a cost to the government, we check that they could have been paid for by a small tax on migrants' income (about 15%).

In the last counterfactuals we study the 2014 Hukou reform initiated by the central government to reduce the demographic pressure on the largest cities by making medium-sized cities more attractive to rural migrants and their families ([Zhang et al. 2018](#)). We use [Zhang et al. \(2018\)](#) index of Hukou stringency pre- and post-reform, and find that many medium-sized cities actually decided to tighten, rather than relax, migration restrictions. We use the fact that the Hukou stringency index correlates well with our estimates of migration costs (once instrumented), to predict how changes in Hukou stringency translate into changes in migration costs, and how these changes, in turn, affect migrant choices. Our simulation shows that the reform lowered slightly overall migration and even increased the migration to mega-cities, the opposite of what the intended effect. In contrast, we show that relaxing Hukou restrictions by one standard deviation outside the mega-cities in the spirit of the reform would result in more migration, but less in those mega-cities, with a moderate increase in family migration and negligible effects on aggregate productivity.

Our findings can be summarized as follows. First, without the ability for migrant households to consume at origin via remittances rural-urban migration would have been 2/3 lower, and aggregate productivity 4.7% lower. Second, housing subsidies at destination could increase substantially family migration to the most productive (and expensive) cities, which would result in substantial rural household welfare and overall productivity gains. Third, uncoordinated migration policies across destinations can lead to undesirable outcomes from both a welfare and productivity standpoint.

Related literature Our paper contributes to the large literature that explores the nature of frictions to labor mobility in developing countries (e.g., [Brandt et al. 2013](#), [Bryan et al. 2014](#), [Ngai et al. 2019](#), [Gai et al. 2021](#), [Adamopoulos et al. 2024](#), among others), and how these frictions shape aggregate outcomes ([Gollin et al. 2014](#), [Bryan and Morten 2019](#), [Tombe and Zhu 2019](#), [Lagakos et al. 2023](#), [Morten and Oliveira 2024](#)). Our paper is closest to structural spatial equilibrium models that quantify migration frictions and their effect on aggregate output ([Bryan and Morten 2019](#), [Tombe and Zhu 2019](#), [Morten and Oliveira 2024](#)). It is also close to [Lagakos et al. \(2023\)](#), who estimate a structural model where rural households can choose between two types of migration (permanent and seasonal), with different costs and benefits. Our contribution is to develop and estimate a spatial equilibrium model of internal migration with two modes of migration: with and without family, which imply a different allocation of income and consumption across space. The model structure with multiple nests allows for rich substitution patterns between migration modes and destinations.⁵ This enables us to consider counterfactuals that go beyond reductions in overall migration frictions and to simulate the effect of education and migration policies that differently affect family and non-family migrants.

Second, our work belongs to the body of research that emphasizes the ties that migrants keep to their origins, through the study of remittances (see [Yang 2011](#), for a review), family left-behind (see [Antman 2013](#), for a review), or temporary migration (see [Dustmann and Görlach 2016](#), for a review). The closest contributions are [Lessem \(2018\)](#), who models the joint life-cycle location choices of spouses to study how wage differentials between Mexico and the U.S. and border enforcement affect the patterns of return migration, and [Albert and Monras \(2022\)](#), who argue that displaced consumption affects the allocation of international migrants across American cities. To our knowledge, neither of these papers, nor the many others surveyed in literature reviews, study how migration frictions may affect family and non-family migrants differently and how the strategic choice to move with or without family affects the spatial distribution of economic activity. Our approach also differs in that most of the previous literature focuses on international migration while we study patterns of internal migration. This context enables us to measure migration flows with and without family between the full set of possible origins and destinations, which is difficult to do in the context of international migration. The Chinese context is important in itself: The [UNICEF \(2018\)](#) estimates that a staggering 59 million children of rural China had been left-behind by at least one par-

⁵Other papers have used models with multiple nests in the context of international migration. [Buggle et al. \(2023\)](#) study the emigration choice of German Jews fleeing Nazi persecutions using one nest for destination choice and one nest for the choice of leaving Germany. [Liu \(2023\)](#) evaluates the equilibrium response to U.S. immigration policies using a model of international migration in which migrants choose between countries, occupations, and modes of entry.

ent in 2005, with well-documented negative effects (Li et al. 2015, Démurger and Wang 2016, Fellmeth et al. 2018, Bai et al. 2018, Gao et al. 2022).

Third, our work relates to the large urban economics literature on the role of agglomeration and dispersion forces in disciplining city size (Tabuchi 1998, Au and Henderson 2006, Desmet and Rossi-Hansberg 2013, 2014). For instance, Au and Henderson (2006), Desmet and Rossi-Hansberg (2013), and Desmet and Rossi-Hansberg (2014) argue, using system-of-cities models, that (productive) cities are too small in China because of barriers to migration. Our contribution is to point out that migration barriers are not just large, but also biased against family migrants, so that productive cities in China are disproportionately populated by migrants without family. The ability to limit their local expenditures and to send remittances back to the family members left behind explains why, despite migration barriers, China saw the emergence of mega-cities.⁶ The urban literature also argue that housing market restrictions in big cities lower aggregate productivity, e.g. in the US context (Hsieh et al. 2019, Duranton and Puga 2023) and Albert and Monras (2022) suggests that international immigrants partly overcome these constraints because they consume less housing to remit income at origin. None of these papers is able to quantify how housing subsidies shape the level and the type of immigration, and the resulting gains in aggregate productivity and welfare, which is what we do in the Chinese context.

1 Data and institutional framework

This section outlines the institutional framework, describes the main data sources, and presents descriptive statistics on migration patterns and living arrangements in cities. Discussion of the empirical strategy and the data used to identify the key parameters governing migration choices is deferred to Section 4.

1.1 Migration barriers in China

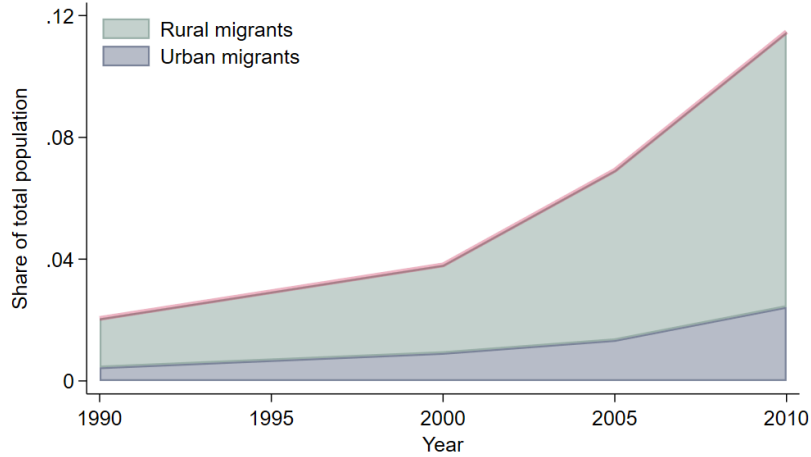
A distinctive feature of the Chinese context is the formal restriction on internal migration imposed by the *hukou* registration system, introduced in 1958. From 1958 to the late 1970s, migration was effectively illegal unless mandated by the government. During the 1980s and 1990s, with the onset of the Reform era, migration was tolerated; however, access to essential services such as food provision remained tied to the place of household registration, which severely curtailed individuals' ability to work outside their place

⁶A report from the United Nations (2018) identifies 33 cities of at least ten million inhabitants, home to about 12% of the world population, of which 27 were in low and middle income countries and 6 were in China (Shanghai, Beijing, Chongqing, Tianjin, Guangzhou, Shenzhen).

of origin for extended periods (see, e.g., [Zhang et al. 2018](#)). Figure 1 shows that inter-prefecture migration remained limited.

From 2000 onward, our study period, food provision and place of registration were decoupled, but *hukou* type continued to condition access to public goods. Agricultural *hukou* holders retained access to land at their place of registration, while non-agricultural *hukou* holders had access to welfare benefits and public services (e.g., enrollment in local schools, healthcare, urban pension plans, and subsidized housing). Owing to the increasing decentralization of migration policy during the Reform era, this general pattern masks substantial cross-location variation in *hukou* stringency. In some cities, migrants had to return to their place of registration to access basic services such as education and healthcare, or faced higher fees at their destination; in others, access to public services was more inclusive ([Song 2014](#)).⁷

Figure 1. Evolution of the migration rate by *hukou* type.



Notes: This figure shows the internal migration rate in China from 1990 to 2010, based on the Population Censuses (1990, 2000, and 2010) and the 2005 Mini-Census. A migrant is defined as an individual whose prefecture of residence differs from her prefecture of household registration. *Rural* and *urban* migrants refer to holders of agricultural and non-agricultural *hukou*, respectively.

1.2 Migration data

Migration flows Our main data source is the 2005 1% Population Survey (hereafter, the “2005 Mini-Census”), which we use to measure migration flows, returns to labor, and local prices.⁸ The sampling frame of the “2005 Mini-Census” is the Public Security

⁷After the end of our study period, in 2014, the central government partly curtailed local discretion over registration policies and initiated a gradual relaxation of migration restrictions in lower-tier cities ([Zhang et al. 2018](#)).

⁸Our analysis also relies on information on population, prices, amenities, and migrant flows from the 1990, 2000, and 2010 Population Censuses, and the “2015 Mini-Census.” These additional sources are primarily used for robustness checks and to construct baseline local characteristics.

Bureau’s 2004 population registry and covers the entire Chinese population, regardless of migration status. We use a random 20% extract of the microdata to characterize individuals’ migration status based on their current place of residence (destination), place of household registration or *hukou* (origin), *hukou* type (agricultural or non-agricultural), and family situation. Information on the date of arrival at the destination allows us to construct a bilateral prefecture-level matrix of migration flows over the period 2000–2005. Throughout the empirical analysis, we define a migrant as an individual residing in a prefecture different from that of their registration.⁹ Under this definition, 5.6% of the Chinese population in 2005 were internal migrants, of whom 80% originated from rural areas. Figure 1 puts these rates in perspective using comparable data from the 1990, 2000, and 2010 Population Censuses. The resulting migration series exhibits a structural break around 2000, coinciding with China’s accession to the WTO (Facchini et al. 2019, Tombe and Zhu 2019), with low and slowly increasing migration rates before 2000 and a sharp rise thereafter.

Living arrangements A key contribution of this paper is to better characterize the diversity of family living arrangements. To identify family structure and living arrangements, we leverage the household roster module of the 2005 Mini-Census. In a first step, we distinguish *families with children* (aged 0–18), irrespective of co-residence, from all *other* households. For simplicity, we assume a single living arrangement for these other households: those without children in this age range are considered to be co-residing in their observed location in the 2005 Mini-Census. In a second step, we focus on families with children and characterize their living arrangements. Families with children in rural areas of origin and with no migrant among the two spouses are classified as stayers. Migrant families with children—defined as those with at least one migrant member among spouses—are further classified into the following categories: (i) those living together at destination (*family moves*), (ii) those in which both adults migrate while children remain at origin (*parents move*), and (iii) those in which only one adult migrates (*mother alone* or *father alone*).¹⁰ We construct comparable measures of living arrangements in alternative data sources covering rural migrants in our setting: the Longitudinal Survey on Rural Urban Migration in China (RUMiC 2008–2009, see Akgüç et al. 2014); and the China

⁹Prefectures are the administrative level between provinces (which are directly below the central government in the Chinese administrative hierarchy) and counties. There were 345 prefectures in China in 2005. Because prefecture boundaries change over time, all data are harmonized to 2005 administrative boundaries. In a sensitivity analysis (Appendix B.7), we explore alternative definitions of migration based on the 2005 “Mini-Census” and 2000 Population Census.

¹⁰This classification is discussed in detail in Appendix A.1. It relies in particular on the fertility module administered to all female respondents aged 15–65, which records all living children and allows us to unambiguously classify mothers as migrants with or without their children.

Migrants Dynamic Survey (CMDS), a nationally representative repeated cross-section conducted by the National Health Commission every year since 2011 (Wang et al. 2021).

Remittances and consumption The living arrangements described above imply that household consumption may occur across multiple locations. We measure remittances and the consumption of non-tradables at destination using RUMiC and CMDS.¹¹ We compute the share of income remitted by dividing annual remittances by annual income. We also aggregate expenditures on housing, education, and other services (e.g., medical care and transport) to construct the share of non-tradable consumption at destination.¹²

1.3 Living conditions in cities and rural locations

Wages We use the “2005 Mini-Census” to measure wages in 2005. Since wage information is not available in the 2000 Census, we rely on average wages from the Statistical Yearbooks to proxy returns to labor at baseline.¹³ The yearbooks distinguish between the wage in the “city” proper (i.e., the urban core of the prefecture) and in the prefecture as a whole (including the rural hinterland). We exploit this distinction to measure destination and origin wages using “city” and prefecture wages, respectively.

Rents and housing conditions We use the 2005 Mini-Census to measure housing costs and characterize housing conditions for both migrants and non-migrants in 2005. The data include a detailed housing module reporting monthly rent and a wide range of dwelling characteristics. We construct a prefecture-level measure of rental prices by averaging monthly rent per square meter across tenants in private accommodation.

Hukou environment One objective is to examine how migration restrictions—both their level and their heterogeneity—affect mobility between rural and urban areas and the allocation of migrants across destinations. To measure *hukou* stringency, we follow Wu and You (2021) and use census data from 2000 and 2010, which report place of birth (at the county level), registration status, and *hukou* type (agricultural or non-agricultural).

¹¹The CMDS survey manual instructs enumerators to include in remittances cash transfers (mailed, wired, or carried home), in-kind transfers, support for left-behind children, parents, or spouses, and purchases of goods for household use. It excludes investment in businesses and savings for future use at origin (Ashraf et al. 2015); as a result, the CMDS measure may provide a lower bound on the share of income remitted. We combine the 2011 CMDS measure with that from RUMiC (2008–2009) to construct remittance shares across 221 destinations (see Appendix A.2).

¹²Additional descriptive statistics on migration patterns, migrant characteristics, consumption behavior, and access to public goods at destination are provided in Appendix A.3.

¹³These data are compiled by the National Bureau of Statistics based on the Reporting Form System on Labor Wage Statistics, the National Monthly Sample Survey System on Labor Force, and the System of Rural Social and Economic Surveys.

We compute a registration probability defined as the share of 15–64-year-old migrants born in another county who hold a local non-agricultural *hukou*. This probability is low on average and increases only modestly, from 9% in 2000 to 12% in 2010.¹⁴

Amenities Amenities are an important component of urban living conditions. As they are largely unobserved, we infer them within the structure of the model (Section 4) and assess the validity of these estimates using data on pollution and commuting in 2015. See Appendix A.2 for details on these additional data sources.

2 Motivating facts

This section presents a set of motivating facts on migration patterns, including location choices, living conditions, and consumption patterns.

2.1 The different modes of migration

As noted in the introduction, most migration models focus on the individual decision to migrate or stay in the origin community. In practice, however, migration often involves households rather than isolated individuals. Some migrants relocate with their entire household, while others split the household across locations. There are, in fact, multiple ways in which migration experiences take place.

In this section, we use the classification introduced above to group migrants into five categories—family moves, parents move, father or mother moves, and other migrants without young dependents—and present descriptive statistics on migration patterns in our context.¹⁵

In Panels A to D of Table 1, we report socio-demographic characteristics from the 2005 Mini-Census for urban residents and rural-urban migrants, separately by migration type. Panel A shows that migrants are younger than urban residents on average and have more children. This partly reflects the One-Child Policy, which allowed rural-*hukou* holders to have a second child if the first was a girl, whereas urban-*hukou* holders were typically limited to one child. Interestingly, the number of children is very similar across migration types: migrants who move with their children have on average 1.6 children, compared to around 1.5 among fathers or mothers who migrate either jointly or alone.

¹⁴In Appendix A.3, we show that this measure correlates closely with indices of *hukou* stringency from Zhang et al. (2018), based on policy documents across 124 Chinese cities. We also use auxiliary survey data to illustrate how migration restrictions affect the lives of rural-urban migrants, particularly in terms of access to public goods.

¹⁵We provide additional descriptive statistics on (i) family migration over time, (ii) the geography of migrant inflows, family migration, and remittances, (iii) the geography of migration restrictions, and (iv) the incidence of return migration in Appendix A.3.

Table 1. Descriptive statistics.

	Urban res.	Rural-urban migrants				
		Family moves	Parents move	Mother alone	Father alone	Other
<i>Panel A: Demographic characteristics (2005 Mini-Census)</i>						
Share of rural-urban migrants	–	18.98%	14.60%	5.30%	10.81%	50.31%
Age	39.968	33.538	32.551	34.080	34.490	28.453
Female (head)	0.305	0.186	0.112	0.831	0.089	0.415
Married	0.809	1.000	1.000	1.000	1.000	0.344
Number of children	1.372	1.649	1.546	1.470	1.522	0.000
Number of children (OCP*)	1.138	1.653	1.547	1.472	1.524	0.000
<i>Panel B: Education (2005 Mini-Census)</i>						
High-school degree	0.569	0.117	0.129	0.071	0.172	0.197
College degree	0.252	0.010	0.014	0.006	0.015	0.024
<i>Panel C: Economic characteristics (2005 Mini-Census)</i>						
Income (head)	1158.959	1091.510	998.630	814.193	1116.440	943.322
Hours worked per week	45.813	55.270	56.214	55.729	55.550	54.932
Housing share	0.315	0.226	0.147	0.287	0.192	0.213
<i>Panel D: Living arrangements (2005 Mini-Census)</i>						
Household size	2.899	3.638	2.186	3.465	4.418	4.152
No kitchen	0.087	0.402	0.532	0.567	0.604	0.586
No toilets	0.180	0.541	0.628	0.547	0.576	0.504
House ownership	0.805	0.171	0.064	0.088	0.055	0.088
<i>Panel E: Expenditures (RUMiC)</i>						
Remittance share	–	0.108	0.200	0.272	0.353	0.145
Non-tradables share	–	0.478	0.421	0.350	0.387	0.428
<i>Panel F: Prospects (CMDS)</i>						
Return migration	–	0.083	0.122	0.126	0.141	–
Hukou conversion	–	0.341	0.355	0.332	0.301	–
Observations (Census)	477,380	20,205	15,546	5,648	11,509	53,572
Observations (RUMiC)	–	840	591	67	341	2,744
Observations (CMDS)	–	40,609	21,684	2,917	5,412	22,973

Notes: In Panels A to D, the sample is restricted to individuals aged 18–64 living in urban areas (2005 Mini-Census). Column (1) reports statistics for urban residents with a local *hukou*. Columns (2)–(6) report statistics for individuals living in urban areas with a rural *hukou*, distinguishing between family arrangements: (2) migrant families in which both parents and at least one child co-reside at destination; (3) migrant parents who move and leave their children behind; (4) migrant mothers who move without their spouse or children; (5) migrant fathers who move without their spouse or children; and (6) migrants without children under 18. Descriptive statistics for *Income (head, RMB)* and *Hours worked per week* are restricted to individuals reporting positive working hours in the past week; *Income (head, RMB)* refers to monthly income. *Number of children* is the predicted number of children (see Appendix A.1, OCP* excludes individuals older than 25 at the time of the introduction of the One-Child Policy). *Housing share* is based on predicted values from a regression of monthly rent (in logs) for respondents in commercial housing on prefecture fixed effects interacted with dwelling characteristics. Panel E relies on the Longitudinal Survey on Rural-Urban Migration in China (2008–2009; see Appendix A); *Remittance share* is the ratio of monthly remittances to monthly income, and *Non-tradables share* is the ratio of consumption at destination (food, rent, and local services) to monthly income. The sample in Panel F is restricted to household heads aged 18–64 living in urban areas in the China Migrants Dynamic Survey (CMDS) in 2011 (and 2016); note that CMDS 2016 does not include information on non-co-resident children. *Migration across provinces* is a dummy equal to 1 if the last migration spell involves crossing a provincial border. *Return migration* is a dummy equal to 1 if the respondent is willing to return to their *hukou* registration place in the future. *Hukou conversion* is a dummy equal to 1 if the respondent is willing to convert their *hukou* to the current destination.

Panel B shows that rural-urban migrants are much less likely to have higher levels of education: between 7 and 20% have at least a high school degree, compared to 57% of urban residents. Panel C shows that migrants earn about 15-20% less than urban residents, despite working substantially longer hours per week. Differences across migration types in education, earnings, and hours worked are, however, relatively small.

Despite lower earnings, migrants tend to spend less on housing than urban residents. When the entire family migrates, the difference in housing expenditure shares is relatively small (20% for migrants versus 17% for urban residents). Mothers who migrate alone also devote a substantial share of their income to housing, likely reflecting their lower earnings relative to fathers who migrate alone. These modest differences in housing expenditure mask large disparities in housing quality. For example, more than 50% of migrants lack access to a kitchen in their dwelling, compared to less than 10% of urban residents. Housing conditions and living arrangements are somewhat better for migrants who move with their entire family. Nevertheless, homeownership rates and overall living conditions remain substantially below those of urban residents, even among migrants with dependent children.

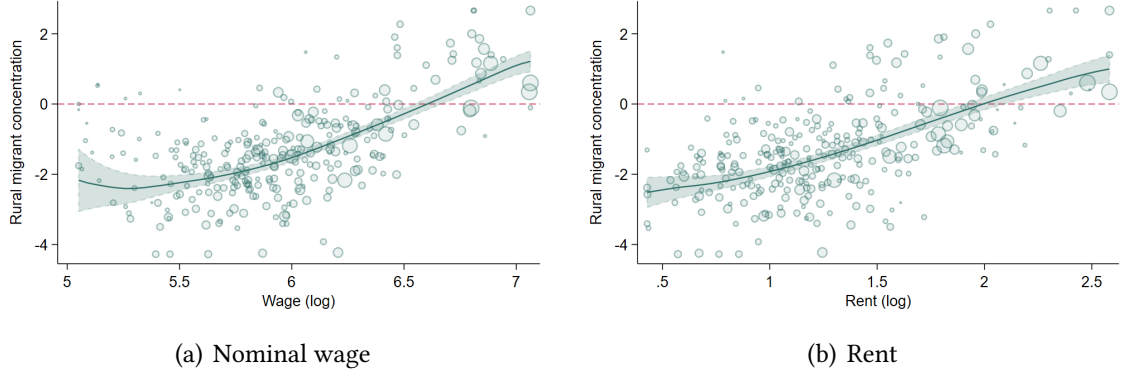
We complement this description with data from RUMiC (2008–2009), which provide information on expenditures and remittances, and from the CMDS (2011–2012), which includes information on past and current migration experiences as well as future intentions. In Panel E of Table 1, we report the average share of monthly income devoted to remittances and to non-tradable consumption (food, rent, and local services) at destination. Migrants who move with their entire family remit less than those whose parents migrate alone, and they spend a larger share of their income on non-tradables at destination.¹⁶ Panel F shows that few migrants plan to return (only about 13% on average), while a larger share (around 30%) express a willingness to convert their *hukou* to their current location if possible.¹⁷ Migrants who move with their children are only slightly less likely to plan to return and somewhat more likely to seek local *hukou*. Migrants without dependent children are younger and remit slightly less, but are otherwise similar in many respects to those with children.

Overall, we find that migrants display markedly different consumption behaviors depending on whether they migrate with or without children. We further investigate these differences in the next section, exploiting geographic variation.

¹⁶The share of migrants living with family is much higher in RUMiC and CMDS (2011–2012, see Table 1) than in the 2005 Mini-Census. This difference does not reflect changes in migration patterns over time (see Appendix A.3) or the geographic coverage of CMDS and RUMiC. A more likely explanation is that surveys are less able to capture migrants who are weakly attached to the destination.

¹⁷One reason migrants are not all willing to convert their *hukou* is that holding a rural *hukou* at origin confers certain advantages, such as land rights (Adamopoulos et al. 2024).

Figure 2. Rural migrant concentration, wages, and rents.



Notes: The y-axis reports the migrant concentration in city c , m_c . In panel (a), the x-axis reports a measure of (log) monthly wage; in panel (b), the x-axis reports a measure of (log) monthly rent per square meter. Wages and rents are both constructed using the 2005 Mini-Census. A bubble is a prefecture of destination and is weighted by its initial urban population in 2000. The lines are local polynomial fits, where each observation is weighted by population.

2.2 Spatial differences between migrants and non-migrants

Spatial concentration Migrants and urban residents differ along several dimensions, as discussed in the previous section. Some of these differences reflect geographic sorting. Migrants move towards destinations that suit their needs and ambitions, sometimes shaped by family considerations, and these may differ systematically from the locations of urban residents. To characterize the allocation of migrants across cities, we construct m_c , a measure of migrant concentration relative to urban residents based on census data:¹⁸

$$m_c = \log \left(\frac{M_c / (\sum_d M_d)}{R_c / (\sum_d R_d)} \right),$$

where R_c denotes the initial population in city c in 2000 and M_c the number of rural migrants arriving in city c between 2000 and 2005. This measure equals zero if migrants are distributed proportionally to the resident population—equivalently, if immigration rates are constant across cities. Figure 2 shows that this is not the case. Panel (a) plots migrant concentration against (log) nominal monthly wages, w_c , in 2005. The relationship is clearly positive: a 1% increase in wages is associated with a 2.44% increase in rural migrant concentration.¹⁹ Panel (b) plots migrant concentration against (log) monthly rents in 2005. The relationship is again positive: a 1% increase in rents is associated with a 1.96% increase in rural migrant concentration. Overall, migrants appear to be attracted

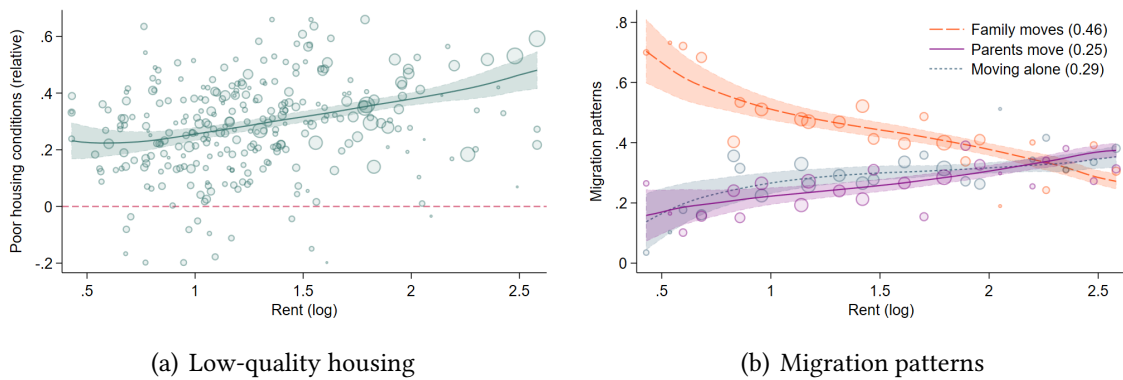
¹⁸We discuss migrant concentration using the so-called Zipf law of city size in Appendix B.1.

¹⁹Appendix B.2 provides more evidence on this relationship. We show that migrants are concentrated in cities where workers work more hours, but are still overly represented in cities with high hourly wages. We also show that rural migrants are concentrated in high-wage, high-rent cities when compared to urban migrants, who may face higher costs of relocating.

by higher nominal wages and are not deterred by higher housing costs.

Living conditions The finding that rural migrants locate in cities with high living costs may appear puzzling, given that they are poorer than urban residents. We therefore examine *how* migrants live in these expensive locations. Panel (a) of Figure 3 shows that they are about 30 percentage points more likely to reside in low-quality housing, based on dwelling characteristics (building materials, kitchen, bathroom, and toilet facilities). Importantly, the share of migrants living in precarious conditions increases sharply with housing prices: rural migrants are about 20 percentage points more likely to live in low-quality dwellings than urban residents in the least expensive destinations, but this gap rises to 50 percentage points in the most expensive cities.²⁰

Figure 3. Migration patterns and housing conditions.

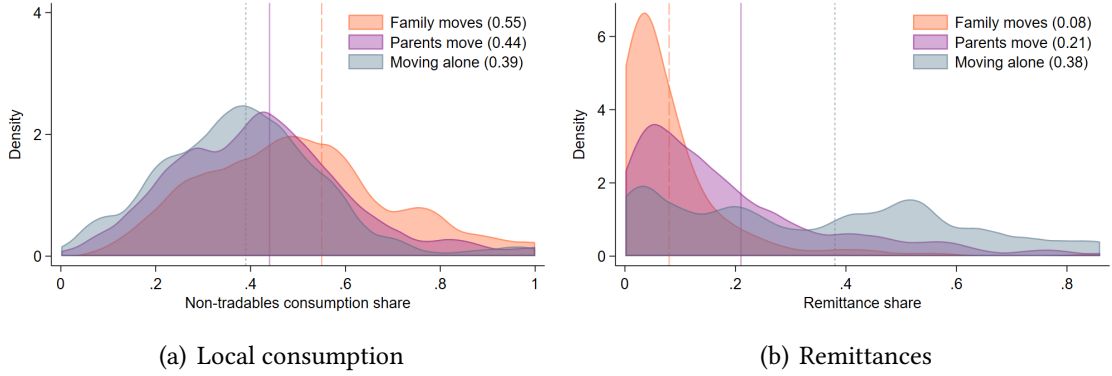


Notes: The x-axis reports a measure of (log) monthly rents constructed using the 2005 Mini-Census. In panel (a), the y-axis reports the difference between the fraction of rural migrants and the fraction of urban residents who live in poor housing conditions, based on their dwelling characteristics measured in the 2005 Mini-Census—the index is based on housing material and the types of kitchen, bathroom, and toilet in the dwelling. In panel (b), the y-axis reports the incidence of the different migration patterns: family moves (orange, long-dash line); parents move (purple, plain); and moving alone (blue, short-dash). A bubble is a prefecture of destination and is weighted by its initial urban population in 2000. The lines are local polynomial fits, where each observation is weighted by population.

To a large extent, these patterns reflect differences in family arrangements. Panel (b) shows the distribution of migrants across family arrangements as a function of local housing costs. Migrants who move with their children are disproportionately located in smaller, lower-cost destinations. By contrast, migrants who leave their dependent children behind are much more likely to settle in high-cost cities, from which they can remit higher incomes to their children.

²⁰Appendix B.3 provides sensitivity analyses using alternative definitions of living arrangements (e.g., co-residing with or without close family members, with or without children, or with or without a spouse). We also study how migrants sort across destinations with different *hukou* stringency, characterize the dynamics of migration arrangements across cities (Appendix B.5), and document migrants' intentions regarding return migration (Appendix B.6).

Figure 4. Remittances and expenditures on non-tradable goods and services in the city.



Notes: In panel (a), we report the distribution of non-tradable expenditures at the main place of residence as a share of income (sources: RUMIC, 2008, 2009). In panel (b), we report the distribution of remittances as a share of income; remittances are extracted from RUMIC (2008, 2009) as described in Appendix A.2. In both cases, we distinguish between: rural migrant households moving as a whole family (orange, long-dash line); rural migrant households moving without children (purple, plain line); and rural migrants moving alone (blue, short-dash line). The vertical lines indicate the average value within each sub-sample—also reported between parentheses in the legend.

Remittances and expenditures on non-tradable goods and services Rural migrants concentrate in high-cost locations despite facing difficult living conditions. What explains this behavior? Figure 4 shows that migrants consume only a small share of their income locally, particularly when they migrate without their family. Differences across living arrangements in local consumption—measured as the ratio of expenditures on non-tradable goods and services to income at destination—amount to roughly 12 percentage points (Panel a). The figure also shows that migrants remit a substantial share of their income to their place of origin (Panel b). Rural migrants without their family remit about 28% of their income, compared to 11% for those living with their family. These transfers are economically large and help explain the prevalence of migration arrangements involving split families: remitting 28% of an urban income is close to financing a full rural income (see Appendix B.4 for a derivation of the rural-urban wage gap and a discussion of housing and education expenditures).

These stylized facts help explain why rural migrants in China are often described as a “floating population.” They select high-wage destinations to maximize nominal income and cope with high living costs by leaving their families behind and reducing local consumption, thereby increasing remittances to their place of origin. In the next section, we develop a spatial equilibrium model of migration that allows migrants to allocate consumption across household members and locations, and to choose whether to migrate with or without spouses or children.

3 Model

In this section, we introduce a quantitative spatial model of location choice in which rural households account for the utility of both parents and children when deciding where family members live and how much they consume.²¹

3.1 Preferences

We assume two exogenous types of households: single-individual households, capturing individuals who have not yet formed a family but are old enough to live independently from their parents; and family households, composed of a father, a mother, and dependent children. Since the novelty of the paper lies in modeling the latter, we focus the discussion on family households. The migration decisions of single individuals are described at the end of the section.

We assume that the utility of a household χ is given by:

$$U = \mathcal{U} \left(u(C^f), u(C^m), u(C^c) \right) - \bar{u} + \varepsilon(\chi)$$

where $\mathcal{U}$ aggregates the utilities of the father (f), the mother (m), and the children (c); u is strictly increasing and concave; C^f , C^m , and C^c denote the consumption bundles of the father, the mother, and the children, respectively; $\bar{u}$ is a direct welfare cost capturing both monetary and non-monetary migration barriers; and $\varepsilon(\chi)$ is an idiosyncratic taste shock, described below. In the baseline model, we assume that household utility is a linear combination of individual utilities:

$$\mathcal{U} \left(u(C^f), u(C^m), u(C^c) \right) = \xi_f u(C^f) + \xi_m u(C^m) + (1 - \xi) u(C^c), \quad \xi = \xi_f + \xi_m,$$

so that the household's problem can be decomposed into three steps: (i) choosing a living arrangement, (ii) allocating the total budget across family members, and (iii) choosing a consumption bundle C^g for each member g . The baseline model further imposes that $u(C) = \ln(C)$.²²

A household residing in r chooses among five living arrangements: (s) staying in the

²¹For simplicity, we assume that urban residents do not move—only rural households are allowed to relocate. We relax this assumption in Appendix C.3. We also assume a single type of labor and extend the model to multiple labor types in Appendix C.4.

²²The baseline model does not explicitly incorporate lifecycle considerations, circular migration, or return migration (see, e.g., Lessem 2018), which are limited in our context (see Appendix B.5). These features could be introduced, for instance, by adding an additional sub-utility capturing parents' utility in retirement, or by interpreting children's utility as part of the parents' future utility stream. Alternatively, one could model both intertemporal choices and coexisting family members within an overlapping-generations framework. We discuss these extensions in Appendix C.2.

rural location, $\ell_m = \ell_f = \ell_c = r$; (m_1) or (m_2) the father or the mother migrating alone to an urban center u , $\ell_f = u$, $\ell_m = \ell_c = r$ or $\ell_m = u$, $\ell_f = \ell_c = r$; (m_3) both parents migrating to u while leaving children at origin, $\ell_f = \ell_m = u$, $\ell_c = r$; and (m_4) relocating the entire family to u , $\ell_m = \ell_f = \ell_c = u$. This choice determines the household's budget constraint through location-specific prices and returns to labor. Indeed, we assume that the consumption bundle C is a Cobb-Douglas aggregate of a (perfectly) tradable good and a non-tradable good. As a result, the household faces different prices depending on where each family member resides. Letting $\ell = (\ell_m, \ell_f, \ell_c)$ denote a given living arrangement, the household's budget constraint can be written as:

$$\mathcal{P}^f(\ell) C^f(\ell) + \mathcal{P}^m(\ell) C^m(\ell) + \mathcal{P}^c(\ell) C^c(\ell) \leq \mathcal{W}(\ell), \quad (1)$$

where $\mathcal{P}^g(\ell)$ is the price index faced by member g under living arrangement ℓ , and each living arrangement comes with a different stream of income, $\mathcal{W}(\ell)$.²³

3.2 Indirect utility

Indirect utility The indirect utility V associated with living arrangement ℓ is:

$$\ln V(\ell) = \xi_f u(C^{f*}) + \xi_m u(C^{m*}) + (1 - \xi) u(C^{c*}) - \bar{u}(\ell), \quad (2)$$

which depends on wages and price indices, themselves determined by the locations of family members. The term $\bar{u}(\ell)$ captures residual welfare costs associated with migration, such as the costs of leaving children behind, travel costs, or land (in)security at origin (see, for instance, Brandt et al. 2013, Gollin et al. 2014, Bryan and Morten 2019, Tombe and Zhu 2019, Gai et al. 2021, Adamopoulos et al. 2024).

Our quantitative model further specifies that, for each household member $g \in f, m, c$, $C^g = (C_t^g)^{1-\alpha-\beta} (C_n^g)^\alpha (C_r^g)^\beta$, where: C_t^g denotes consumption of the tradable good, C_n^g consumption of the non-tradable good at the place of residence, and C_r^g residual consumption of the non-tradable good at origin. The latter captures the empirical fact that even households migrating with family dependents remit a small share of their income.

Income streams depend on the living arrangement: $\mathcal{W}(r, r, r) = 2w_r$, $\mathcal{W}(u, r, r) = \kappa_1(w_u^f + w_r)$, $\mathcal{W}(r, u, r) = \kappa_2(w_u^m + w_r)$, $\mathcal{W}(u, u, r) = \kappa_3(w_u^f + w_u^m)$, and $\mathcal{W}(u, u, u) = \kappa_4(w_u^f + w_u^m)$, where w_u^f and w_u^m denote the father's and mother's urban income, respectively, and w_r denotes rural income. These assumptions allow us to derive the allocation of consumption across goods, household members, and locations (see Appendix C.1).

²³ $\mathcal{W}(\ell)$ may depend on the current labor income of the father and the mother, as well as on arrangement-specific remittance, labor supply, or education costs. Leaving children behind may adversely affect their educational attainment (Du 2023) and their future ability to migrate to cities (Gao et al. 2022). Conversely, human capital accumulation may be more costly in urban than in rural areas.

Omitting additive constants and assuming that the tradable good is the numeraire, the indirect utility of a household staying in location r is:

$$\ln V_r^s = \ln(2w_r) - (\alpha + \beta) \ln p_r.$$

The indirect utility of a household in which only the father migrates from origin r to destination u is:

$$\ln V_{ru}^{m_1} = \ln(w_u^f + w_r) - \xi_f \alpha \ln p_u - [\beta + (1 - \xi_f)\alpha] \ln p_r + \ln \kappa_1 - \bar{u}_{ru}^{m_1}.$$

The indirect utility of a household in which only the mother migrates from origin r to destination u is:

$$\ln V_{ru}^{m_2} = \ln(w_u^m + w_r) - \xi_m \alpha \ln p_u - [\beta + (1 - \xi_m)\alpha] \ln p_r + \ln \kappa_2 - \bar{u}_{ru}^{m_2}.$$

The indirect utility of a household in which both parents migrate from r to u while children remain at origin is:

$$\ln V_{ru}^{m_3} = \ln(w_u^f + w_u^m) - \xi \alpha \ln p_u - [\beta + (1 - \xi)\alpha] \ln p_r + \ln \kappa_3 - \bar{u}_{ru}^{m_3}.$$

The indirect utility of a household migrating from r to u with the entire family is:

$$\ln V_{ru}^{m_4} = \ln(w_u^f + w_u^m) - \alpha \ln p_u - \beta \ln p_r + \ln \kappa_4 - \bar{u}_{ru}^{m_4}.$$

These expressions show that bilateral migration costs, $\{\bar{u}_{ru}^{m_j}\}$, and effective income differences, $\{\ln \kappa_j\}$, have a similar effect on the common components of household utilities, allowing us to define effective migration costs as $\{\tau_{ru}^{m_j} = \bar{u}_{ru}^{m_j} - \ln \kappa_j\}$, which we use throughout the rest of the paper.

Intuition The preceding assumptions on preferences and production allow us to compare welfare across different choices and to highlight the trade-offs underlying migration decisions, as shaped by spatial and gender differences in returns to labor, prices of non-tradables across space, and effective migration costs. First, consider the welfare difference between both parents migrating without children (m_3) and staying at origin (s):

$$\ln V_{ru}^{m_3} - \ln V_r^s = \ln(w_u^f + w_u^m) - \ln(2w_r) - \xi \alpha \ln \frac{p_u}{p_r} - \tau_{ru}^{m_3},$$

where parents trade off wage gains, $\ln(w_u^f + w_u^m) - \ln(2w_r)$, against higher living costs, $\xi \alpha \ln \frac{p_u}{p_r}$, and other (potentially non-monetary) migration costs, $\tau_{ru}^{m_3}$. Second, migrants

who leave family dependents behind have stronger incentives to locate in high-wage, high-price destinations, as illustrated by the relative utility of bringing children when migrating:

$$\ln V_{ru}^{m_4} - \ln V_{ru}^{m_3} = -(1 - \xi)(\alpha + \beta) \ln \frac{p_u}{p_r} - [\tau_{ru}^{m_4} - \tau_{ru}^{m_3}],$$

where the monetary gains from leaving children behind and shifting consumption to lower-cost locations are weighed against the non-monetary costs of such arrangements. Third, gender differences in returns to labor, preference weights, and migration costs shape the relative incidence of father-only versus mother-only migration:

$$\ln V_{ru}^{m_2} - \ln V_{ru}^{m_1} = \ln \frac{w_u^m + w_r}{w_u^f + w_r} - (\xi_m - \xi_f)\alpha \ln \frac{p_u}{p_r} + \tau u_{ru}^{m_2} - \tau u_{ru}^{m_1},$$

where a gender wage gap, $\ln \frac{w_u^m + w_r}{w_u^f + w_r}$, discourages female migration and offsets the fact that mothers typically consume less at destination, $(\xi_m - \xi_f)\alpha \ln \frac{p_u}{p_r}$.

The following lemma provides the intuition behind some of the trade-offs faced by households in choosing their location(s) and living arrangements.

Lemma 1. *Households migrate if real wage differences are high between origin and destination, relative to effective migration costs. When migrating, migrants leave family dependents at origin when the effective migration cost for doing so is relatively low and the difference in the cost of living between rural and urban locations is high—the strength of the latter effect depending on the consumption share allocated to children $(1 - \xi)$ and the share of non-tradables in overall consumption $(\alpha + \beta)$.*

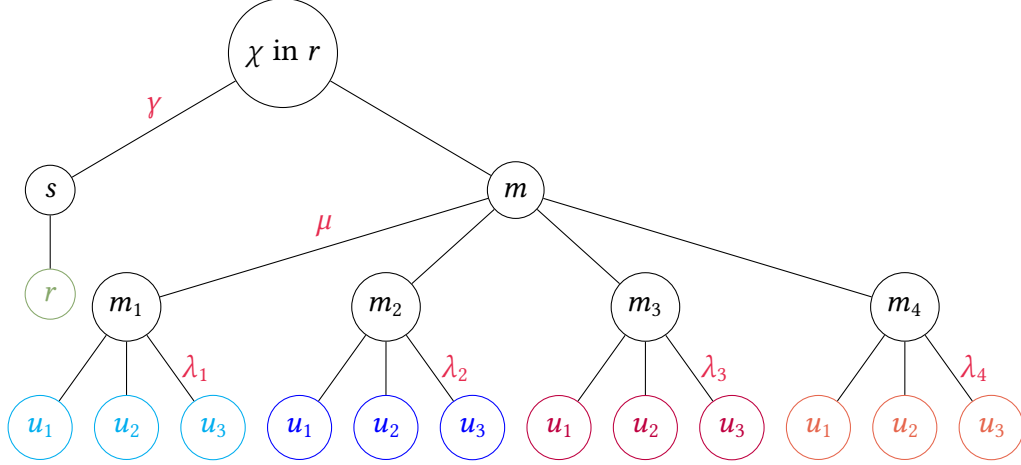
3.3 Location choice

A nested structure The indirect utility representation allows us to formulate a discrete, nested choice model capturing whether, how, and where households relocate. The problem is given by:

$$\max_{\mathbf{k}} \{\ln V_{\mathbf{k}} + \varepsilon_{\mathbf{k}}\},$$

where we recast the choice of living arrangement as $\mathbf{k}$, a vector capturing whether ($m \in \{0, 1\}$), how ($j \in \{1, 2, 3, 4\}$), and where (u) to relocate. For instance, the choice $\ell = (\ell_f, \ell_m, \ell_c) = (u, r, r)$ corresponds to migration ($m = 1$) with the father moving alone ($j = 1$) to destination u . We assume that idiosyncratic preferences for relocation, $\varepsilon_{\mathbf{k}}$, follow a generalized extreme value (GEV) distribution with a nested structure (see Figure 5). The upper nest captures the decision to relocate, with shape parameter γ ; the intermediate nest captures the choice of migration mode, with shape parameter μ ; and the lower nests capture location choices, with shape parameters $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$. Given

Figure 5. The nested logit structure.



Notes: The Figure represents the nested structure induced by our assumptions on the distribution of idiosyncratic preferences for relocation, ϵ_k . The parameter γ is the shape parameter for the upper nest of the generalized extreme value distribution; the parameter μ is the shape parameter for the intermediate nest of the generalized extreme value distribution; and the parameters $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ are the shape parameters for the lower nests of the generalized extreme value distribution.

this structure, the probability that household i “ j -relocates” from r to u is given by:

$$\pi_{jru} = \left(\frac{V_r^m}{V_r} \right)^{1/\gamma} \left(\frac{V_r^{m_j}}{V_r^m} \right)^{1/\mu} \left(\frac{V_{ru}^{m_j}}{V_r^{m_j}} \right)^{1/\lambda_j}. \quad (3)$$

By the law of large numbers, π_{jru} is also the share of households who choose to “ j -move” from r to u , which can be decomposed into: the share of households who are born in r and migrate elsewhere, $(V_r^m/V_r)^{1/\gamma}$; the fraction of households that migrate with living arrangement j among those who migrate, $(V_r^{m_j}/V_r^m)^{1/\mu}$; and the fraction of those movers choosing city u , $(V_{ru}^{m_j}/V_r^{m_j})^{1/\lambda_j}$.²⁴

The migration of single individuals Our framework also includes a second (exogenous) type of households, modeled as single individuals for simplicity. We allow the share of each exogenous type to vary by location of origin, and we allow for them to have different preferences across goods of different nature. Relative to family households, the problem faced by individual migrants is simpler, as they do not face within-household living arrangement decisions.

First, consistent with the previous notation, the indirect utility of moving to destination u depends on wages, w_u , prices at destination and origin (with a residual share

²⁴The expected value of being born in r , V_r , is given by $V_r = \left[(V_r^m)^{1/\gamma} + (V_r^s)^{1/\gamma} \right]^\gamma$, where V_r^s is the value of (fully) staying in location r . The expected value of relocating from r is $V_r^m = \left[\sum_{j \in \{1,2,3,4\}} (V_r^{m_j})^{1/\mu} \right]^\mu$; and the expected value of “ j -relocating” from r is $V_r^{m_j} = \left[\sum_u (V_{ru}^{m_j})^{1/\lambda_j} \right]^{\lambda_j}$.

β_O remitted back), and effective migration costs, τ_{ru}^O . Second, the choice set involves only whether ($m \in \{0, 1\}$) and where (u) to relocate. The corresponding nested structure omits the intermediate nest and includes only a lower nest (location choice, with shape parameter λ_O) and an upper nest (migration decision, with shape parameter γ_O).

3.4 Production and equilibrium

We now turn to the production side of the economy. Rural areas produce a freely traded agricultural good according to a linear production function, $Y_r = Z_r L_r$. Profit maximization implies that $w_r = Z_r$, where w_r denotes the wage (i.e., the price of labor) in rural area r .

Second, the production of tradable output at u leads to the following (inverse) labor demand equation:

$$\ln w_u = \ln A_u - \frac{1}{\sigma} \ln L_u + \frac{1}{\sigma} \ln Y_u, \quad (4)$$

where A_u is location-specific productivity, L_u is labor, and σ governs the elasticity of labor demand.

Third, we assume that non-tradable output, which we interpret as housing services, is produced by combining tradable inputs and land:

$$\ln Y_u^H = \ln T_u^H + \eta \ln p_u$$

where T_u^H denotes land available for housing, η is the housing-supply elasticity, and p_u is the unit price of the non-tradable good. We can now define an equilibrium:

Definition 1. *The equilibrium is defined by: (a) households decide whether, how, and where to migrate, and migrant flows are given by Equation (3); (b) households decide how much to consume; (c) intermediate producers maximize profits and determine the demand for labor, so that the wage w_u is determined by Equation (4); and (d) markets for the non-tradable good clear to determine the price p_u in each urban location by equating demand and supply.*

3.5 Extensions and reinterpretations of the model

Our baseline model is static.²⁵ However, it can be reinterpreted in a dynamic framework. For instance, parents may care about both their own utility and the expected future utility of their children:

$$\mathcal{U}(u(C^f), u(C^m), \mathbb{E}[u(C^c)]) = \xi_f u(C^f) + \xi_m u(C^m) + (1 - \xi_f - \xi_m) \mathbb{E}[u(C^c)]$$

²⁵This section summarizes a more detailed discussion in Appendix C.2.

We can also allow income to include both current parental wages and the expected future wages of children, all depending on location choices. In this extension (E1), expectations about future wages and about future migration prospects shape current migration decisions: for example, higher expected urban wage growth encourages family migration, whereas higher expected rural wage growth discourages migration and increases the likelihood that migrants leave their children at origin.

We have so far abstracted from return migration and family reunification. A simple 2-period extension, in which parents can change family arrangements across the two periods, would allow us to capture these dynamics.

Finally, we abstract from the role of migration in human capital accumulation. In extension (E3), preferences take the form:

$$\mathcal{U}(u(C^f), u(C^m), u(C^{c_1}), u(C^{c_2}), u(C^{c_2})) = \xi_f u(C^f) + \xi_m u(C^m) + \xi_1 u(C^{c_1}) + \xi_2 u(C^{c_2}),$$

where $C^{c_2} = \gamma H$ denotes children's future consumption, assumed to be linear in human capital H , which is chosen by the household. In this setting, (i) migration increases the resources available for investment in human capital, and (ii) leaving children behind allows households to benefit from lower rural education costs.²⁶

4 Estimation

In this section, we estimate the key parameters of the model: (i) preference parameters; (ii) the shape parameters of the location choice model, which govern the responsiveness of migration flows to destination characteristics, migration modes, and origin conditions; and (iii) the elasticities governing the response of wages and rents to migrant inflows at destination. The estimation exploits several sources of exogenous variation, which we introduce here and describe in greater detail in Appendix D.

4.1 Estimation of the preference parameters

We begin by describing how we recover the components entering the indirect utilities. Under Cobb-Douglas preferences, the optimal allocation of consumption across goods and family members implies that fixed shares of income are devoted to each component of the consumption bundle.²⁷ In the four-arrangement version of the model, data on

²⁶Although education quality may be higher in urban areas, rural migrants often lack access to public schools and must rely on informal, lower-quality private institutions or pay substantial fees (Goodburn 2009, Dang et al. 2020). As a result, education is effectively more costly in urban areas for rural children, partly due to *hukou* policies.

²⁷We estimate the elasticity of substitution between non-tradable consumption across locations in Appendix D.1. We find limited substitutability across locations; however, the elasticity between consumption

expenditures and remittances allow us to identify the non-tradable expenditure share (α), the remittance share that persists even for migrants living with their children (β), the overall weight placed on parents relative to children (ξ), and the relative weight of the father within parental utility (ξ_f).

Using the RUMiC 2008 and CHIP 2007 surveys, we compute four key moments: (i) the share of income spent on non-tradables by migrants living with their children, $\alpha + \beta = 0.627$; (ii) the share of income remitted by migrants living with their children, $\beta = 0.128$; (iii) the share of income remitted by migrant parents without children, $(1 - \xi)\alpha + \beta = 0.195$; and (iv) the share of income remitted by migrant fathers living alone, $(1 - \xi_f)\alpha + \beta = 0.349$. These moments imply the following preference parameters (see Appendix C.1 in which we also derive preference parameters for single individuals):

$$\alpha = 0.500, \quad \beta = 0.128, \quad \xi = 0.864, \quad \xi_f = 0.556.$$

We then combine these parameters with observed wages and prices across locations to construct real wages for the four migration arrangements: father alone, mother alone, both parents with children left behind, and whole-family migration.

4.2 Estimation of the location choice model

The location choice model features three nests. The lower nests govern substitution across destinations within each migration arrangement and are characterized by the parameters $\{\lambda_j\}_{j \in \{1,2,3,4\}}$. The middle nest governs substitution across migration arrangements and is summarized by μ , while the upper nest governs the migration-versus-staying decision and is summarized by γ . We estimate these parameters sequentially.

The lower nests (λ_j) For each migration arrangement $j \in \{1, 2, 3, 4\}$, Equation (3) implies:

$$\ln \pi_{jru}^c = \ln \frac{M_{jru}}{M_{jr}} = \delta_{jr} + \frac{1}{\lambda_j} [\ln w_u - \alpha \ln \mathcal{P}_{jru}] + \varepsilon_{jru}, \quad (5)$$

where r denotes the origin, u the destination, and j the migration arrangement. The dependent variable is the probability of choosing destination u , conditional on arrangement j and origin r . The fixed effect δ_{jr} captures the arrangement-specific inclusive value of leaving origin r .

We estimate Equation (5) using Poisson pseudo-maximum likelihood on the largest connected set of origin $\times$ destination pairs, instrumenting destination real wages with

at origin and destination is close to one. For this reason, we adopt a Cobb-Douglas structure of preferences in the baseline model.

a trade-induced labor demand shock following [Facchini et al. \(2019\)](#). The resulting estimates of the lower-nest parameters are reported in [Table 2](#).

Table 2. The lower nests (λ_j).

	Father alone (1)	Mother alone (2)	Both parents (3)	Whole family (4)
Value at destination	14.093 (2.654)	16.741 (2.585)	10.145 (1.611)	6.424 (1.229)
Observations	74,031	74,031	74,031	74,031
Migration mode	$j = 1$	$j = 2$	$j = 3$	$j = 4$
F-stat	19.41	25.60	18.83	15.72

Notes: The unit of observation is an origin-destination prefecture pair in 2005. The specification is estimated using a two-stage Poisson pseudo-maximum likelihood (PPML-IV) estimator, with population weights at origin in 2000. Standard errors (reported in parentheses) are two-way clustered at the origin and destination levels. The explanatory variable is the model-implied value at destination under migration arrangement j , instrumented using a trade-induced labor demand shock following [Facchini et al. \(2019\)](#). Controls include origin fixed effects, (log) population at destination in 2000, and (log) geodesic distance between origin and destination prefectures.

These estimates indicate that migrants who leave their children behind are more responsive to destination real wages than those who migrate with their entire family. Responsiveness is highest for mothers and fathers migrating alone, lower for both-parents migration, and lowest for whole-family migration. This arrangement-specific sensitivity to destination conditions—relative to idiosyncratic preferences—shapes the sorting of family arrangements across locations, complementing the model’s central mechanism—migrants who leave dependents behind are less exposed to destination living costs and therefore sort more strongly toward high-wage destinations.

The middle nest (μ) The middle nest governs substitution across migration arrangements:

$$\ln \pi_{jr}^c = \ln \frac{M_{jr}}{M_r} = \frac{1}{\mu} \ln(V_r^{m_j}) + \iota_j + \theta_r + \varepsilon_{jr},$$

where M_{jr} denotes the number of migrants from origin r choosing arrangement j , and M_r the total number of migrants from origin r . The terms ι_j and θ_r are arrangement and origin fixed effects, respectively. The values $V_r^{m_j}$ are derived from the model structure and the lower-nest estimates.

We estimate the equation using Poisson pseudo-maximum likelihood and report the results in [Table 3](#). In column (2), we instrument $V_r^{m_j}$ using predicted rents based on land-supply constraints at potential destinations, which disproportionately affect migration arrangements involving larger families at destination.

Our preferred estimate implies a relatively small value of μ , around $1/5.419 \approx 0.18$, indicating that households are highly responsive to differences in the relative values of

Table 3. The middle nest (μ).

	PPML (1)	PPML-IV (2)
Value of j -migrating	1.483 (0.484)	5.419 (2.745)
Observations	771	771
F-stat	–	52.31

Notes: The unit of observation is an origin $r \times$ migration arrangement j . Standard errors (reported in parentheses) are clustered at the origin level. The specification is estimated using Poisson pseudo-maximum likelihood (PPML), with population weights at origin in 2000. Controls include origin and migration-arrangement fixed effects. The dependent variable is the share of migrants from origin r choosing arrangement j . The explanatory variable is the model-implied inclusive value of arrangement j from origin r . In column (2), this variable is instrumented using predicted rents based on land-supply constraints at potential destinations.

migration arrangements. In other words, changes in living costs or migration barriers that affect the relative attractiveness of moving alone, migrating as parents, or relocating with children translate into large shifts in the family arrangements of migrants.

The upper nest (γ) The upper nest links the relative value of migration to the emigration rate:

$$\ln \left(\frac{\pi_r}{1 - \pi_r} \right) = \frac{1}{\gamma} \ln \left(\frac{V_r^m}{V_r^s} \right) + \varepsilon_r,$$

where V_r^m denotes the inclusive value of migration from origin r , and V_r^s the value of staying, constructed from rural wages and prices. To identify γ , we instrument origin conditions using agricultural shocks based on local cropping patterns and agricultural commodity price movements, following [Imbert et al. \(2022\)](#).

Table 4. The upper nest (γ).

	PPML (1)	PPML-IV (2)
Relative value of migration (V_r^m/V_r^s)	0.082 (0.286)	2.669 (1.169)
Observations	257	257
F-stat	–	17.67

Notes: The unit of observation is a prefecture of origin. Robust standard errors are reported between parentheses. The specification is estimated using Poisson pseudo-maximum likelihood (PPML), with population weights at origin in 2000. In column (2), the instrument is a local agricultural output shifter following [Imbert et al. \(2022\)](#).

Our preferred estimate implies $\gamma = 1/2.669 \approx 0.375$, indicating that emigration responds strongly to changes in the relative value of migrating versus staying.

The bilateral migration costs (τ) An important determinant of household location choice is the distribution of bilateral migration costs $\tau_{ru}^{m_j}$, which depend on the origin r , the destination u , and the migration arrangement j . We recover these costs from the model-implied migration probabilities:

$$\pi_{jru} = \left(\frac{V_r^m}{V_r} \right)^{1/\gamma} \left(\frac{V_r^{m_j}}{V_r^m} \right)^{1/\mu} \left(\frac{V_{ru}^{m_j}}{V_r^{m_j}} \right)^{1/\lambda_j},$$

together with the estimated lower-, middle-, and upper-nest parameters and the monetary components of V_r^s and $V_{ru}^{m_j}$. Conditional on a normalization, we recover the residual non-monetary costs that rationalize the observed migration matrix.

The pattern of estimated bilateral costs is consistent with the qualitative evidence in Section 2: non-monetary costs are highest for mothers migrating alone and lowest for whole-family migration, while in less attractive cities parents may still prefer to migrate without their children. These reduced-form costs also correlate strongly with migration policies, particularly *hukou* restrictions and the probability of obtaining local registration.

Table 5. Migration costs (τ_{jru}) and migration policies.

	Father alone	Mother alone	Both parents	Whole family
Migration costs (τ_{jru})	(1)	(2)	(3)	(4)
<i>Panel A: stringency index</i>				
Hukou stringency	1.079 (0.426)	1.619 (0.464)	1.446 (0.629)	1.670 (0.803)
Observations	2,334	2,080	2,000	2,118
F-stat	7.50	9.82	6.61	5.73
<i>Panel B: conversion probability</i>				
Hukou conversion	-8.440 (3.184)	-10.205 (3.317)	-9.730 (3.885)	-9.502 (3.769)
Observations	2,894	2,551	2,486	2,694
F-stat	7.30	9.67	7.09	10.88

Notes: The unit of observation is an origin-destination prefecture pair in 2005 with finite migration costs (τ_{jru}). Standard errors (reported in parentheses) are clustered at the destination level. The explanatory variable is the model-implied migration cost (τ_{jru}) under migration arrangement j . The instrument is the relative level of grain reserves in 2000, g_u . Controls include origin fixed effects and (log) geodesic distance between origin and destination prefectures.

One component of migration costs (τ_{jru}) reflects restrictive migration policies. These policies are influenced by factors that are imperfectly observed and may directly affect migrants' choices (e.g., local labor market conditions, the quality of public goods, housing policies, or the state of local finances). To isolate exogenous variation in migration

policies, we follow [Zhang et al. \(2020\)](#) and use the level of grain reserves in 2000, g_u , as an instrument, allowing us to identify their causal impact on the spatial distribution of migration barriers.²⁸

In Table 5, we report the effect of migration policies on inferred migration barriers across the four family arrangements, using two proxies. The first is the *hukou* stringency composite index developed by [Zhang et al. \(2018\)](#), which combines restrictions related to *hukou*-dependent access to public services, housing, and labor markets (Panel A). The second proxy is the probability of *hukou* conversion between 2000 and 2010, following [Wu and You \(2021\)](#) (Panel B). More stringent *hukou* policies are associated with significantly higher inferred migration costs, with somewhat weaker effects for fathers migrating alone. To illustrate the implied welfare differences across destinations, consider the median city in terms of restrictions (Zhuzhou, Hunan) versus the most restrictive city (Beijing). The composite index is 0.60 lower in Zhuzhou. Our estimates imply that such a policy gap translates into substantial welfare differences, equivalent to a 172% increase in real wages for whole-family migrants and a 91% increase for fathers migrating alone.²⁹

4.3 Labor demand and housing supply

The final block of the model concerns production at destination. The labor-demand elasticity governs the response of wages to migrant inflows, while the housing-supply elasticity governs the response of rents. Following Equation (4), we estimate the labor-demand elasticity using:

$$\Delta \ln w_u = -\frac{1}{\sigma} \Delta \ln L_u + \varepsilon_u,$$

which we identify using variation in migrant inflows driven by agricultural push factors at origin. We estimate the housing-supply elasticity from local market clearing:

$$\Delta \ln p_u = \frac{1}{1 + \eta_u} \left(\frac{\Lambda}{\alpha} - \frac{1}{\sigma} \right) \ln \left(1 + \frac{M_u}{N_u} \right) + \varphi_u,$$

using the same source of exogenous variation.

²⁸The rationale dates back to constraints on the non-market allocation of resources in a poor country with limited communication and state capacity (see, e.g., [Riskin 1981](#)). The opening of the Chinese economy in the 1990s and 2000s was expected to generate significant migration flows, and the initial disparity in *hukou* stringency reflected the capacity of a prefecture to sustain its population without external intervention ([Cai et al. 2001](#)). The legacy of the Great Famine—and its management by the central government—may have reinforced these self-sufficiency concerns among local policymakers, given the perceived risks of relying on external grain supplies ([Meng et al. 2015](#)).

²⁹These estimates are based on the following calculations: a 0.60 reduction in the composite index lowers migration barriers, $\tau_{ru}^{m_j}$, by 0.647 for fathers alone and 1.002 for whole-family migrants. Given our modeling assumptions, $\tau_{ru}^{m_j}$ is expressed in (log) units of real wages, and these increases would correspond to a $(\exp(0.647) - 1) \times 100 \approx 91\%$ increase in real wages for fathers alone and a $(\exp(1.002) - 1) \times 100 \approx 172\%$ increase for whole-family migrants.

Table 6. Labor demand and housing supply.

	Wage		Rent	
	(1)	(2)	(3)	(4)
Immigration rate	-0.248 (0.060)	-0.460 (0.224)		
Rented units			0.222 (0.035)	0.078 (0.132)
Observations	289	289	269	269
F-stat	—	23.28	—	21.17

Notes: The unit of observation is a prefecture. Robust standard errors are reported between parentheses. In columns (1) and (2), the dependent variable is (log) wages, and the explanatory variable is the relative immigration rate between 2000 and 2005, computed using the migration module of the “2005 Mini-Census.” In columns (3) and (4), the dependent variable is (log) rents, and the explanatory variable is the change in rented housing units between 2000 and 2005. Columns (1) and (3) report OLS estimates. In columns (2) and (4), the explanatory variable is instrumented using agricultural shocks between 2000 and 2005 at the typical origin associated with each prefecture (following [Imbert et al. 2022](#)).

Our preferred 2SLS estimates imply a wage elasticity with respect to immigration of about $1/\sigma = -0.46$ and a housing-supply elasticity of $\eta \approx 1/0.078$. These estimates indicate that migrant inflows exert moderate downward pressure on wages and upward pressure on rents, a key mechanism for evaluating the general-equilibrium effects of migration policies and different migration arrangements.

5 Counterfactuals and discussion

In this final section, we quantify the contribution of price differentials between origins and destinations, family arrangements (allowing for family splits), gender wage gaps, housing policies, and *hukou* policies to migration patterns, as well as the welfare of rural and urban residents and to aggregate productivity.

In each counterfactual experiment, we follow a common five-step procedure, and we: (i) translate the experiment into changes in model parameters; (ii) compute counterfactual emigration rates for each family arrangement and from each origin to each destination; (iii) aggregate these flows—accounting for the number of individuals in each arrangement—at the destination level to obtain counterfactual immigration rates; (iv) update urban wages and rents in response to immigration using the estimates from [Table 6](#); and (v) recompute migration values and choices. We iterate these steps until convergence, defined as a maximum change in wages and rents across destinations below 0.1% between successive iterations. We then report migration flows (in number of individuals), both overall and for large cities (population above 5 million), the prevalence of different family arrangements, and the resulting changes in rural and urban welfare and aggregate productivity.

5.1 Displaced consumption and family splits

In counterfactual (1a), we shut down the key mechanism of displaced consumption by simulating a remittance tax that forces migrants to consume all their income at destination prices. The results of this experiment are reported in the first column of Table 7. Without the ability to shift consumption to cheaper origins, migration becomes substantially less attractive: the number of migrants declines from 30.5 million between 2000 and 2005 to about 13 million, and migration to large cities falls even more sharply, from 13.6 million to around 4 million. This decline is accompanied by a marked shift in *how* families relocate. The baseline distribution—roughly balanced across father- or mother-alone migration, both parents migrating with children left behind, and whole-family migration—gives way to predominantly unsplit families. As a result, the welfare of rural-born households declines by about 2.11%, while the welfare of urban-born households increases by 3.16%, reflecting lower congestion costs.³⁰ Finally, aggregate productivity declines significantly (-3.17%), highlighting the economic value of migrants’ ability to work in high-productivity cities without bearing the full cost of living at destination.

In counterfactuals (1b) and (1c), we instead explore the consequences of making certain family splits prohibitively costly (e.g., due to social norms or limited child-rearing capacity at origin). Counterfactual (1b) eliminates all arrangements except those in which all household members reside in a single location, effectively ruling out about two-thirds of baseline migration choices. This experiment reduces family migration overall, but also induces substantial reshuffling across arrangements: the 11.6 million family migrants initially spread across arrangements become 8.36 million migrants relocating as intact families. Counterfactual (1c) instead prohibits *mothers* from leaving children behind, eliminating two arrangements: mothers migrating alone, and couples migrating while leaving children at origin (typically with grandparents). In this scenario, the share of father-alone migrants rises from 22.75% to 41%, while whole-family migration increases from 33% to 59%. Interestingly, these reallocations affect different destinations: father-only migrants concentrate in large cities, whereas migrating families are more likely to relocate to smaller destinations. Overall, both counterfactuals affect welfare and productivity in the same direction as in (1a), although the magnitudes are substantially smaller.

³⁰There is substantial heterogeneity across cities: welfare remains largely unchanged in smaller cities, while residents of large cities experience gains of about 5%. Appendix E.1 explores these distributional effects by documenting changes in migrant concentration, inflows, and welfare across cities relative to the baseline.

Table 7. The role of consumption patterns and migration frictions—counterfactual experiments.

	Base. C0	Pri. C1a	Spli. C1b	Moth. C1c	Gap C2	Subsidy		Hukou	
						C3a	C3b	C4a	C4b
Migrants (millions)									
<i>All</i>	30.57	13.21	27.80	28.18	34.20	28.05	36.37	29.21	36.93
<i>Mega-cities</i>	13.64	4.20	12.30	12.62	14.32	11.49	24.22	14.69	11.52
Of all migrants (%)									
<i>Family migrants</i>	38.03	35.84	30.98	31.80	46.10	37.75	37.18	34.63	38.17
Of family migrants (%)									
<i>Father alone</i>	22.75	2.35	0.00	41.07	10.69	24.28	19.69	25.16	18.89
<i>Mother alone</i>	11.67	0.31	0.00	0.00	21.16	13.83	8.81	13.97	13.24
<i>Parents move</i>	32.61	26.78	0.00	0.00	33.13	31.80	34.72	30.97	31.33
<i>Family moves</i>	32.97	70.56	100.00	58.93	35.02	30.10	36.78	29.89	36.55
Change in ... (%)									
<i>Rural welfare</i>	0.00	-2.11	-0.57	-0.22	0.44	-0.38	0.67	-0.43	0.52
<i>Urban welfare</i>	0.00	3.16	0.45	0.41	-0.68	0.56	-0.85	-1.10	-1.20
<i>Productivity</i>	0.00	-3.17	-0.34	-0.29	0.36	-0.64	1.80	-0.54	-0.07

Notes: This table reports statistics on the extent, nature, and consequences of migration flows in the baseline (0) and across the following counterfactual experiments. Counterfactual (1a) simulates a remittance tax that forces migrants to consume all income at destination prices, while counterfactuals (1b) and (1c) impose prohibitive costs on family-splitting arrangements and mother-child separations (see Section 5.1). Counterfactual (2) increases female wages in urban areas to match those of males (Section 5.2). Counterfactuals (3a) and (3b) respectively implement the observed changes in city-level housing subsidies between 2000 and 2010 and a policy that maintains subsidy rates while extending access to migrants at the level of urban residents with a local *hukou* (Section 5.3). Counterfactuals (4a) and (4b) respectively evaluate the 2014 *hukou* reform (based on observed changes documented in Zhang et al. 2018) and a reform aligned with its stated objective (Section 5.4). Additional statistics—covering wages, rents, remittances to rural origins, and measures of migrant concentration, family arrangements, and distributional effects across cities—are reported in Appendix E.1.

5.2 Gender wage gap

Our framework allows for gender differences in returns to labor, preference weights, and migration costs, each of which shapes the prevalence of different family arrangements. In counterfactual (2), we isolate these effects by increasing female wages in urban areas to match those of males. This experiment leads to an increase of about 3.6 million migrants and a reversal in solo migration patterns: whereas fathers are twice as likely to migrate alone as mothers in the baseline, the opposite holds under wage equalization. This seemingly counterintuitive result reflects the respective contributions of preference weights and migration costs. Although the costs of leaving children behind are higher for women, mothers account for a smaller share of household consumption. As a result, migrating mothers can remit a larger fraction of household resources to the origin, where prices are substantially lower.

The role of such price differentials is therefore central; we next turn to policy experiments that directly affect these relative prices.

5.3 Housing policies

In this section, we explore the consequences of subsidizing housing for migrants.³¹ In our model, the spatial distribution of living costs shapes migration incidence, family arrangements, the direction of migration flows, and, ultimately, welfare and productivity outcomes.

In counterfactual (3a), we evaluate the impact of the decline in housing subsidies observed between 2000 and 2010 across cities. We construct a measure of subsidy changes by combining variation in access to subsidies and subsidy rates across locations. The resulting decrease in subsidy generosity reduces migration flows—particularly toward large cities—and shifts migration patterns away from whole-family moves toward solo migration.

Counterfactual (3b) considers the opposite policy: maintaining subsidy rates while extending access to migrants at the same level as urban residents with a local *hukou*. Under this more inclusive policy, the number of migrants increases by about 6 million, accompanied by substantial changes in spatial allocation. Migration to mega-cities nearly doubles, reflecting both their high living costs and the more restrictive housing policies they impose in the baseline. As a result of increased migration to the most productive locations, aggregate productivity rises by nearly 1.80%. This gain implies that the productivity benefits of such policies—evaluated to be around \$10 billion annually—would exceed their fiscal costs by more than a factor of two.

5.4 Migration frictions

The previous counterfactuals illustrate how price differentials across space, family arrangements (including family splits), gender wage gaps, and housing subsidies shape the incidence, nature, and direction of migration. We now examine how *hukou* policies affect these migration patterns and their normative implications.

Studying the effects of migration policy is challenging for at least two reasons. First, migration policy is endogenous, leading most of the literature to rely on structural approaches. Second, even within a structural framework, one must determine how policy affects the model’s fundamentals. In this section, we take the stance that migration policy directly influences mode-specific migration frictions, $\{\tau_{ru}^{m_j}\}$, and argue that our estimated model is well suited to address these challenges. Recovering credible migration frictions requires a realistic model that accounts for the existence and substitutability of different migration modes. Moreover, by leveraging exogenous variation in the strin-

³¹By leveraging spatial differences in the price of non-tradable goods, these results also speak to other comparable barriers faced by migrants at destination, including access to subsidized public services such as healthcare and schooling.

gency of *hukou* policies, the specifications reported in Table 5 allow us to isolate the policy-induced component of arrangement-specific migration frictions.

Counterfactual (4a) exploits changes in the *hukou* composite index documented in Zhang et al. (2018)—and the mapping between stringency and migration costs identified in Panel A of Table 5—to evaluate the 2014 *hukou* reform. Although central government guidelines aimed to tighten restrictions in large cities and relax them in medium-sized cities to alleviate demographic pressures, implementation was decentralized, and many medium-sized cities instead tightened restrictions. The reform led to a modest decline in overall migration, but an increase in migration to the largest cities. Strikingly, by reshaping migration costs across destinations, the reform reduced the welfare of both rural- and urban-born households, as well as aggregate productivity.

In counterfactual (4b), we follow the spirit of the 2014 reform—abstracting from local local idiosyncrasies—and relax the *hukou* composite index by one standard deviation outside mega-cities. Under this counterfactual, migration increases by about 6 million individuals, with flows shifting away from mega-cities toward smaller destinations, in line with the reform’s original objective. Migration patterns also tilt toward more family-based arrangements, generating non-negligible welfare gains for rural-born households. However, because migration is redirected toward smaller labor and housing markets, the welfare of urban residents declines, and the overall effect on aggregate productivity is close to neutral (-0.07%).

5.5 Discussion

This section summarizes our main findings on: (i) the normative implications of displaced consumption and migration frictions with a focus on their redistributive effects; (ii) model extensions allowing for agglomeration and congestion externalities; and (iii) the quantitative and qualitative insights induced by our precise modeling of migration and consumption choices relative to alternative, standard models of location choice.³²

Normative implications and redistribution The export-led growth in China of the 2000s induced wide disparities in living standards across space—between rural and urban areas, and between booming cities along the coast and cities in the interior. In Appendix E.1, we quantify the welfare implications of the possibility for rural-urban migrants to displace part of their consumption and of migration frictions. We show that the ability to displace consumption acted as a strong progressive force, benefiting a large population of rural households from the hinterlands of booming urban centers at

³²The reader interested in redistributive effects, the role of externalities, and modeling alternatives can refer to Appendix E.1, E.2, and E.3, respectively.

the expense of the residents in such cities. More specifically, it reduced welfare inequalities between urban-born and rural-born households and between residents of different cities; it however increased the welfare disparities across rural-born households, benefiting rural households in the hinterlands of expensive cities. By contrast, equalizing migration frictions across living arrangements (counterfactual 3a) has far more modest welfare implications.

Introducing externalities The heterogeneous growth of cities is possibly disciplined by agglomeration and congestion forces. Our baseline quantitative model only features such forces through the modest (equilibrium) adjustment of labor and housing markets.

In Appendix E.2, we show that agglomeration economies (e.g., through positive production spillovers from rural migrants, as in Combes et al. 2015) or congestion externalities (e.g., through negative spillovers on local amenities) would affect our quantification exercise at the margin. Agglomeration forces imply a slightly larger effect of relaxing migration restrictions on migrant inflows, with more positive welfare implications across the board. Congestion forces have the exact opposite effect. Both externalities have limited bite, because their effects are dwarfed by the other drivers of migration, i.e., the large rural-urban income gaps and the significant bilateral migration costs. The most interesting extension allows for remittances to boost production at origin (as in Pan and Sun 2024, Khanna et al. 2022). This extension increases the social returns to migration, while decreasing its private return. In such a framework, migrants migrate less following the relaxation of restrictions than with our baseline model, even though the policy would have larger welfare effects, and migration subsidies might be welfare-enhancing.³³

Sensitivity analysis and alternative migration models In this paper, we develop a location choice model that allows potential migrants to choose whether to migrate or not, how to migrate, and where, and to transfer part of their consumption across household members and across space. In Appendix E.3, we illustrate the quantitative and qualitative insights gained through the adoption of those two novel features by estimating three alternative models: (1) a multinomial model of location choice (as in Bryan and Morten 2019, Tombe and Zhu 2019); (2) a nested model where rural households can decide whether to migrate or not, and where; and (3) a nested model adding the possi-

³³In Appendix E.2, we focus on productivity and amenity spillovers. We could also augment our model with alternative policy-relevant channels based on elasticities to immigration estimated in the literature, e.g., future income losses for left-behind children (Gao et al. 2022), the provision of non-traded services (Cortes 2008), criminality (Abramitzky et al. 2024), or slums (Marx et al. 2013). Particularly interesting applications would allow for externalities that are specific to the migration mode. For instance, from Figure 3, we would expect non-family migration to lower demand for housing quality in destination cities, which may be seen as a negative externality.

bility for migrants to remit (as in [Albert and Monras 2022](#)). In all these models, we treat family and non-family migrants as two separate populations, thus estimating separate elasticities. We show that these alternative models produce misspecified migration costs and severely underestimate the impact of relaxing migration barriers on the *composition* and direction of migration flows. Even the closest model to ours—model 3—would not capture the substitution across migration modes and the changes in the allocation of migrants across space due to changes in the nature of migration flows.

6 Conclusion

This paper offers a new perspective on migration. While most of the literature has studied individual migration choices, we argue, instead that family considerations is a big part of the migration experience.

We make this argument in the context of China, where internal migration policies limit the extent to which rural families can fully migrate to urban areas so that rural migrants are described as a “floating population”. We establish that many migrants leave their dependent children behind. Sometimes, only the father or the mother moves, sometimes both of them move jointly, leaving the children with grandparents. Oftentimes, though, the entire household relocates. These different forms of migration occur in different places. When the primary objective for migrating is sending remittances, migrants tend to leave dependents behind and tend to concentrate in large and expensive cities where they live in poor housing conditions. When instead, migrants bring the dependent children with them, they tend to spread much more across destinations. We provide a model that captures these ideas in a parsimonious way. We use the estimated model to argue that remittances allow the concentration of migration and economic activity into the large and expensive cities, which explains part of the aggregate productivity gains observed in China in the early 2000s. This comes, though, at substantial costs for rural migrants. Through the lenses of the model we identify housing as one of the main barriers faced by migrants, especially limiting family migration.

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Floating population: migration with(out) family and Chinese economic development

Online Appendix

This appendix **(A)** describes the data, **(B)** presents additional motivating evidence, and provides supplementary material for **(C)** the model, **(D)** the estimation, and **(E)** the counterfactual exercises.

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A Data description

This section complements Section 1 by providing: (i) a detailed presentation of our procedure for characterizing family arrangements in census data; (ii) a brief description of additional data sources; and (iii) an extended discussion of migrant allocation, migration barriers, and split families across space and over time.

A.1 Family arrangements

The 2005 1% Population Survey (“Mini-census”) records individuals’ places of *hukou* registration, allowing us to identify migrants at destination and construct an origin-destination matrix. However, data collected at the place of residence do not always allow us to unambiguously determine family arrangements. Households can be complex, and information on migration decisions is limited; in particular, we do not observe the locations of a migrant’s spouse and children unless they co-reside.

In this section, we describe our empirical approach to classifying households into different family arrangements and present additional descriptive statistics on these arrangements.

Characterizing family arrangements Determining family arrangements in census-like data is challenging because relationships between household members are only partially observed. In addition, we have limited information on non-co-resident relatives: for men, we do not observe whether they have children living elsewhere; for women, a fertility module indicates whether they have children, but not their age or location.

To address the first challenge, we begin by recoding the variable capturing the relationship to the household head to ensure that—whenever possible—the head is a prime-age (18–64) adult. We then define eligible adults as married prime-age individuals and determine whether their spouse co-resides based on the (recoded) relationship to the household head. In most cases (over 98% of married prime-age adults), this can be done with little ambiguity. For example, if the head’s son and daughter-in-law are present, they almost certainly form a couple, whereas the head’s brother and sister do not. In more ambiguous cases, we identify plausible matches and assign probabilities. For instance, if a household contains exactly two eligible individuals of opposite sex unrelated to the head, we assume they form a couple; if there is one woman and two men, we assign the woman a probability of 100% and each man a probability of 50% of co-residing with a spouse. We then assign children to adults based on inferred relationships. Nearly 99% of children can be allocated in this way; the remainder are classified as not co-residing with either parent, and the corresponding adults are treated as living without their children.

This procedure allows us to characterize family arrangements for over 99% of married prime-age household members. The resulting categories are: father alone, mother alone, couple alone, father with children, mother with children, and couple with children.

The second challenge arises from the distinction between household members and co-residents. Households in which the head, spouse, and children (if any) live together are treated as co-residing in their observed location in the 2005 Mini-Census. By contrast, household members of migrant families—the focus of our study—may be spatially separated and therefore not observed within the same household, complicating the characterization of their living arrangements. For example, married adults observed at the destination may not be co-residing with their children under age 18. In our baseline approach, we assign probabilities that married adults at destination have left household members behind. This procedure relies on two sources of information: (i) the fertility module, which provides information on each prime-age woman’s children, and (ii) the 2010 Population Census, which records whether registered household members reside elsewhere, along with their location, age, and other characteristics.

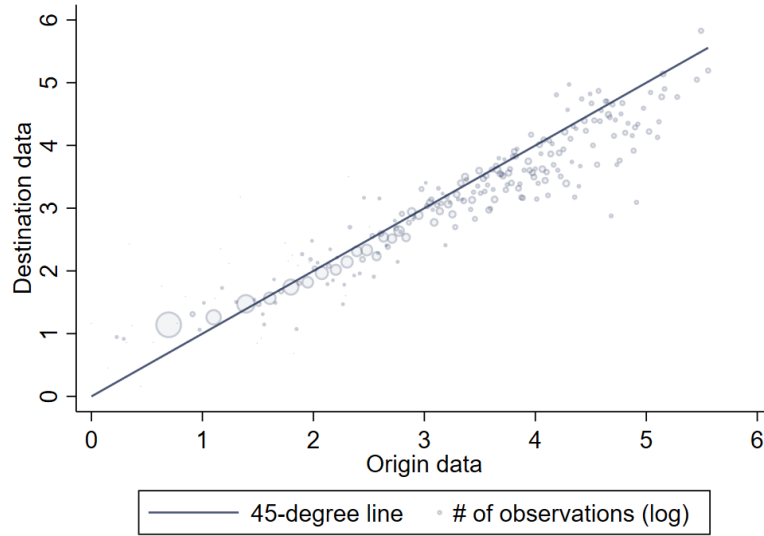
More specifically, we estimate—using *origin-household data* on emigrants—the probability that a married man who would be categorized as “father alone” at destination has a spouse and at least one child at origin, allowing this probability to vary across detailed age bins.³⁴ We apply the same approach to other family arrangements and use the estimated coefficients to classify “possible parents” and “possible spouses” into parents leaving children behind, spouses leaving family behind, and an “other” category. The latter also includes non-married migrants and migrant couples without children under 18, who do not face the same migration decisions as parents with young children.

Figure A1 evaluates the accuracy of the probabilistic classification of family arrangements. For each origin $\times$ destination $\times$ family arrangement cell, we compare the number of migrants measured using origin- and destination-household data in the 2010 Census, applying the regression-based correction to the latter. Despite the “missing migrant” patterns documented in destination data (Ebenstein and Zhao 2015), most observations lie on or close to the 45-degree line. This provides reassuring evidence that destination data capture migrant flows reasonably well and that family arrangements can be reliably recovered.

Descriptive statistics Panel A of Table A1 reports the shares of rural-urban migrants by family arrangement in the 2005 data *prior to* reclassifying “false parents” and “false spouses.” Four arrangements dominate: father alone, mother alone, couple alone, and

³⁴Eligible migrants are all married, by definition. Divorce was a rare event in the 2000s, due to social stigma (Cai et al. 2025); this procedure therefore allows us to recover migrants’ underlying family situations.

Figure A1. Family arrangements using origin- and destination-household data.



Notes: This figure shows the concordance in migrant counts between origin- and destination-household data across origin $\times$ destination $\times$ family arrangement cells in the 2010 Population Census. The family arrangements considered are: parents moving without children; father moving alone; mother moving alone; father with children only; mother with children only; and other. (The “family moving” category is excluded, as such migrants are typically not reported in origin households and are therefore only observed at destination.) Destination counts are corrected using the reclassification procedure described in the text. The bubbles represent averages across 300 bins of the (log) number of observations (i.e., the unweighted number of migrants in the census data) per cell, as measured in origin data; bubble size is proportional to the total number of observations in each bin. The solid line corresponds to the 45-degree line.

couple with children. However, more than 8% of married prime-age migrants are observed in other arrangements—namely, without a co-residing spouse but with children. As discussed above, this classification does not account for the limitations of destination-based data on migrants.

In Panel B, we implement the full procedure to recover family arrangements. Relative to Panel A, the share of migrants in the mother-alone category declines markedly, as many women in that group do not have children under 18 and were therefore misclassified. Reclassification using origin data indicates that parents rarely live with children at destination without their spouse; accordingly, we reassign such cases to the couple-with-children category throughout the paper.³⁵

Consistent with this evidence, our model and main empirical analysis focus on four family arrangements (or migration modes): father alone, mother alone, couple alone, and couple with children.

³⁵These cases likely correspond to effectively separated “spouses” who are both migrants, implying that they are not reported in remaining origin households and are only observed at destination.

Table A1. Non-adjusted family arrangements for migrants in the 2005 1% Population Survey.

		Parents' locations			Total
		Father at dest. Mother at orig.	Mother at dest. Father at orig.	Both mig. same city	
Panel A: non-adjusted					
Children's locations	Origin	23.66	17.95	29.95	71.56
	Destination	4.18	3.77	20.49	20.49
	Total	21.72	27.84	50.44	100.00
Panel B: adjusted					
Children's locations	Origin	21.83	10.94	31.15	63.91
	Destination	1.65	1.67	36.09	36.09
	Total	12.61	23.48	67.23	100.00

Notes: This table reports the relative frequencies (in percent) of different migration modes, defined by the locations of parents and children. In Panel B, figures are based on reclassified family arrangements, following the procedure described in the text. Parents' locations fall into three categories: (i) the husband migrates while the wife remains at origin; (ii) the wife migrates while the husband remains at origin; and (iii) both migrate. Children's locations are classified into two categories: (i) all remain at origin; or (ii) all reside with their parents at destination. Source: 2005 1% Population Survey. The table is based on 29.9 million individuals in Panel A and 21.6 million in Panel B.

A.2 Living conditions in cities

In addition to the data on wages, housing rents, living arrangements, and housing characteristics from the “2005 Mini-Census”—described at length in Section 1—we further characterize urban living conditions using several complementary sources: data on remittances and consumption patterns from household surveys, pollution measures derived from satellite imagery, commuting data from the “2015 Mini-Census”, and additional wage data for years other than 2005 from Statistical Yearbooks. We defer the description of data used for identification to Appendix D.

Remittances and consumption We use the Longitudinal Survey on Rural Urban Migration in China (RUMiC, 2008–2009) and the China Migrants Dynamic Survey (CMDS, 2011) to measure remittances and the consumption of non-tradables at destination. Both surveys target migrants, as reflected in their sampling strategies and questionnaires, but they differ in coverage and detail. CMDS is nationally representative, although its questionnaire is less comprehensive, whereas RUMiC covers only 15 cities but provides highly detailed information tailored to the migrant experience in China.³⁶ The measurement of remittances also differs across surveys. In CMDS, the relevant question can be translated as “The total amount of money you gave to your hometown in the past year”, whereas RUMiC elicits this information through several components, including “the total net value of money and commodities brought or remitted back to your hometown

³⁶Given the absence of a sampling frame, RUMiC selects migrant respondents as follows: (i) it randomly samples enumeration areas within each city; (ii) it lists all workplaces within these areas; (iii) it collects information on the number of employees and migrant workers in each workplace; and (iv) it randomly selects migrant workers to participate in the survey (Meng and Manning 2010).

this year”, to which displaced education fees and other “children-raising” fees can be added. As a result, remittances measured in CMDS tend to be lower than in RUMiC. We therefore rescale CMDS remittances across 221 destinations to match RUMiC levels in the 14 overlapping destinations.

RUMiC has the advantage of decomposing consumption and investment into 20 categories, which we exploit to characterize the consumption of non-tradables across different migration arrangements. Finally, RUMiC includes two questions that we do not use for estimation but that provide supportive evidence for the key mechanisms underlying our quantitative approach: (i) “If you were still in your home village, how much do you estimate you could earn every month?”, which allows us to construct a measure of the rural-urban wage gap (see Appendix B.4, and the associated reference in Section 2.2); and (ii) “What is the main reason why your children do not live with you?”, with common responses including “high living cost in city”, “high cost of attending school/kindergarten”, “school or kindergarten cannot be found here”, or “education in hometown is better than here”.

Pollution and commuting Pollution data are drawn from TEMIS satellite images covering the period 1997–2015, with a spatial resolution of 20–25 km. We map raster data on NO₂ concentration—capturing industrial and vehicular emissions—to Chinese prefectures to construct prefecture-by-year measures of pollution. These measures can be interpreted as proxies for air quality. We also compute average commuting times at the prefecture level using a random 20% micro extract of the 2015 1% Population Survey. These data provide a proxy for congestion.

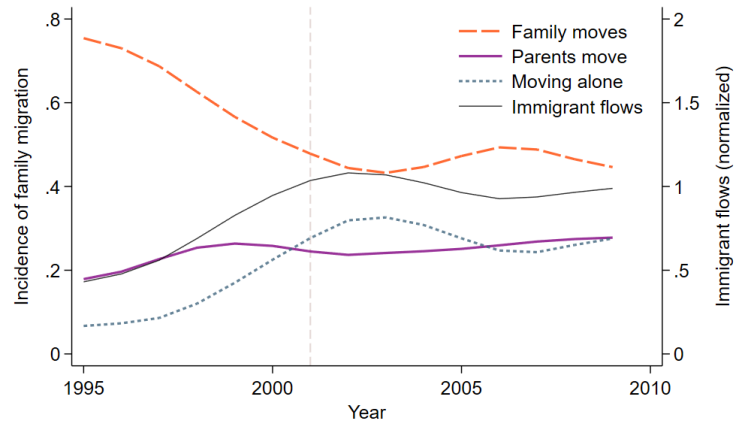
Statistical Yearbooks We use aggregate data compiled by the National Bureau of Statistics—based on the Reporting Form System on Labor Wage Statistics, the National Monthly Sample Survey System on Labor Force, and the System of Rural Social and Economic Surveys—to construct baseline wage measures in 2000.

A.3 Descriptive statistics

This section provides additional descriptive statistics complementing those presented in Section 2.1.

Migrant inflows and family migration over time Figure A2 shows the composition and magnitude of migrant inflows to urban areas between 1995 and 2010. Inflows accelerate around the time of WTO accession, coinciding with broader reforms that pushed migrants from rural hinterlands toward expanding metropolitan areas. After 2000–2001,

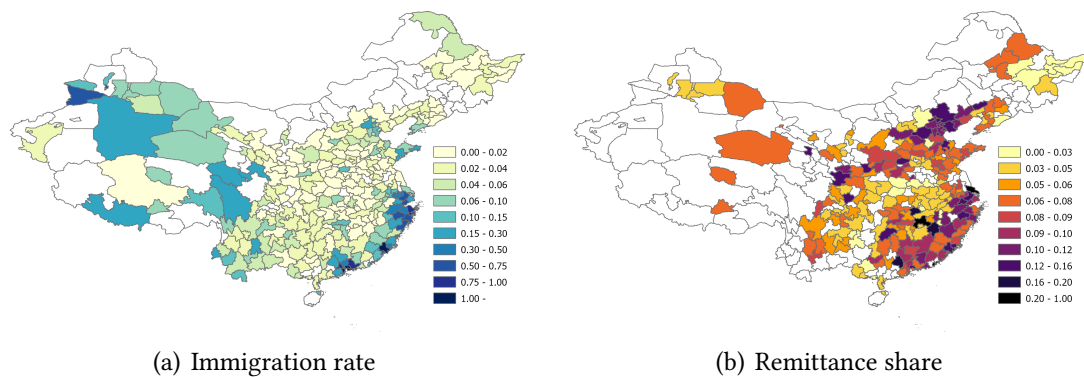
Figure A2. Migrant inflows and migration patterns over time.



Notes: This figure shows the composition and magnitude of migrant inflows to urban areas between 1995 and 2010, using the Population Censuses (2000 and 2010) and the “2005 Mini-Census.” A migrant is defined as an individual whose prefecture of residence differs from her prefecture of household registration. Migration patterns follow our baseline classification: family moves (orange, long-dashed line), parents move (purple, solid line), and solo migration (blue, short-dashed line). The vertical dashed line marks China’s WTO accession in 2001. Two differences relative to Figure 1 are worth noting. First, migration incidence is measured using annual flows. Second, migrant inflows are normalized by the contemporaneous urban population and set equal to 1 in 2000.

urban populations grow steadily, but, more importantly for our purposes, the composition of migrant inflows remains remarkably stable over time: about 60% of new migrants to cities relocate without their children.

Figure A3. Migrant inflows and remittances across prefectures in 2005.

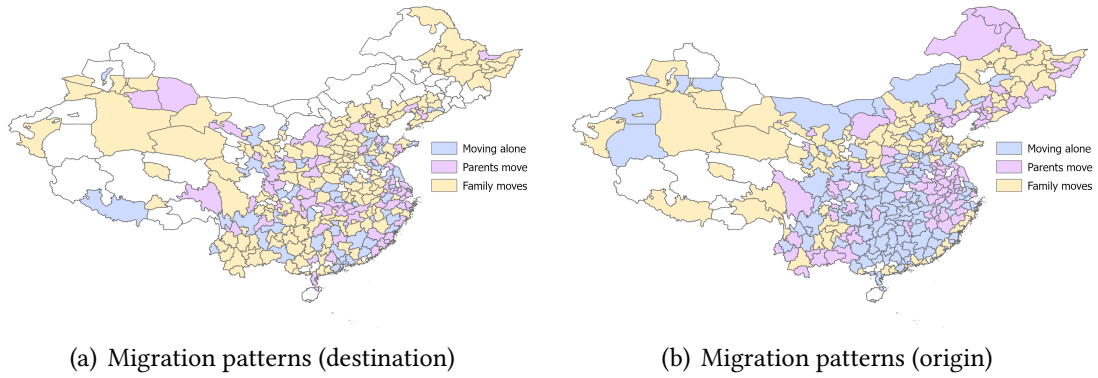


Notes: Panel (a) displays the share of rural-urban migrants in the 2005 1% Population Survey (“2005 Mini-Census”) across urban prefectures. The sample is restricted to urban areas, and rural-urban migrants are defined as individuals with a rural *hukou* residing in these locations. Higher shares in Western regions largely reflect lower population density. Panel (b) shows the share of income devoted to remittances across destinations, based on CMDS (2011–2012).

Migrant inflows and remittances across space Figure A3 illustrates the spatial distribution of migration to cities in China: panel (a) shows the allocation of migrants across

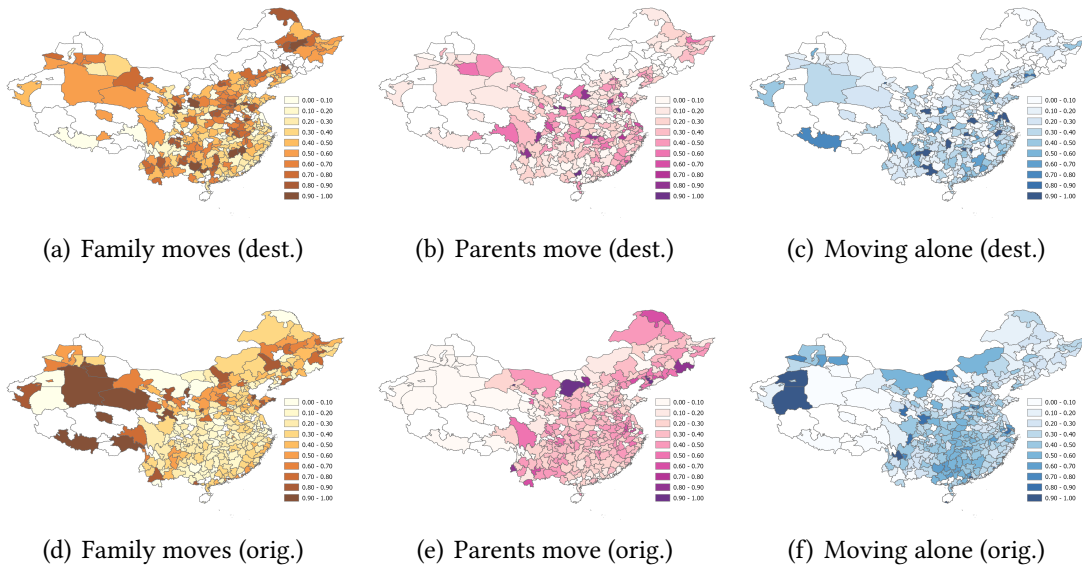
destinations in 2005, while panel (b) reports remittance shares by destination. Excluding sparsely populated Western regions, migrants are concentrated in large cities (e.g., Beijing and Shanghai) and in major exporting centers such as Tianjin, Fuzhou, and Shenzhen/Guangzhou in the South. Migrants in these destinations also remit a larger share of their income (panel b).

Figure A4. Migration patterns across destinations and origins in 2005.



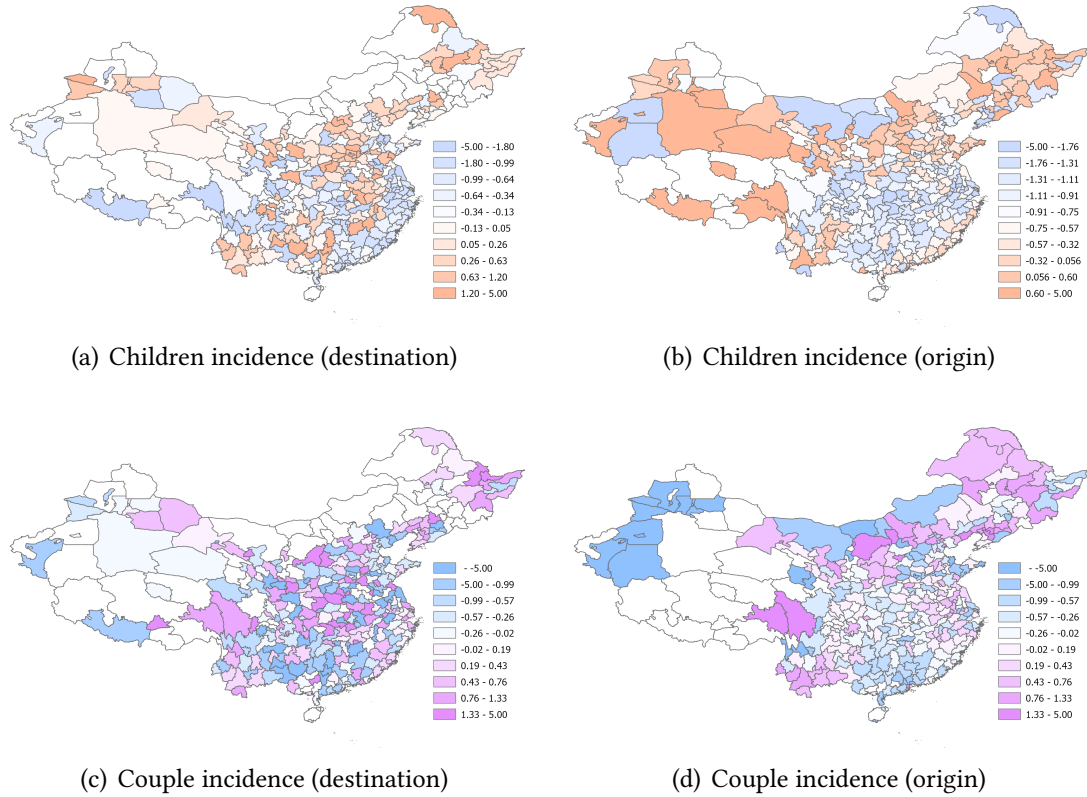
Notes: Panel (a) shows the main modes of migration across *destinations*: rural migrant households moving as a whole family (orange), moving without children (purple), and individuals migrating alone (blue). Panel (b) presents the same breakdown across *origins*. These statistics are derived from the 2005 Mini-Census and are based on migrant flows between 2000 and 2005.

Figure A5. Migration patterns across destinations and origins in 2005—details.



Notes: The top panels show migration patterns across *destinations*: the share of rural migrant households moving as a whole family (panel a, light to dark orange), moving without children (panel b, light to dark purple), and individuals migrating alone (panel c, light to dark blue). The bottom panels present the same breakdown across *origins* (panels d–f). These statistics are derived from the 2005 Mini-Census and are based on migrant flows between 2000 and 2005.

Figure A6. Migration patterns across destinations and origins in 2005—children and couple incidence.

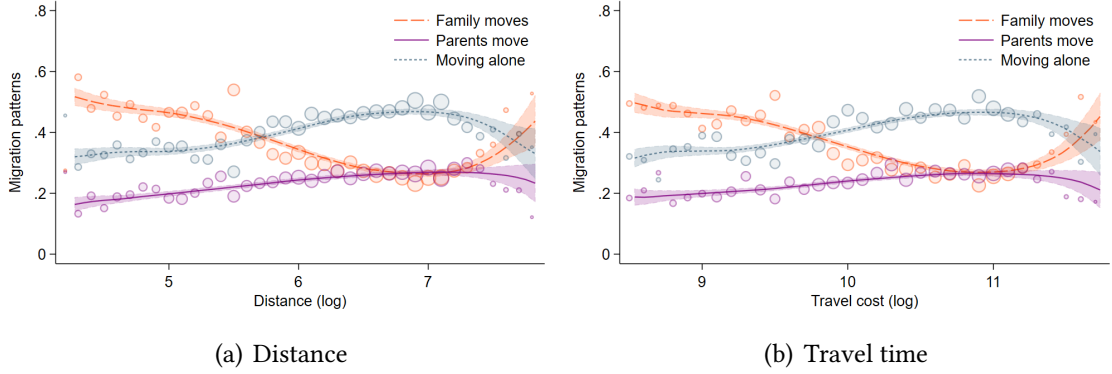


Notes: Panels (a) and (b) show the relative (log) incidence of migrating with children (orange) versus without children (blue) across *destinations* and *origins*. Panels (c) and (d) show the relative (log) incidence of migration by both parents (purple) versus a single parent (blue, most often the male) across *destinations* and *origins*. These statistics are derived from the 2005 Mini-Census and are based on migrant flows between 2000 and 2005.

Migration patterns Figure A4 shows that migration patterns vary markedly across space. First, the share of migrants living with their families at destination is negatively correlated with both migrant incidence and remittance intensity: in large cities and major exporting centers, migrants are less likely to co-reside with their families—see panel (a). Second, the previous observation, coupled with the gravity of migration flows, generates disparities across origins in the share of migrants who move with their families, and thus in the prevalence of family members left behind by primary earners—see panel (b). These geographic differences are pronounced and highlight significant spatial heterogeneity in migration patterns across Chinese cities.

The gravity of migration flows has two distinct implications for the decisions of families to move jointly or remain split across locations. First, proximity to congested destinations with strong barriers to family migration increases the incidence of split families for a given distance, as shown in panel (b) of Figure A4. Second, distance between origins and destinations also helps explain variation in migration patterns (Figure A7). However,

Figure A7. Migration patterns and distance in 2005.



Notes: Panel (a) shows how migration arrangements—family moves; parents move; moving alone—vary with geodesic (“as-the-crow-flies”) distance between origins and destinations. Panel (b) uses an alternative measure of distance based on travel time along the transportation network.

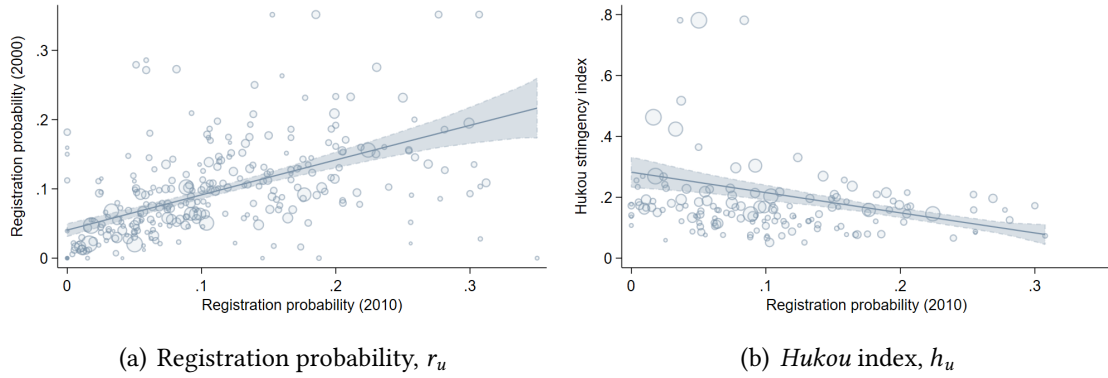
the first mechanism is quantitatively more important. Since most of the population is concentrated in Central China and along the coast, not so far from typical migration destinations, the higher incidence of family migration observed for very distant prefectures (see the right tail of Figure A7) accounts for less than 1% of all migration spells.

Migration barriers A key determinant of the spatial allocation of migrants and their families is the stringency of local migration barriers (see Section 1.1). In this section, we first describe and compare available measures of such barriers. We then show how these measures relate to migrants’ experiences at destination and discuss the spatial distribution of *hukou* stringency across cities.

When quantifying the effects of migration policies (see Section 4.2), we proxy the local regulatory environment affecting immigration using the probability of *hukou* conversion. To construct this measure, we follow Wu and You (2021) and use data from the 2010 Population Census to compute the share of migrants aged 15–64 who moved for work-related reasons, were born in another county, and are locally registered with a non-agricultural *hukou*. This yields a city-level measure of the probability that migrants can convert their household registration at destination, denoted by r_u .

We compare this measure of the *hukou* environment with alternative proxies in Figure A8. Panel (a) plots the registration probability in 2000 against that in 2010 (one of our main measures), using data from the Population Censuses. The two measures are strongly positively correlated, indicating substantial persistence in local legislation despite the rapid growth in migration over this period (see Figure 1). At the same time, most prefectures lie below the 45-degree line, suggesting a general relaxation of *hukou* conversion restrictions between 2000 and 2010. This measure of the *hukou* environment

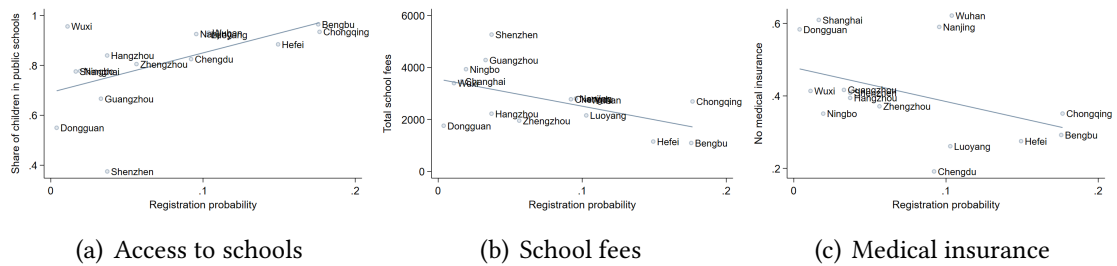
Figure A8. Measures of the *hukou* environment.



Notes: Panel (a) shows the correlation between census-based measures of local household registration probabilities in 2000 and 2010, following [Wu and You \(2021\)](#). Panel (b) shows the correlation between the pre-2014 *hukou* stringency index developed by [Zhang et al. \(2018\)](#) and the household registration probability in 2010. Each bubble represents a destination prefecture and is weighted by its urban population in 2000. The lines correspond to local polynomial fits, with observations weighted by population.

is nonetheless an equilibrium object, as it is based on observed—and therefore selected—migration outcomes. Panel (b) instead relates the 2010 registration probability to the composite index from [Zhang et al. \(2018\)](#), denoted by h_u (our other main measure). This index is constructed from policy and legislative documents and captures the ease with which migrants can obtain local household registration, independently of observed migration flows. The two measures are strongly negatively correlated, suggesting that they capture, respectively, the leniency and stringency of local *hukou* policies.

Figure A9. Access to public goods and the *hukou* environment.



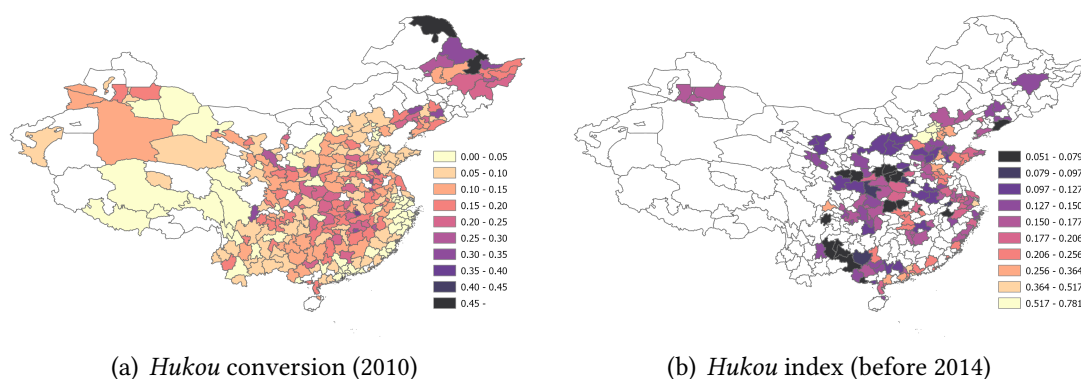
Notes: This figure shows the correlation between measures of access to public goods from RUMiC (2008) and our measure r_u of *hukou* leniency, defined as the city-level probability of household registration conversion computed from the 2010 Population Census, following [Wu and You \(2021\)](#). Each dot represents a destination prefecture. The lines correspond to ordinary least squares regression fits. “Share of children in public schools” is the proportion of migrant households’ children attending public schools, conditional on residing at destination. “Total school fees” include tuition, food, remedial classes, and other expenses (e.g., uniforms), but exclude sponsorship, boarding, and selection fees. “No medical insurance” denotes the share of migrant household heads without any form of medical coverage. Similar correlations are obtained using the alternative measure of *hukou* stringency, h_u , from [Zhang et al. \(2018\)](#).

An important caveat of both census- and legislation-based measures is that they rely on *hukou* conversion, which remains a rare event for rural migrants—especially for the typical low-income, low-education worker. Cities generally condition local registration

on stringent criteria, such as investing more than one million RMB in a business or holding a college degree. To assess whether our measure of the *hukou* environment captures migrants' actual experiences at destination, we use data from RUMiC (2008), a survey of rural-urban migrants and their households covering 15 cities in nine provinces. Figure A9 reports correlations based on the census-based registration probability measure; similar results (with opposite sign) are obtained using the legislation-based *hukou* stringency index.

The left panel shows that the probability that migrants' children (conditional on residing at destination) attend public schools is higher in cities with more lenient *hukou* environments. The middle panel indicates that, conditional on school attendance, children of migrants face lower schooling costs in more liberal environments. Finally, the right panel shows that migrants in more restrictive *hukou* environments are significantly less likely to have medical insurance, highlighting limited access to healthcare in urban China.

Figure A10. Migration barriers across prefectures.

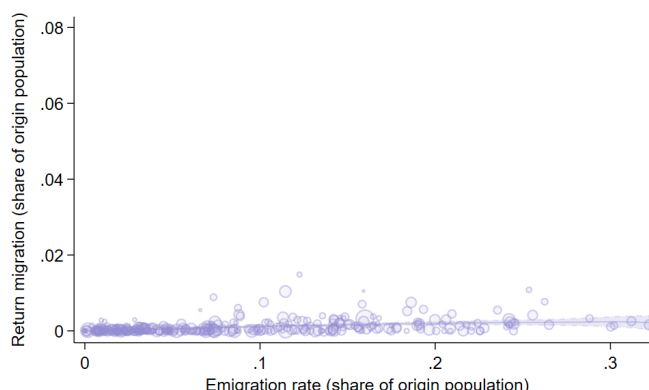


Notes: Panel (a) shows the variation in *hukou* conversion, r_u —a measure constructed following the procedure developed in Wu and You (2021). Panel (b) uses the composite index capturing the difficulty with which migrants could obtain a local urban *hukou* before 2014, h_u (Zhang et al. 2018).

We conclude by examining the spatial distribution of barriers to internal migration in Figure A10. Panel (a) reports the *hukou* conversion measure from the 2010 Census, while panel (b) shows the composite index capturing the ease with which migrants could obtain a local urban *hukou* prior to 2014 (Zhang et al. 2018). Migration barriers broadly align with the geography of economic growth during the Reform period. In particular, restrictions on access to public services reflect local fiscal constraints and, historically, concerns about food security in the presence of migration inflows. These constraints are closely linked to expected migration—strongly correlated with local growth prospects—as well as to fiscal capacity and grain reserves. In Sections 4.2 and 5.4, we exploit variation in

the latter to identify exogenous differences in migration barriers across space.

Figure A11. The incidence of return migration.



Notes: This Figure compares the number of migrants having departed from their origins after 2000 (x-axis) with the number of those having returned between 2004 and 2005 (y-axis) across prefectures.

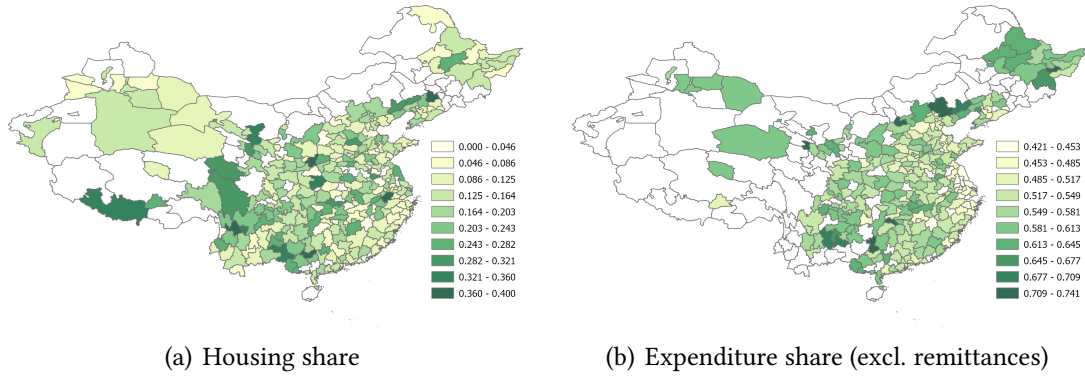
Return migration An intriguing feature of rural-urban migration in China, given the institutional constraints on permanent settlement in cities, is the low incidence of return migration. One possible explanation is the limited availability of non-agricultural employment opportunities in rural areas (despite the effects of remittances documented in [Pan and Sun 2024](#)). We quantify return migration in Figure A11 by comparing the number of migrants who left their origin after 2000 with those who returned between 2004 and 2005. In rural areas where roughly 10% of the population out-migrated during this period, only about 0.3% returned.³⁷ We further discuss return migration and migrants’ prospects in Appendix B.6, where we show that most migrants prefer to remain at destination, even when their families remain at origin.

Robustness and alternative definitions We conclude with a few robustness checks. First, we provide a sensitivity analysis of Figure A3 by considering alternative measures of (displaced) consumption in Figure A12. WPanel (a) reports the share of income devoted to housing and shows that in favored destinations—where migrants remit larger fractions of their income—housing expenditures are relatively lower. More broadly, migrants consume less overall in these locations: panel (b) shows that the ratio of total consumption to income is lower in the most attractive destinations.

In the paper, we use a baseline classification of migration spells that distinguishes migrants living with their entire family at destination, those leaving children behind

³⁷[Imbert et al. \(2022\)](#) further studies the patterns of return migration in the “2005 Mini-Census,” e.g., allowing them to infer the extent of return migration between 2000 and 2005. The conclusion remains that return migration is one order of magnitude lower than migration flows.

Figure A12. Consumption of non-tradables and expenditure shares of migrants across space.



Notes: Panel (a) displays the share of income devoted to housing expenditures across destinations (from CMDS, 2011–2012; the measure includes the employer contribution if housing is provided by the employer). Panel (b) displays the ratio of expenditures (from CMDS, 2011–2012; excluding remittances) to income.

(typically with grandparents), and solo migrants. In practice, however, migration arrangements are more diverse and may involve other dependents.

In Table A2, we replicate Table 1 using four alternative classifications: (i) migrants living with versus without children; (ii) female migrants living with their children at destination versus those leaving children at origin (focusing on women with children, identified using the fertility module of the “2005 Mini-Census”); (iii) migrants living with any relative versus none; and (iv) migrants living with a spouse versus without a spouse. The results are broadly consistent with those obtained under our baseline classification.

Table A2. Descriptive statistics—living arrangements.

	Children		Children (f)		Spouse		Relatives	
	With	Without	With	Without	With	Without	With	Without
<i>Panel A: Demographic characteristics</i>								
Age	36.08	30.12	35.61	35.40	35.20	29.86	35.13	28.97
Female (head)	0.220	0.397	1.000	1.000	0.095	0.467	0.188	0.471
Married	0.987	0.609	0.992	0.988	0.999	0.566	0.961	0.515
Number of children	1.691	1.428	1.687	1.605	1.433	1.499	1.592	1.437
Number of children (OCP*)	1.643	1.354	1.643	1.530	1.333	1.436	1.508	1.379
<i>Panel B: Education</i>								
High school (at least)	0.133	0.188	0.129	0.105	0.151	0.186	0.146	0.196
College (at least)	0.011	0.022	0.010	0.007	0.014	0.022	0.014	0.024
<i>Panel C: Economic characteristics</i>								
Income (head, RMB)	1,196	1,024	1,186	969	1,175	1,012	1,154	1,000
Hours worked per week	55.32	55.53	55.31	56.11	55.83	55.34	55.61	55.41
Housing share	0.232	0.213	0.226	0.234	0.162	0.230	0.199	0.224
<i>Panel D: Living arrangements</i>								
Co-residents	3.106	2.779	3.281	2.203	2.747	2.893	2.673	2.967
No kitchen	0.388	0.594	0.385	0.541	0.465	0.583	0.444	0.618
No toilets	0.545	0.573	0.544	0.613	0.593	0.556	0.576	0.561
House ownership	0.174	0.047	0.172	0.071	0.115	0.059	0.130	0.039
<i>Panel E: Location characteristics</i>								
City income (RMB)	711	862	715	822	762	857	754	879
Observations	12,952	46,231	9,999	15,388	17,234	41,949	23,525	35,658
	0.218	0.782	0.393	0.607	0.291	0.709	0.397	0.603

Notes: The sample is restricted to household heads aged 15–64 living in urban areas (2005 1% Population Survey). Columns (1) and (2) distinguish individuals who moved with children from those who did not (for women with children in columns 3 and 4); columns (5) and (6) distinguish those who moved with a spouse from those who did not; and columns (7) and (8) distinguish those who moved with any relative from those who moved alone. Descriptive statistics for *Monthly income (RMB)* and *Hours worked per week* are restricted to individuals reporting positive working hours in the past week. *Number of children alive* is predicted using the 2010 Population Census (see Appendix A.1; OCP* excludes women older than 25 at the time of the One-Child Policy). *Housing share* is based on predicted values from a regression of log monthly rent for individuals renting commercial housing on prefecture fixed effects interacted with dwelling characteristics.

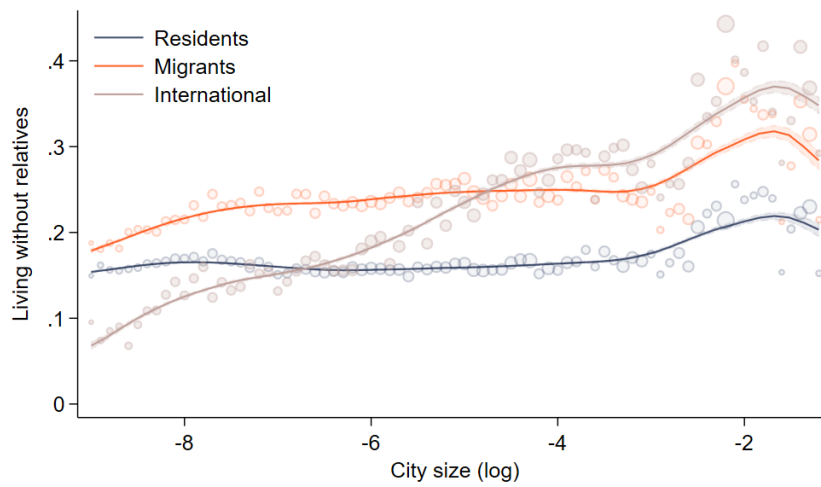
B Complements to the empirical analysis

This section provides additional evidence complementing Section 2.

B.1 Rural-urban migration across countries

We argue that the migration patterns observed in China share important similarities with those documented in other countries.

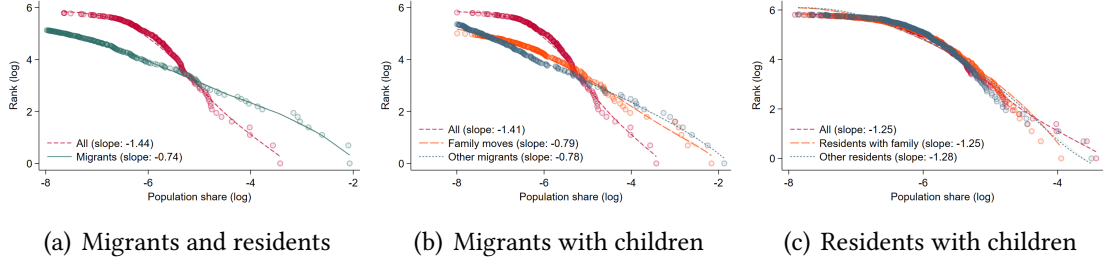
Figure B1. Living without relatives in urban settings (residents, rural-urban migrants, and international migrants).



Notes: This figure relies on 149 censuses from IPUMS, for which we observe second-level administrative units, respondents' location (rural or urban), migration status (based on birthplace or residence 10 years prior), and living arrangements at destination. The y-axis measures the propensity to live without relatives at destination (i.e., alone or as a couple, without children). The lines correspond to local polynomial fits, with observations weighted by population. The dots represent averages across 1,000 bins of (log) relative city size, defined as city population divided by the total urban population within each census wave. The sample covers the following countries: Argentina, Benin, Bolivia, Brazil, Cambodia, Cameroon, Chile, Colombia, Costa Rica, Cuba, Dominican Republic, Ecuador, Egypt, El Salvador, Fiji, Ghana, Greece, Guatemala, Guinea, Haiti, Honduras, India, Indonesia, Iran, Iraq, Israel, Kyrgyz Republic, Lesotho, Malawi, Malaysia, Mali, Mauritius, Mexico, Mongolia, Mozambique, Myanmar, Nepal, Nicaragua, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Rwanda, Senegal, Sierra Leone, South Africa, Thailand, Togo, Uganda, United States, Uruguay, Venezuela, Vietnam, Zimbabwe.

Living without family Figure B1 uses data from 149 censuses across 55 countries (IPUMS) to show that migrants—defined as individuals who moved within the past 10 years or, in some censuses, inferred from place of birth—are more likely than non-migrants to live without family members. In a subset of censuses with fertility information, we can verify that this pattern does not reflect differences in having children, but rather the decision to leave them behind in rural areas. The figure also reveals a clear gradient with respect to city size: migrants are less likely to co-reside with relatives in larger cities. Finally, these patterns are considerably stronger for international migrants (purple curve).

Figure B2. The concentration of migrants across Chinese cities.



Notes: The x-axis reports (log) population by type (e.g., total population, migrants) across prefectures using the “2005 Mini-Census,” with each type normalized to sum to 1 across all prefectures. The y-axis shows the corresponding (log) rank of prefectures. Letting M_c denote the number of rural migrants in city c and $m_c = M_c / \sum_d M_d$ their population share, we estimate $\log R(m_c) = \alpha + \beta \log(m_c)$. The Zipf law predicts $\beta = -1$.

The concentration of migrants across cities We document the extent to which migrants, particularly those moving without family, concentrate in a limited number of cities. To do so, we rely on the Zipf law of city size, which posits a linear relationship between (log) population and (log) rank, with a slope of -1 .

Panel (a) of Figure B2 illustrates this relationship in China for all urban residents (green dots and line) and for rural migrants (red dots and line). While the Zipf law appears to hold for the overall urban population, rural migrants are substantially more concentrated: the relative size of the migrant population in the largest city is about three times that observed for the average urban resident. Panel (b) shows that migrants without family are even more concentrated. By contrast, this gradient is much less pronounced for urban residents with a local *hukou* (panel c).

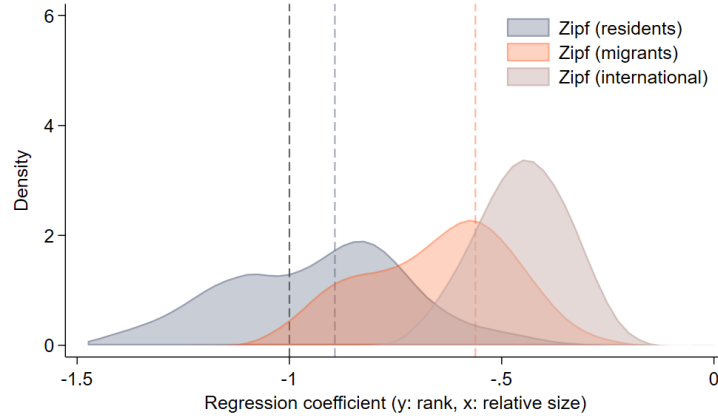
The high concentration of migrants is not unique to China. Figure B3 shows that migrants tend to concentrate more in large cities than residents across many countries, with international migrants exhibiting even stronger concentration than rural-urban migrants. The dashed lines indicate that China lies close to the modal values of these distributions.

B.2 The sorting of migrants across cities

The evidence presented in Section 2 shows that migrants tend to sort into cities with higher monthly wages.

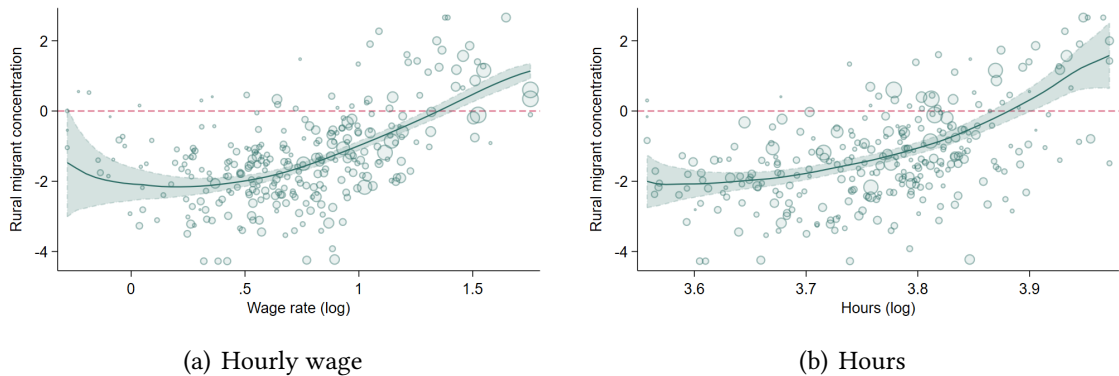
Wage rates and hours Figure B4 decomposes this finding into two distinct effects: (i) migrants sort into cities with higher wage rates (i.e., wages adjusted for hours worked), and (ii) migrants sort into cities where workers supply more hours. The latter effect is economically meaningful, as workers in the highest-wage locations work 25-30% more

Figure B3. The concentration of migrants across cities in other contexts.



Notes: This figure reports the distribution of Zipf coefficients, estimated from regressions of (log) population on the (log) rank of cities. The analysis relies on the same 149 censuses from IPUMS, for which second-level administrative units are observed (see Figure B1). The dashed lines indicate the position of China in the corresponding distributions. The sample includes the following countries: Argentina, Benin, Bolivia, Brazil, Cambodia, Cameroon, Chile, Colombia, Costa Rica, Cuba, Dominican Republic, Ecuador, Egypt, El Salvador, Fiji, Ghana, Greece, Guatemala, Guinea, Haiti, Honduras, India, Indonesia, Iran, Iraq, Israel, Kyrgyz Republic, Lesotho, Malawi, Malaysia, Mali, Mauritius, Mexico, Mongolia, Mozambique, Myanmar, Nepal, Nicaragua, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Rwanda, Senegal, Sierra Leone, South Africa, Thailand, Togo, Uganda, United States, Uruguay, Venezuela, Vietnam, Zimbabwe.

Figure B4. Rural migrant concentration, hourly wage, and hours worked.

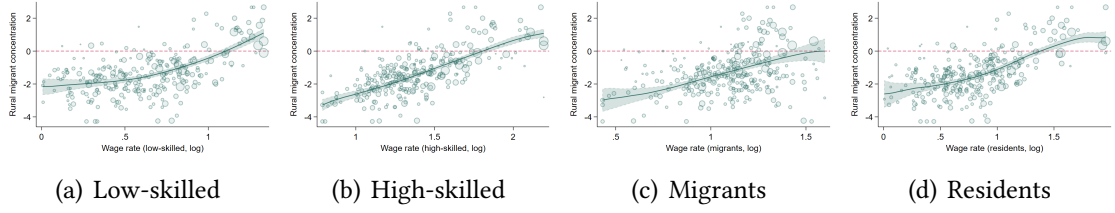


Notes: The y-axis reports migrant concentration in city c , m_c , as defined in Section 2. In panel (a), the x-axis shows the (log) hourly wage; in panel (b), it shows the (log) number of hours worked in a typical week. Hours and wages are constructed by aggregating individual responses from the 2005 1% Population Survey. Each bubble represents a destination prefecture and is weighted by its urban population in 2000. The lines correspond to local polynomial fits, with observations weighted by population.

hours than those in the lowest-wage locations.³⁸

³⁸One explanation is that the substitution effect dominates the income effect for low-income workers in our context. Another likely explanation is a compositional effect, both in terms of occupations and worker characteristics. For instance, migrants tend to work longer hours and are overrepresented in high-wage locations.

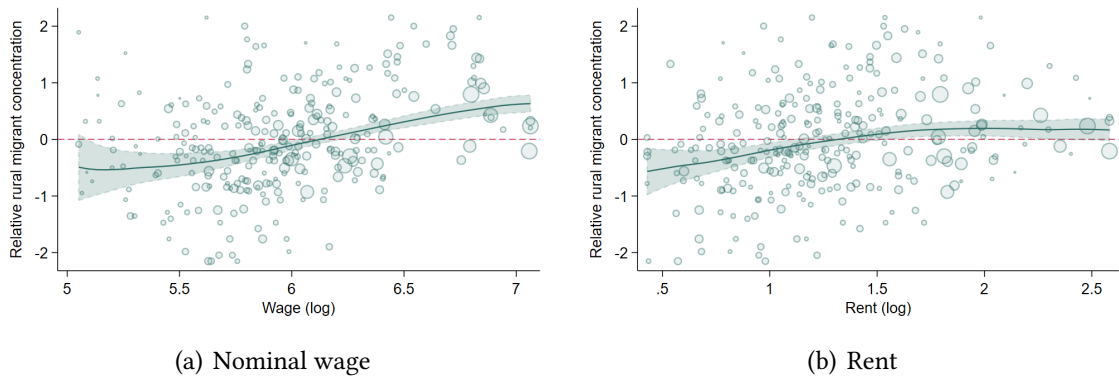
Figure B5. Rural migrant concentration and various measures of wages.



Notes: The y-axis reports migrant concentration in city c , m_c , as defined in Section 2. The x-axis reports alternative measures of (log) monthly wages constructed from the “2005 Mini-Census”: (i) average wages for low-skilled workers in panel (a), defined as workers without a high-school degree; (ii) average wages for high-skilled workers in panel (b), defined as workers with a high-school degree; (iii) migrant wages in panel (c); and (iv) resident wages in panel (d).

Alternative measures of wages Figure B5 further examines the relationship between migrant concentration and returns to labor using four alternative wage measures from the “2005 Mini-Census”: a measure of low-skilled wage in panel (a); a measure of high-skilled wage in panel (b); average wages for rural migrants in panel (c); and average wages for residents in panel (d). These measures are strongly correlated and convey a consistent message: migrant concentration is higher in cities with higher wages.

Figure B6. Relative migrant concentration and living conditions in cities.



Notes: The y-axis reports relative migrant concentration in city c , rm_c . In panel (a), the x-axis shows (log) monthly wages; in panel (b), the x-axis reports a measure of (log) rents. Wages and rents are constructed by aggregating individual responses from the 2005 1% Population Survey. Each bubble represents a destination prefecture and is weighted by its urban population in 2000. The lines correspond to local polynomial fits, with observations weighted by population.

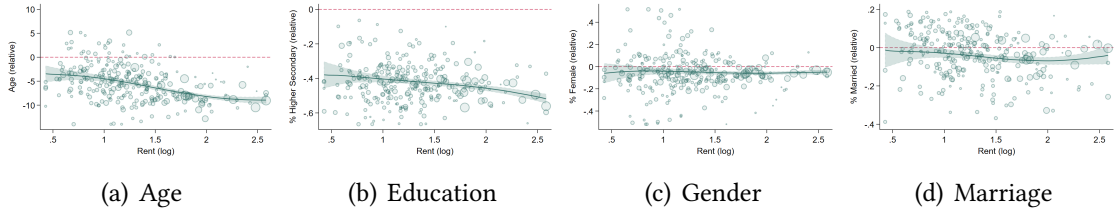
Urban migrants Rural migrants may face lower mobility costs than urban residents when relocating *across cities*. Urban residents are already settled and benefit from access to local services that would be forfeited upon moving to another city, even if returns to labor are higher. A natural implication is that urban migrants should be fewer in number and exhibit different location choices than rural migrants. To document this

fact, we construct a measure of relative migrant concentration in city c , rm_c , defined as:

$$rm_c = m_c - \log \left(\frac{U_c / (\sum_d U_d)}{R_c / (\sum_d R_d)} \right) = \log \left(\frac{M_c / (\sum_d M_d)}{U_c / (\sum_d U_d)} \right),$$

where U_c denotes the number of urban migrants in city c who arrived between 2000 and 2005. This measure equals zero if migrants are allocated identically across cities regardless of their registration type (rural or urban). Panel (a) of Figure B6 shows that relative migrant concentration is positively associated with nominal wages: rural migrants sort more strongly into high-return locations than urban migrants. A 1% increase in nominal wages is associated with a 0.5% increase in the relative share of rural migrants. Panel (b) presents the corresponding relationship with rents.

Figure B7. The selection of migrants relative to residents in expensive cities.



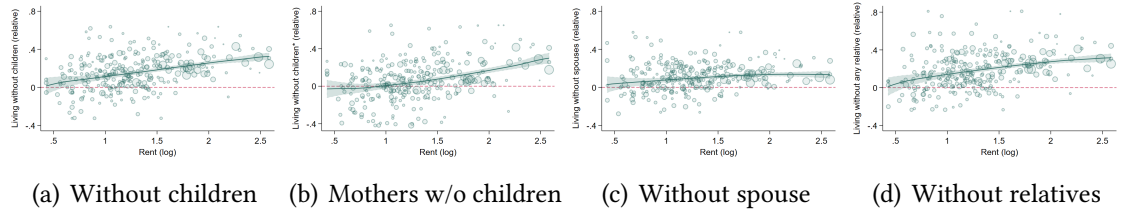
Notes: The x-axis reports (log) monthly rents constructed from the “2005 Mini-Census.” In panel (a), the y-axis shows the difference in average age between rural migrants and urban residents. In panel (b), the y-axis reports the difference between the proportion of migrants and the proportion of urban residents who have at least higher secondary education. In panel (c), the y-axis reports the difference in the share of females. In panel (d), the y-axis reports the difference in the share of married individuals.

B.3 The selection of migrants across cities

Selection along other characteristics Section 2 shows that the selection of rural migrants differs from that of urban residents across cities with varying living conditions. For example, migrants are less likely to live in adequate housing and to reside with their children in high-wage, high-rent locations.

Figure B7 further documents this selective sorting across destinations. We find that migrants are younger than residents, with this difference more pronounced in expensive locations (panel a); they are also much less likely to have completed high school (panel b); there is no strong relationship between migrant gender composition and rents, although migrants are slightly more likely to be male in more expensive cities (panel c); and likewise, we find no clear gradient with respect to marital status, although migrants are somewhat less likely to be married than residents in high-cost locations (panel d).

Figure B8. Migrants and family—sensitivity analysis.



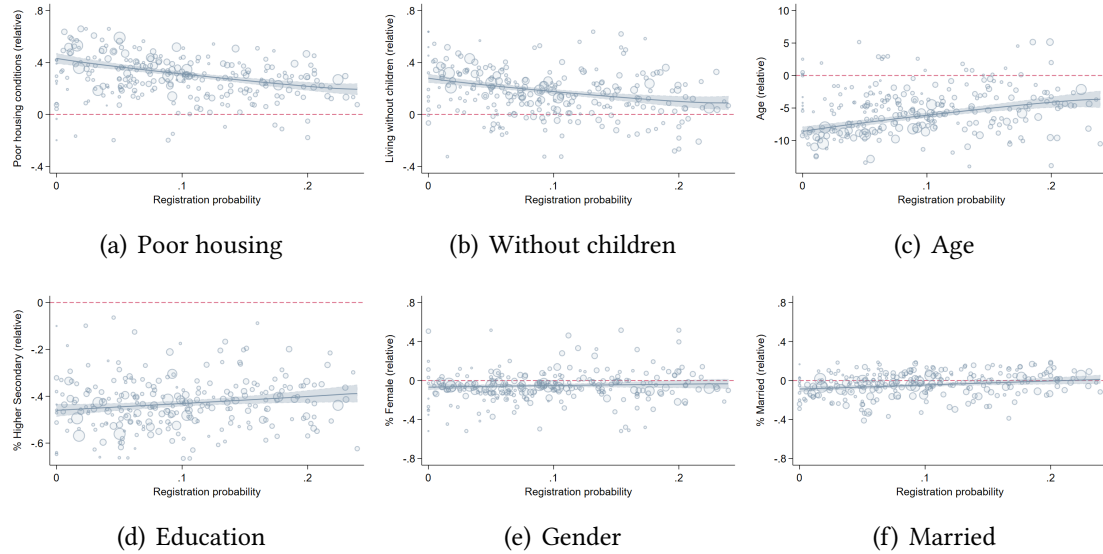
Notes: The x-axis reports (log) monthly rents constructed from the “2005 Mini-Census.” In panel (a), the y-axis shows the difference between the share of rural migrants and urban residents living without their children; panel (b) restricts the sample to women reporting having children. In panel (c), the y-axis shows the difference in the share living without a spouse, and in panel (d), the difference in the share living without any relatives. Each bubble represents a destination prefecture and is weighted by its urban population in 2000. The lines correspond to local polynomial fits, with observations weighted by population.

Other measures of living arrangements There is evidence that migrants with different characteristics sort into different cities. In the main text (Section 2.2), we focus on the *choice* of migrating with or without family and its interaction with location decisions. We now provide a sensitivity analysis in Figure B8.

First, panel (a) replaces the baseline distinction of living with versus without family by living with versus without children. Second, one concern raised by Figure B7 is that expensive cities may attract non-family migrants simply because individuals without dependent children (e.g., unmarried migrants) or those less involved in childcare (e.g., men) are more likely to select into such locations. To address this, we focus on women with children and examine whether they bring their children to more expensive destinations. Panel (b) shows that rural migrant mothers are 8–10 percentage points more likely than urban resident mothers to live with their children in the least expensive locations, but 8 percentage points less likely to do so in the most expensive cities. Panel (c) reports the relative probability of living without a spouse across destinations. Panel (d) broadens the definition of family to include any relative and shows similar patterns: rural migrants are more likely to live without relatives in expensive locations, while they are as likely as residents to co-reside with relatives in less expensive cities.

Selection along *hukou* restrictions We conclude by providing evidence on the selection of migrants and migration patterns across cities with varying registration restrictions. To do so, we rely on our main measure of *hukou* stringency from the 2010 Census: the share of migrants aged 15–64, who moved for work-related reasons, were born in another county, and are locally registered with a non-agricultural *hukou* (following Wu and You 2021). We then replicate Figures 3 and B7, replacing the x-axis with this *hukou* conversion probability. Panels (a) and (b) of Figure B9 show that differences in living arrangements between migrants and residents are smaller in cities with more lenient *hukou*

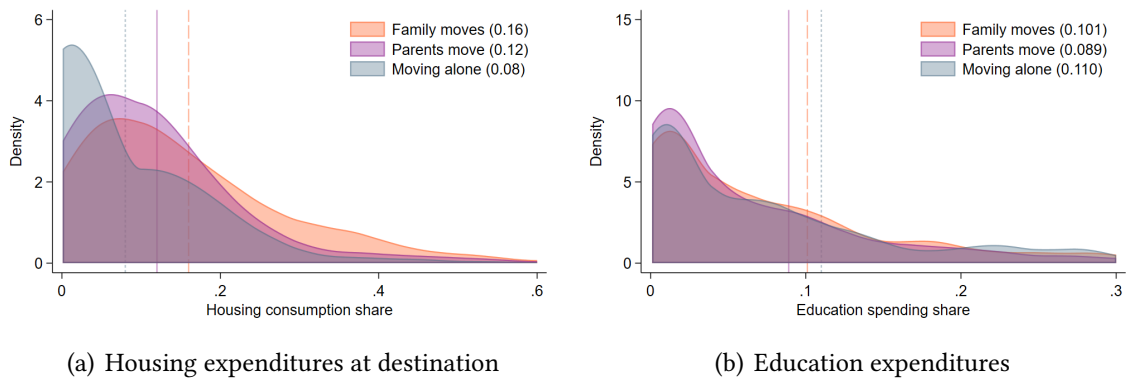
Figure B9. The selection of migrants relative to residents across cities with different local restrictions.



Notes: The x-axis reports the probability that rural migrants convert their *hukou* registration, computed from the 2010 Census. In panel (a), the y-axis shows the difference between the share of rural migrants and urban residents living in poor housing conditions, based on dwelling characteristics from the “2005 Mini-Census.” In panel (b), the y-axis reports the difference between the share of rural migrants and the share of urban residents who live without children. In panel (c), the y-axis reports the difference between the average age of rural migrants relative to that of urban residents. In panel (d), the y-axis reports the difference between the proportion of migrants and the proportion of urban residents who have at least higher secondary education. In panel (e), the y-axis reports the difference in the share of females. In panel (f), the y-axis reports the difference in the share of married individuals. Each bubble represents a destination prefecture and is weighted by its urban population in 2000. The lines correspond to local polynomial fits, with observations weighted by population.

policies (i.e., higher conversion probabilities). However, the education gap remains large across all destinations, regardless of the level of migration restrictions (panel d).

Figure B10. Migrants and expenditures on housing and education.



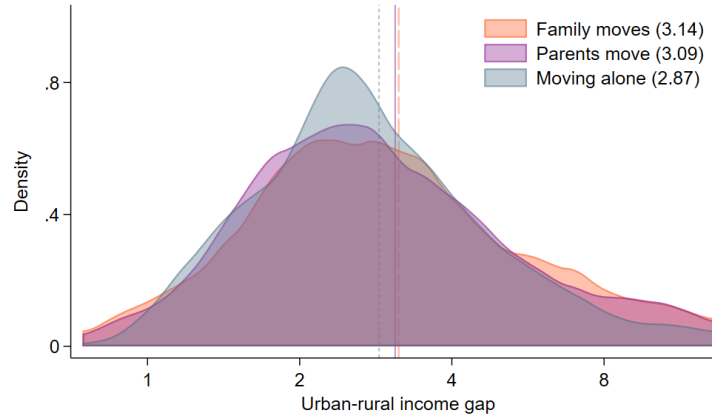
Notes: Panel (a) reports the distribution of housing expenditures as a share of income at destination (source: RUMiC, 2008–2009). Panel (b) reports the distribution of education expenditures as a share of income, including expenses incurred at both destination and origin (e.g., fees and other education-related costs). In both panels, we distinguish between migrants living with their family at destination (orange) and those who have left their family at origin (blue).

B.4 Remittances, expenditures, and the rural-urban wage gap

In Section 2.2, we document the share of income that migrants allocate to remittances, distinguishing between those living with their family and those living apart. Migrants living with their family are found to remit less. We now provide complementary evidence on migrants' consumption patterns across living arrangements using data from the Longitudinal Survey on Rural Urban Migration in China (RUMiC, 2008–2009).

Remittances and expenditures on non-tradables Panel (a) of Figure B10 shows the distribution of housing expenditures at destination for migrants living with and without their family. We find that roughly half of the difference in spending on non-tradable goods and services across these groups is accounted for by housing expenditures. Panel (b) focuses on education expenditures, comparing spending at destination for migrants living with their children to spending at origin for those who leave their children behind. Expenditures are broadly similar, suggesting that migrants who bring their children often rely on lower-cost, informal schooling options of lower quality.

Figure B11. Migrants and the hypothetical rural-urban wage gap.

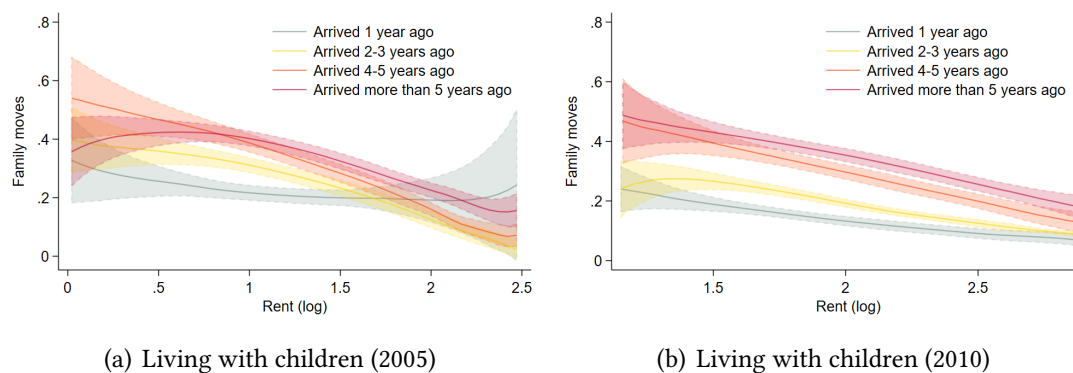


Notes: This figure displays the distribution of a hypothetical rural-urban wage gap, constructed by comparing migrants' actual income at destination with a counterfactual income at origin. The latter is based on responses to the question: "If you were still in your home village, how much do you estimate you could earn every month?" We distinguish between migrants living with their family at destination (orange) and those who have left their family at origin (blue). The x-axis is shown on a logarithmic scale.

The rural-urban wage gap Figure B11 highlights a central driver of rural-urban migration: the wage gap between rural and urban areas, which offsets higher living costs at destination and the utility cost of family separation. Using a hypothetical question from the Longitudinal Survey on Rural Urban Migration in China, we approximate the income migrants would earn if they remained in their home village. We find that the wage gap is similar across migration arrangements: migrants earn more than three times the rural

wage. Consistent with this, remittances of about 30% of urban income—documented in Section 2.2—amount to nearly a full rural wage.

Figure B12. Living with children throughout the migration spell.



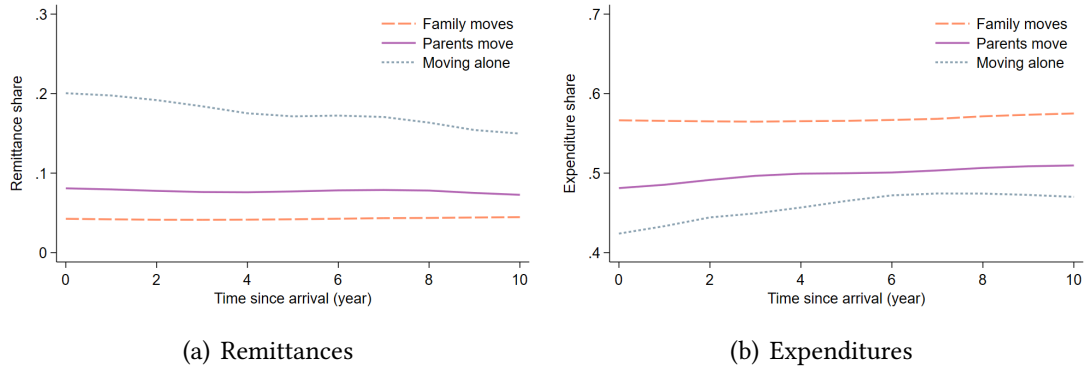
Notes: The x-axis reports a measure of (log) monthly rents constructed using the “2005 Mini-Census.” The y-axis reports the share of migrants living with family at destination (in 2005 for panel a, in 2010 for panel b). The lines correspond to local polynomial fits, with observations weighted by population. The green line represents migrants who arrived one year prior to the census (after 2004 in panel a, after 2009 in panel b); the yellow line is computed for migrants arrived between 2 and 3 years prior to the census; the orange line is computed for migrants arrived between 4 and 5 years prior to the census; and the red line represents arrivals more than 5 years prior.

B.5 The dynamics of migration arrangements across cities

The evidence in Section 2 abstracts from dynamic adjustments in migration arrangements over migrants’ life cycles. While Appendix A.3 documents the stability of migrant inflows, we now examine how living arrangements evolve over time. Figure B12 reports the incidence of family migration as a function of time since arrival at destination, using data from 2005 (panel a) and 2010 (panel b). A natural concern is that split arrangements—such as leaving children behind—may be temporary, reflecting an initial phase during which migrants accumulate resources and information before bringing their family to destination. We find little evidence supporting this hypothesis. If anything, time since arrival matters primarily in less expensive cities, and the relationship between migration arrangements and destination prices becomes even steeper after 4-5 years.

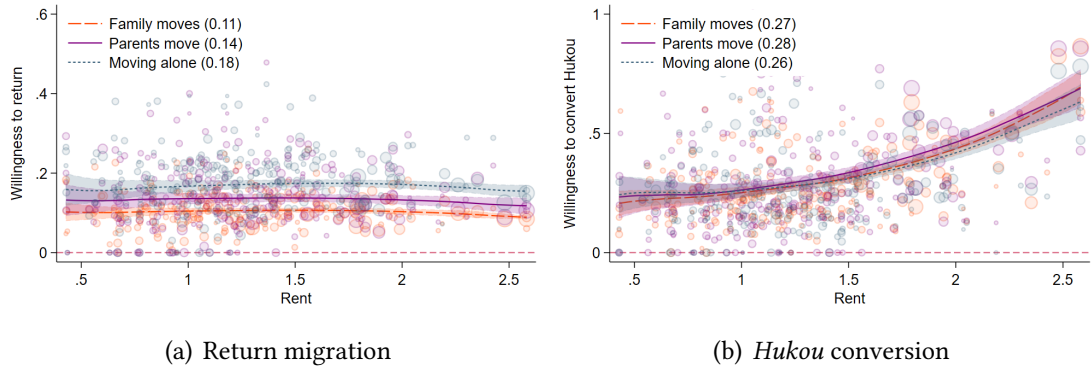
Figure B13 shows the consumption patterns of migrants with and without family as a function of time since arrival. While some adjustment occurs over the migration spell, the gap between the two groups remains large and fairly stable (or converges only slowly), whether measured through remittance behavior (panel a) or consumption at destination (panel b).

Figure B13. Remittances throughout the migration spell.



Notes: The x-axis reports time since arrival for migrants interviewed in CMDS (2011). In panel (a), the y-axis shows the remittance share for migrants living with their family at destination (orange line) and those living without family (blue line). In panel (b), the y-axis shows the ratio of expenditures (excluding remittances) to income for the same groups.

Figure B14. Future prospects across migration spells.



Notes: The x-axis reports (log) monthly rents constructed from the “2005 Mini-Census.” In panel (a), the y-axis shows the average willingness to return (from CMDS) for migrants living with their children (orange) and those without children (blue). In panel (b), it shows the average willingness to convert *hukou* at destination (from CMDS) for the same groups. Each bubble represents a destination prefecture and is weighted by its urban population in 2000. The lines correspond to local polynomial fits, with observations weighted by population.

B.6 Prospects, return migration, and *hukou* conversion

Figure B14 documents heterogeneity in migrants’ prospects across destinations and living arrangements. Specifically, we use data from the China Migrants Dynamic Survey (CMDS) on migrants’ willingness to return to their origin and their willingness to convert their *hukou* at destination (irrespective of actual eligibility).

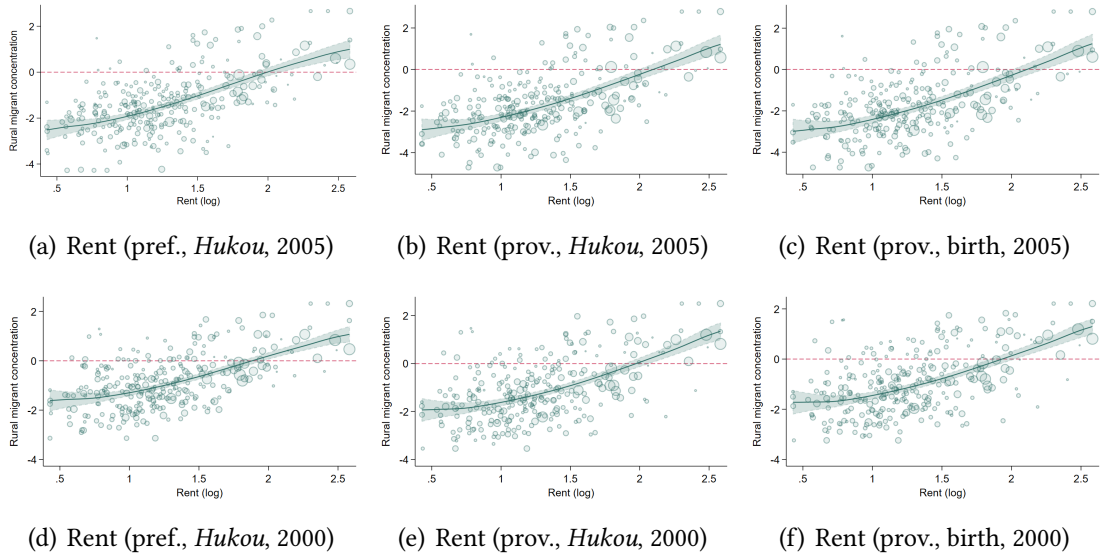
We find that the share of migrants willing to return is relatively small (panel a): about 16% among migrants living without family, compared to 11% among those living with family at destination. In contrast, roughly 40% of migrants express willingness to convert their *hukou* to their destination location—a prospect that was unlikely around 2000 but

became more realistic with gradual policy changes, culminating in the 2014 reforms. This evidence supports our modeling choice to abstract from dynamic features such as return migration.

B.7 A sensitivity analysis of migrant definitions

Our measure of migration is based on discrepancies between place of household registration and place of residence. Because some migrants are able to convert their *hukou* and register at destination, this approach may misclassify a subset of rural-urban migrants as urban residents. Moreover, the pronounced underrepresentation of migrants in less expensive cities observed in Figure 2 may partly reflect higher *hukou* conversion probabilities in those locations.

Figure B15. Rural migrant concentration—alternative migration definitions.



Notes: The x-axis reports (log) monthly rent per square meter, constructed from the “2005 Mini-Census” (2005) or the 2000 Population Census. In panels (a)–(f), the y-axis shows migrant concentration in city c , m_c (see Section 2.2). The panels vary the definition of migration along three dimensions: province- versus prefecture-level aggregation, data source (2005 1% Population Survey versus 2000 Population Census), and *hukou*- versus birthplace-based definitions. Each bubble represents a destination prefecture and is weighted by its urban population in 2000. The lines correspond to local polynomial fits, with observations weighted by population.

In this section, we use additional information from the 2000 and 2005 censuses to construct alternative measures of migration and assess the robustness of our stylized facts. Our baseline measure relies on three components: (i) the discrepancy between place of residence and place of household registration; (ii) information on *hukou* type; and (iii) the timing of migration (within the past five years). We retain (iii) and complement it with information on place of birth (available in 2000) or on place of residence

five years prior to the census (used as a proxy for birthplace in 2005).³⁹ Because these alternative origin measures are available only at the province level, we also reproduce our main stylized facts using *hukou*-based migration defined across provincial boundaries.

Figure B15 reproduces selected results from Figures 2 and 3 using these alternative definitions. The variations span three dimensions: the reference year (2005 in the baseline or 2000), the spatial unit (prefecture versus province), and the definition of migrants (*hukou*-based versus birthplace-based). Across all specifications, we obtain gradients that are nearly identical to our first stylized fact. While the level and slope of the fitted polynomials vary slightly, our second stylized fact remains robust to these alternative definitions.⁴⁰

³⁹The latter is an acceptable proxy of birthplace or the place of *hukou* registration before conversion if step migration is limited. [Imbert et al. \(2022\)](#) show that this was indeed the case in 2000–2005 in China.

⁴⁰Additional results (not reported) confirm that Figure 2(a) and Figure 3(b) are robust to these changes.

C Complements to the model

This section complements Section 3 by deriving indirect utilities and consumption patterns across goods and family members, and by discussing several model extensions in three steps.

C.1 Migration, living arrangements, and consumption bundles

Consumption across members and across goods We suppose that the consumption bundle for each household member is defined as:

$$C^g = (C_t^g)^{1-\alpha-\beta} (C_n^g)^\alpha (C_r^g)^\beta,$$

where C_t^g is the consumption of the tradable good, C_n^g is the consumption of the non-tradable good for household member g , and C_r^g is a residual consumption of the non-tradable good at origin. The latter is introduced to best capture that all migrant households remit back to the origin, as illustrated in Section 2.2. In this sub-section, we omit the subscript associated with the living arrangement, ℓ .

Under the assumptions that the sub-utilities of each household member are additive and separable, the overall household program can be decomposed into the following steps. First, the household chooses a consumption bundle C^g for each household member and for a given expenditure E^g ,

$$\max_{C_t^g, C_n^g, C_r^g} (C_t^g)^{1-\alpha-\beta} (C_n^g)^\alpha (C_r^g)^\beta$$

subject to $P_t C_t^g + P_n C_n^g + P_r C_r^g \leq E^g$. Second, the output of the previous program can be used to define a price index $\mathcal{P}^g$ such that $\mathcal{P}^g C^g = E^g$, and the household allocates expenditures across household members such as to maximize:

$$\max_{C^f, C^m, C^c} \xi_f u(C^f) + \xi_m u(C^m) + (1 - \xi) u(C^c)$$

where $\xi = \xi_f + \xi_m$ and subject to $\mathcal{P}^f C^f + \mathcal{P}^m C^m + \mathcal{P}^c C^c \leq \mathcal{W}$. Third, the previous computations provide us with a consumption bundle across goods and across household members, and an indirect utility $\ln V = \xi_f u(C^{f*}) + \xi_m u(C^{m*}) + (1 - \xi) u(C^{c*})$, which can be expressed as a function of prices. With the latter, one can solve for the discrete, nested choice model described in Section 3.3.

The choice of consumption bundles $\{C^g\}_{g \in \{f, m, c\}}$ leads to the following price indices:

$$\mathcal{P} = \left(\frac{P_t}{1 - \alpha - \beta} \right)^{1 - \alpha - \beta} \left(\frac{P_n}{\alpha} \right)^\alpha \left(\frac{P_r}{\beta} \right)^\beta,$$

and the following consumption allocations:

$$P_t C_t^g = (1 - \alpha - \beta) E^g, \quad P_n C_n^g = \alpha E^g, \quad P_r C_r^g = \beta E^g,$$

The allocation of expenditures across household members implies that $E^f = \xi_f \mathcal{W}$, $E^m = \xi_m \mathcal{W}$, and $E^c = (1 - \xi) \mathcal{W}$. These equations, combined with Equation (2), allow us to derive the expressions used in Section 3 for indirect utilities.

Indirect utilities To derive indirect utilities, we need to specify the household budget across the different living arrangements. We assume that the streams of wages are given by $\mathcal{W}(r, r, r) = 2w_r$, $\mathcal{W}(u, r, r) = \kappa_1(w_u^f + w_r)$, $\mathcal{W}(r, u, r) = \kappa_2(w_u^m + w_r)$, $\mathcal{W}(u, u, r) = \kappa_3(w_u^f + w_u^m)$, and $\mathcal{W}(u, u, u) = \kappa_4(w_u^f + w_u^m)$, where w_u^f and w_u^m are the father's and mother's urban income. Assuming that the tradable good is the numeraire, i.e., $P_t = 1$, and omitting constants, we can combine the previous calculations to derive the indirect utility of a household staying in location r :

$$\ln V_r^s = \ln(2w_r) - (\alpha + \beta) \ln p_r.$$

A father moving from origin $r \in R$ to destination $u \in U$, while leaving the rest of the family behind, would generate the indirect utility:

$$\ln V_{ru}^{m_1} = \ln(w_u^f + w_r) - \xi_f \alpha \ln p_u - [\beta + (1 - \xi_f) \alpha] \ln p_r + \ln \kappa_1 - \bar{u}_{ru}^{m_1},$$

when a mother moving alone would generate:

$$\ln V_{ru}^{m_2} = \ln(w_u^m + w_r) - \xi_m \alpha \ln p_u - [\beta + (1 - \xi_m) \alpha] \ln p_r + \ln \kappa_2 - \bar{u}_{ru}^{m_2}.$$

The indirect utility of a household in which both parents migrate from r to u while children remain at origin is:

$$\ln V_{ru}^{m_3} = \ln(w_u^f + w_u^m) - \xi \alpha \ln p_u - [\beta + (1 - \xi) \alpha] \ln p_r + \ln \kappa_3 - \bar{u}_{ru}^{m_3}.$$

Finally, the indirect utility of a household migrating from r to u with the entire family is:

$$\ln V_{ru}^{m_4} = \ln(w_u^f + w_u^m) - \alpha \ln p_u - \beta \ln p_r + \ln \kappa_4 - \bar{u}_{ru}^{m_4}.$$

In the paper, we do not devote much attention to the problem of single-individual households. Indeed, facing similar bundles as family households,

$$C^O = (C_t^O)^{1-\alpha_O-\beta_O} (C_n^O)^{\alpha_O} (C_r^O)^{\beta_O},$$

but with one main residence and one single preference weight, their choice sums up to choosing between staying,

$$\ln(w_r) - (\alpha_O + \beta_O) \ln p_r.$$

or migrating to destination u ,

$$\ln(w_u) - \alpha_O \ln p_u - \beta_O \ln p_r + \ln \kappa_O - \bar{u}_{ru}^O.$$

Expenditure shares and preference parameters In order to derive the allocation of household members across space, we need to estimate the set of preference parameters and recover the relative weight of each household member $(\xi_f, \xi_m, 1 - \xi)$ and the consumption shares (α, β) . In this subsection, we show how observing (i) the total consumption allocated to non-tradables, (ii) the remittance share for households migrating without children, and (iii) the remittance share for households migrating with children is sufficient to identify a unique set of parameters $(\alpha, \beta, \xi_f, \xi_m)$.

First, with a structure of “overlapping” Cobb-Douglas preferences, the optimal allocation of consumption across the different goods and household members implies that a fixed share of income is devoted to each good/member. All members allocate a share $\alpha + \beta$ of their respective budget to the non-tradable good, such that the overall share of income allocated to the non-tradable good is $x = \alpha + \beta$, irrespective of the location of such consumption. In RUMiC (see Section 2.2 for an illustration of the underlying data), we find that households spend two thirds of their budget on the consumption of localized goods and services, either at destination or origin ($x = 0.66$).

Second, assuming that remittances only cover the consumption of the non-tradable good, father alone transfer a share $y_2 = \beta + (1 - \xi)\alpha = 0.35$ back to the origin, migrant households transfer a share $y_2 = \beta + (1 - \xi)\alpha = 0.28$ back to the origin when migrating without their children and a share $z = \xi\beta = 0.11$ when migrating with their children (see Section 2.2, where we display the distribution of remittance shares by living

arrangement). Overall, the previous equations form the following system:

$$\begin{cases} x = \alpha + \beta \\ y_1 = \beta + (1 - \xi_f)\alpha \\ y_2 = \beta + (1 - \xi)\alpha \\ z = \xi\beta, \end{cases}$$

which can be inverted to get $\alpha = x - z/(1 - (y_2 - z)/x) = 0.500$, $\beta = z/(1 - (y_2 - z)/x) = 0.128$, $\xi_f = 1 - (y_1 - z)/x = 0.864$, and $\xi = 1 - (y_2 - z)/x = 0.556$.⁴¹

A similar logic applied to single-individual households would bring $\alpha_O = 0.459$ and $\beta_O = 0.166$, thus only differing marginally from family households in terms of their overall consumption across goods.

C.2 Extensions and reinterpretations of the model

A dynamic reinterpretation of the model While we have introduced a model where the location choice is static, it is easy to reinterpret the model in a dynamic manner. For instance, we can assume that $\mathcal{U}(u(C^f), u(C^m), u(C^c)) = \xi_f u(C^f) + \xi_m u(C^m) + (1 - \xi) \mathbb{E}[u(C^c)]$, i.e., parents care about their utility and the expected, future utility of their children. Similarly, we can flexibly parametrize $\mathcal{W}(\ell_m, \ell_f, \ell_c)$ to capture the fact that the stream of wages is equal to the parents' wages and the expected wages of the younger generation, both depending on their respective location choice. Under this reinterpretation, we gain additional insights: (i) higher expected urban wages encourage family migration, whereas higher expected rural wages discourage migration overall and tilt migration toward migrants moving without family; and (ii) the importance of location as determining the capacity for children to later migrate to urban areas is also crucial.

A model with return migration and family reunification We have, so far, ignored the possibility that rural migrants return to their origins, or that non-family migrants in urban areas manage to bring some or all family dependents to the city. A simple 2-period extension of the model can accommodate these ideas, allowing migrants to change their status across periods, for instance, reunifying the family at destination or returning at origin in the second period. Each of these options would come with a different stream of income. The insights we obtain from these extended models are, however, very similar to the ones already discussed. The gain is that the extended model allows us to discuss

⁴¹In these calculations, we assume that households do not remit amounts that will serve the consumption of the tradable good at origin. An alternative approach would consider that remittances do include such expenses.

family reunification and return migration policies.⁴²

A model with human capital A final alternative is to explicitly model human capital accumulation. In this interpretation, we assume that the choice is static—the household maximizes the following preferences:

$$\mathcal{U}(u(C^f), u(C^m), u(C^{c_1}), u(C^{c_2}), u(C^{c_2})) = \xi_f u(C^f) + \xi_m u(C^m) + \xi_1 u(C^{c_1}) + \xi_2 u(C^{c_2}),$$

where C^{c_1} is the current consumption of the young generation, and C^{c_2} is the future consumption of the young generation.⁴³ We assume for simplicity that the future consumption of the young generation is linear in their acquired human capital, H , i.e., $C^{c_2} = \phi H$. In this alternative model, the household faces a budget constraint in the present period, and of the following form:

$$\mathcal{P}^f C^f + \mathcal{P}^m C^m + \mathcal{P}^c C^{c_1} + QH \leq \mathcal{W},$$

where prices $\mathcal{P}^g$ and Q depend on where household members reside. If the household stays, all prices are at the origin. If migrants moves without the young generation, $(\mathcal{P}^c, Q)$ are at origin. If the household moves as a unit, all prices are at destination.

Under these assumptions, the household will choose consumption baskets and human capital accumulation under the different living arrangements as before, except that the share devoted to children will be divided into:

1. Expenditure on consumption and education of the young, for households who stay at origin:

$$\ln \left(\frac{p_r C_r^{c_1}}{W_\ell} \right) = \ln \xi_1, \quad \ln \left(\frac{Q_r H_r}{W_\ell} \right) = \ln \xi_2$$

2. Expenditure on consumption and education of the young, for households who leave kids at origin:

$$\ln \left(\frac{p_r C_r^{c_1}}{W_\ell} \right) = \ln \xi_1, \quad \ln \left(\frac{Q_r H_r}{W_\ell} \right) = \ln \xi_2$$

3. Expenditure on consumption and education of the young, for households who migrate with the kids:

⁴²Having multiple periods could allow us to consider the role of credit constraints. For instance, credit constraints may encourage non-family migration in ways that are similar to high education costs in cities.

⁴³Descriptive statistics extracted from RUMiC (2008–2009) indicate that $\xi \approx 0.85$, $\xi_1 + \xi_2 \approx 0.15$, and $\xi_2 \approx 0.08$, which is based on the fact that education expenditures are stable across living arrangements at around 8–10% of the household income.

$$\ln \left(\frac{p_u C_u^{c_1}}{W_\ell} \right) = \ln \xi_1, \quad \ln \left(\frac{Q_u H_u}{W_\ell} \right) = \ln \xi_2$$

One can see that leaving children behind provides advantages relative to staying or moving with those children: (i) migration allows the household to allocate a much higher income across the different consumption pots, and (ii) migration *without children* allows the household to face origin prices when choosing the consumption (price p_r) and human capital accumulation (price Q_r) of the young generation. For instance, one can derive the relative gain in human capital accumulation for households who migrate without their children as:

$$\ln \left(\frac{H_r}{H_u} \right) = - \ln \left(\frac{Q_u}{Q_r} \right),$$

where the latter plays a similar role as our previous κ 's in that it creates an education-driven wedge between the indirect utilities across living arrangements of movers.

While education quality is probably higher in urban centers than in rural areas, rural migrants do not have free access to urban schools and end up entering low-quality public schools or informal, migrant-run private schools (Chen and Feng 2013). In short, we find that $Q_u > Q_r$ for rural children and the gap to be strongly linked to *hukou* policies.

C.3 Model with mobile urban dwellers

In our baseline model, we assume for simplicity that urban residents are immobile. In practice, there is some urban-urban migration in China, even if, as Figure 1 makes clear, it is much less important than rural-urban migration.⁴⁴ In this section, we expand our model so that urban residents are mobile across locations, which allows us to determine the initial allocation of urban residents as a function of location fundamentals and model parameters.

Urban residents decide where to live based on the following utility function:

$$\ln V_u = (1 - \alpha - \beta) \ln C_t + (\alpha + \beta) \ln C_n,$$

subject to a budget constraint,

$$C_t + p_u C_n \leq w_u,$$

where we use the same notation as the main text. In this context, utility maximization results in the following indirect utility for each individual i with origin u and destination

⁴⁴The fact that there is not much urban to urban mobility around the year 2000 in China, as documented in Figure 1, probably reflects the fact that the gain from moving is much lower for urban residents than for rural ones, rather than limits to mobility.

$v \in U$:

$$\ln V_{uv}^m + \varepsilon_{iuv} = \ln w_v - (\alpha + \beta) \ln p_v - \tau_{uv} + \varepsilon_{iuv}.$$

This maximization problem results in the following share of workers across locations:

$$\frac{N_{uv}}{N_u} = \left(\frac{V_{uv}^m}{V_u^m} \right)^{1/\lambda},$$

where $\frac{N_{uv}}{N_u}$ is the probability for inhabitants of u to migrate to v , conditional on migrating to any other city in U . In this case, the marginal mover between any two urban locations is indifferent across locations, as is normal in spatial equilibrium models.

We can use this labor supply equation together with the Cobb-Douglas version of the labor demand equation to solve for the initial distribution of urban residents across locations, which, in the baseline model, we took as exogenous:

$$w_u = w(Z_u, N_u)$$

and:

$$p_u = p(T_u, N_u).$$

Hence, we can substitute these two equations into V_u^m to obtain that:

$$V_u^m = \left[\sum_u \left[w(Z_u, N_u) / p(T_u, N_u)^{\alpha+\beta} \right]^{1/\lambda} \right]^\lambda.$$

These equations define a system of U equations and U unknowns ($N_u, \forall u \in U$) that uniquely determines the distribution of urban residents N_u as a function of fundamentals $\{Z_u, T_u\}$ and the main elasticities of the model, as formally shown in [Allen and Arkolakis \(2014\)](#). Note that we can even get closed-form solutions for the distribution of urban residents as a function of fundamentals.

C.4 Model with multiple skills

In our baseline model, we assume, for simplicity, that there is only one labor type. In practice, labor may be heterogeneous. We discuss here how to incorporate multiple factor types at destination in a framework that follows [Amior and Manning \(2021\)](#), and we investigate how this affects the local labor and housing markets.⁴⁵

⁴⁵Considering multiple skills is probably more important from the perspective of recipient locations than from sending rural communities. It is quite natural to think that, in urban locations, there are many highly qualified jobs that are different in nature than jobs that require fewer/other types of skills.

Local production As in the main text, we assume that tradable output in location u is produced with the following production function:

$$Y_u = Z_u \left[(1 - \theta) K_u^{\frac{\sigma-1}{\sigma}} + \theta L_u^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}},$$

where Z_u is the local (exogenous) productivity, K_u denotes capital or land, the parameter σ denotes the elasticity of substitution between labor and the other factor, but labor, L_u , is now a composite of different types of workers that can be expressed as:

$$L_u = \left[\sum_e \theta_e (L_{ue})^{\frac{\sigma_e-1}{\sigma_e}} \right]^{\frac{\sigma_e}{\sigma_e-1}}.$$

This production function allows us to apply the results of [Amior and Manning \(2021\)](#), assuming that each factor can be decomposed between urban residents and (rural) migrants as $L_{ue} = N_{ue} + M_{ue}$. Defining the fraction of urban residents and migrants in each (e, u) cell as $v_{ue} = N_{ue}/N_u$ and $\mu_{ue} = M_{ue}/M_u$, the labor aggregate can be written as a function of aggregate labor across types:

$$L_u = F(N_{ue} + M_{ue}, \forall e) = \left[\sum_e \theta_e (v_{ue} N_u + \mu_{ue} M_u)^{\frac{\sigma_e-1}{\sigma_e}} \right]^{\frac{\sigma_e}{\sigma_e-1}} = F(v_{ue} N_u + \mu_{ue} M_u).$$

Defining $\mathcal{L}(M_u, N_u) = F(v_{ue} N_u + \mu_{ue} M_u)$ and holding the immigrant distribution across factor types fixed, an inflow of migrants results in the following:

$$\frac{\partial \mathcal{L}(M_u, N_u)}{\partial M_u} = \sum_e \mu_{ue} \frac{\partial F_e(N_{ue} + M_{ue}, \forall e)}{\partial M_{ue}}.$$

The effect of a migrant shock will be the weighted average of the effect of migrants to each factor type. Under perfect competition in the labor market, this can be interpreted as the average effect on wages in the location.

Hence, the counterfactuals that we performed should be interpreted as holding the distribution of migrants across skill types fixed in each location.

Local housing markets Having multiple factor types also affects the housing market. With multiple skills, there are multiple wage levels. These different wage levels enter the demand for housing, which is reflected in the market clearing condition of the housing sector:

$$T_u p_u^\eta = \sum_e \frac{w_{ue}}{p_u} [(\alpha + \beta) N_{ue} + \Gamma M_{ue}].$$

We can rewrite this expression as:

$$\ln p_u = \frac{1}{1+\eta} \ln \left[(\alpha + \beta) N_u \left(\sum_e w_{ue} v_{ue} \right) + \Gamma M_u \left(\sum_e w_{ue} \mu_{ue} \right) \right] - \frac{1}{1+\eta} \ln T_u,$$

where Γ is the effective income share allocated to the non-tradable good by migrants. In turn, this expression can be rewritten as:

$$\ln p_u = \frac{1}{1+\eta} \ln \left[(\alpha + \beta) N_u \bar{w}_u^N + \Gamma M_u \bar{w}_u^M \right] - \frac{1}{1+\eta} \ln T_u.$$

This expression takes into account that the average wage of urban residents and immigrants may be different because natives and immigrants may be differently distributed over factor types. The intuition is that an immigrant inflow will increase the demand for housing, thereby putting upward pressure on housing prices. At the same time, however, the immigrant shock may affect wages in the city, which in turn, affects the demand for housing. Which of these two forces dominates is, in general, ambiguous.

D Complements to the model estimation

This section provides complements to Section 4: (i) we estimate an elasticity of substitution between consuming at origin and at destination to rationalize the use of Cobb-Douglas preferences; (ii) we identify the shape parameters of the location choice model and describe how we extract exogenous variation in the relative value of emigrating with family; (iii) we estimate the labor demand and housing supply elasticities; and (iv) we shed light on the distribution of bilateral migration costs.

D.1 Elasticity of substitution between origin and destination

This section estimates elasticities of substitution between consuming the non-tradable good across locations. As we will see, the estimates encourage us to simplify the analysis and assume a Cobb-Douglas structure of preferences in the baseline model. We estimate the following empirical relationship between the average (log) expenditure share on remittances in city u , $\ln(\mathcal{E}_u^{m_j})$, for each living arrangement $\{\lambda_j\}_{j \in \{1,2,3,4\}}$ and for single-individual migrant households, and local housing prices, p_u , constructed respectively from the China Migrants Dynamic Survey (2011) and the “2005 Mini-Census”:

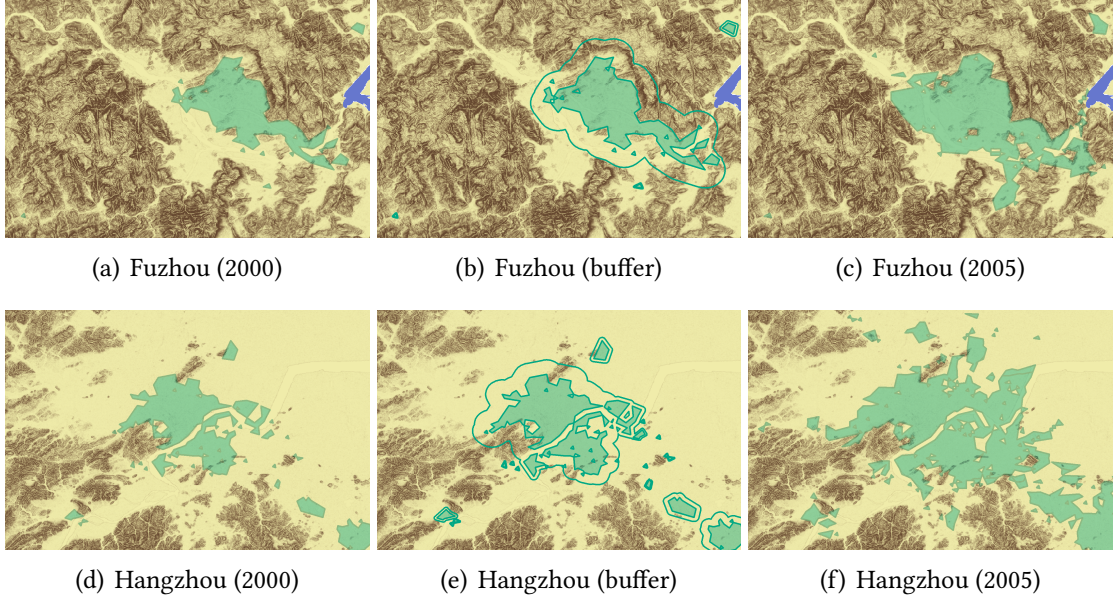
$$\ln(\mathcal{E}_u^{m_j}) = a + b_j \ln p_u + e_u^{m_j},$$

where the estimate b_j would be related to the elasticity of substitution between consuming the non-tradable good across locations (“ $b_j = \rho - 1$ ”). An issue with estimating the previous equation is that the residual, $e_u^{m_j}$, might capture migrant composition and unobserved local amenities, inducing an endogeneity concern with omitted variation jointly affecting prices and expenditure shares.

Exogenous variation in housing supply We exploit exogenous variation in housing supply across destinations to predict variation in the price of non-tradables (proxied by the price of housing services). To do so, we identify the shape of cities before our episode of mass migration, and we characterize topography in their immediate hinterlands.

We proceed in three steps. In a first step, we draw on the identification of impervious areas by the Beijing City Lab in 2000 to identify the urban extent of each city within a given prefecture. In a second step, we construct a city-specific buffer, the extent of which is calibrated to ensure that all cities grow proportionally and homogeneously in all directions (Harari 2020). In a third step, we identify water coverage and the local ruggedness within this buffer of potential urban sprawl. In our baseline strategy, we calculate the share of non-developable land within this land stretch for city u , s_u , by classifying a pixel of $30\text{m} \times 30\text{m}$ as “non-developable” if the average slope is above 5

Figure D1. An example of our procedure with Fuzhou and Hangzhou.



Notes: The delineation of impervious areas, as identified from Landsat satellite imagery, is provided by the Beijing City Lab—see <https://www.beijingscitylab.com/>—and impervious areas are indicated as plain green areas (2000 in panels a–b and d–e, 2005 in panels c and f). The green line in panels b and e corresponds to urban sprawl, as predicted by a uniform growth across cities and within cities in all directions.

degrees. Figure D1 provides insight about the construction of the instrument and the variation that it induces across urban areas. Fuzhou and Hangzhou are two historical cities. As shown in panels (a) and (d) of Figure D1, they markedly differ in constraints to their expansion before mass migration: Fuzhou is in a valley along the Min River and is surrounded by steep hills (especially in the north), while Hangzhou is located in a plain with a few scattered hills. Fuzhou would need to build on a very large share of “non-developable” land if it were to expand in all directions and as much as the average Chinese city (panel b). Hangzhou, on the other hand, would face very limited constraints (panel e). In 2005, we find indeed that Fuzhou experienced an unbalanced urban sprawl concentrated toward the south, while Hangzhou sprawled massively in every direction.

Elasticity of substitution between consuming at origin and destination We use the previous instrument, i.e., the share of developable land at the fringe of cities, to identify the elasticity of substitution between consuming at origin and destination. We rely on the following specification:

$$\ln(\mathcal{E}_u^{m_j}) = a + b_j \ln p_u + \mathbf{X}_u \beta + e_u^{m_j},$$

Table D1. Elasticity of substitution between consuming at origin and destination—first-stage.

	Rent (log)
Share of non-developable land	1.237 (0.348)
Share of non-developable land $\times$ TFP	0.976 (0.524)
Observations	173

Notes: A unit of observation is a prefecture. Robust standard errors are reported between parentheses. The specification uses population weights in 2000. The dependent variable is the (log) rent, computed using the housing module of the “2005 Mini-Census.” The (log) rent is instrumented by (i) the share of non-developable land as induced by local geography around city borders in the baseline period 2000–2005 (an instrument based on the work of [Saiz 2010](#), [Harari 2020](#)) and (ii) its interaction with manufacturing Total Factor Productivity in 2000 ([Imbert et al. 2022](#)). The set of controls consists of: manufacturing Total Factor Productivity in 2000; and (log) population in 2000.

where u is a city, $\ln(\mathcal{E}_u^{m_j})$ is the average (log) expenditure share on remittances associated with a living arrangement $\{\lambda_j\}_{j \in \{1,2,3,4\}}$ (also adding single-individual migrant households), and p_u is the average rent, inferred from CMDS (2011) and from the “2005 Mini-Census,” respectively. We use our previous geographical variation to construct two instruments: (i) s_u , the share of non-developable land as induced by local geography around city borders in the baseline period, and (ii) its interaction with (log) manufacturing Total Factor Productivity in 2000 ([Imbert et al. 2022](#)). The specifications reported in Table D2 thus include (log) manufacturing Total Factor Productivity in 2000 and (log) population at destination as separate controls. Table D1 shows the first-stage estimates.

Table D2. Elasticity of substitution between consuming at origin and destination.

	Family households				Single-ind.
	Father	Mother	Couple	Whole	
Remittance share	(1)	(2)	(3)	(4)	(5)
Rent (log)	0.205 (0.363)	-0.098 (0.476)	0.692 (0.275)	0.122 (0.220)	0.163 (0.307)
Observations	162	142	172	173	173
Migration mode	$j = 1$	$j = 2$	$j = 3$	$j = 4$	O
F-stat	7.22	6.88	7.70	7.74	7.74

Notes: A unit of observation is a prefecture. Robust standard errors are reported between parentheses. The specification uses population weights in 2000. The dependent variable is the (log) expenditure share spent on remittances (CMDS, 2011) and the explaining variable is the (log) rent, computed using the housing module of the “2005 Mini-Census.” In column (1)–(4), the expenditure share on remittances is calculated for fathers migrating alone ($j = 1$), mothers migrating alone ($j = 2$), couples migrating without children ($j = 3$), and migration with the entire family ($j = 4$); and the expenditure share on remittances is calculated for single-individual migrant households in column (5) (O). The (log) rent is instrumented by (i) the share of developable land as induced by local geography around city borders in the baseline period (2000–2005), an instrument based on the work by ([Saiz 2010](#), [Harari 2020](#)), and (ii) its interaction with manufacturing Total Factor Productivity in 2000 ([Imbert et al. 2022](#)). The set of controls consists of: manufacturing Total Factor Productivity in 2000; and (log) population in 2000.

The estimates of b are reported in Table D2, using family households in columns (1)–(4) and single-individual migrant households in column (5). Our estimates are consistent

with an average elasticity $\rho = (b + 1) \in [0.8, 1.6]$ centered around 1.20, i.e., close to (and not statistically significantly different from) 1. There is some limited substitution between consuming at destination and at origin, and high housing prices induce migrants to displace more of their consumption to rural hinterlands—which is consistent with our findings about living conditions in expensive cities (see Section 2).⁴⁶

D.2 Estimation of the location choice model

The location choice model of Section 4.2 is characterized by three nests and three associated specifications.

The lower nest (λ_j) and its gravity structure In Section 4.2, we estimate a simple model of location choice across destinations for each migration arrangement $j \in \{1, 2, 3, 4\}$ (see Table 2). The identification of the lower nest relies on a productivity shifter that impacts real wages at destination, i.e., the trade shock induced by accession to the World Trade Organization (Facchini et al. 2019). The latter is a Bartik-style instrument at the prefecture level capturing the reduction in trade policy uncertainty faced by Chinese exporters of different sectors to the United States.

Table D3. The lower nest (λ_j)—first-stage.

Value at destination	Father alone (1)	Mother alone (2)	Both parents (3)	Whole family (4)
Trade shock	0.957 (0.217)	0.776 (0.153)	1.342 (0.309)	1.273 (0.321)
Observations	74,031	74,031	74,031	74,031
Migration mode	$j = 1$	$j = 2$	$j = 3$	$j = 4$

Notes: A unit of observation is a pair of origin/destination prefectures in 2005. The specification uses population weights at origin in 2000. Standard errors are reported between parentheses and are two-way clustered at the level of origins and destinations. The dependent variable is the value at destination calculated for: father alone at destination ($j = 1$, column 1), mother alone at destination ($j = 2$, column 2), both parents with children left behind ($j = 3$, column 3), and whole-family migration ($j = 4$, column 4). The set of controls consists of: origin fixed-effects; (log) population at destination in 2000; and (log) geodesic distance between the origin and destination prefectures. The explanatory variable is a trade shock computed following Facchini et al. (2019).

We provide the first-stage specification underlying the estimations of the lower nest in Table D3. The trade shock is a strong predictor of real wages across destinations—irrespective of living arrangements and the manner in which real wages are computed

⁴⁶The identification assumption is that local geography at the fringe of cities only affects expenditures on housing through local housing prices, and that it does so more acutely in highly productive cities. One concern is that local geography could affect the type of housing arrangements (dorms, informal housing, etc.) and that local housing prices might be contaminated by such a variation. In unreported robustness checks, we further correct for housing arrangements in the construction of local housing prices without finding any significant differences in our final estimates.

(i.e., how the composite price index, $\mathcal{P}$, weighs local non-tradable goods at destination versus at origin).

The middle nest (μ) The middle nest of our nested structure can be identified through the decision of moving with or without family, given the value of migrating from each origin r and across different arrangements, $V_r^{m_j}$. More specifically, the value of family migration across origins r can be written as follows:

$$\ln(V_r^{m_j}) = \ln \left(\left[\sum_{u \in U} (V_{ru}^{m_j})^{1/\lambda_j} \right]^{\lambda_j} \right).$$

We use the parameters $(\alpha, \xi, \beta, \lambda_1, \lambda_2, \lambda_3, \lambda_4)$ and the residual migration costs $(\tau_{ru}^{m_j})$ from the lower nest estimation to compute the value of migrating across the different arrangements, $V_r^{m_j}$ following [Buggle et al. \(2023\)](#). These values combine the composite attractiveness of destinations for different residential choices $(V_{ru}^{m_j})$, which contain many different, unobservable factors. Our baseline strategy is to identify the parameter μ using,

$$\ln(\pi_{jr}^c) = \frac{1}{\mu} \ln \left(\frac{V_r^{m_j}}{V_r^m} \right),$$

where π_{jr}^c is the emigration rate of migrants of mode j from origin r , conditional on emigrating from r .

The values of moving, $V_r^{m_j}$, might be contaminated by measurement error or by omitted variation, for instance, if the lower nest is mis-specified. In our baseline strategy, we rely on exogenous price variation in nearby cities to extract differential variation in the composite values $V_r^{m_j}$. Our approach consists in extracting a backward-looking, exogenous predictor of local prices, which would differentially affect migrants across different arrangements through their heterogeneous reliance on local non-tradable goods. To this end, we use the “non-developable” land within a buffer of potential urban sprawl in 2000, s_u , constructed in [Appendix D.1](#), which strongly affects housing prices. We combine this variation s_u with the distance from an origin r to possible destinations u , $\zeta_{ru} = 1/d_{ru}$, in a gravity structure to construct instruments, z_{rj}^k , for the relative value of migration under arrangement k :

$$z_{rj}^k = \sum_{u \in U} \zeta_{ru} s_u \times \mathbb{1}_{j=k}.$$

We provide the first-stage specification underlying the estimations of the middle nest in [Table D4](#). The gravity-weighted pressure on local prices at destination is a strong predictor of the relative value of family migration in particular (the living arrangement with the largest reliance on destination prices).

Table D4. The middle nest (μ)—first-stage.

	Value of migration
Exposure to living standards (z_{rj}^1 , mode 1)	-0.067 (0.187)
Exposure to living standards (z_{rj}^4 , mode 4)	-0.423 (0.042)
Observations	771

Notes: A unit of observation is a prefecture of origin r and a migration arrangement j . Standard errors are clustered at the origin level and are reported between parentheses. The specification uses population weights at origin in 2000, and the set of controls only includes origin fixed effects and migration-mode fixed effects. The dependent variable is the model-computed value of migration from each origin and family arrangement. The explanatory variables are gravity-based measures based on the share of developable land at the fringe of cities at destination, s_u (see D.1 for a description of the construction of s_u), interacted with the migration mode k .

The upper nest (γ) The identification of the upper nest (γ) of the location model relies on the following equation:

$$\ln \left(\frac{\pi_r}{1 - \pi_r} \right) = \frac{1}{\gamma} \ln \left(\frac{V_r^m}{V_r^s} \right).$$

We consider the following empirical counterpart:

$$\ln \left(\frac{\pi_r}{1 - \pi_r} \right) = a + b \ln \left(\frac{V_r^m}{V_r^s} \right) + \varepsilon_r,$$

where V_r^m is the value of migrating from location r :

$$\ln V_r^m = \ln \left[\sum_{j \in \{1,2\}} (V_r^{mj})^{1/\mu} \right]^\mu,$$

and V_r^s is the real wage in origin location r :

$$\ln V_r^s = \ln w_r - (\alpha + \beta) \ln p_r.$$

We instrument the relative value, $\ln V_r^m / V_r^s$, with exogenous shocks to agricultural productivity across possible origins by combining international commodity prices with local cropping patterns (in the manner of [Imbert et al. 2022](#)). We first collect Agricultural Producer Prices data (APP, 1991–2016) from the FAO: The data report producer prices at the farm gate in each producing country. For any given crop, we aggregate these country-specific prices into a yearly, international producer price as a weighted average across

countries using the baseline share in crop-specific exports as the country/crop weight.⁴⁷ We then clean these (log) international producer prices from long-run trends by applying a HP filter (see [Imbert et al. 2022](#)) and isolating the residual, d_{ct} , for any given year t and commodity c .

Table D5. The upper nest (γ)—first-stage.

	Relative value of migration (V_r^m/V_r^s)
Agricultural revenue shock	-3.309 (0.787)
Observations	257

Notes: A unit of observation is a prefecture (of origin). Robust standard errors are reported between parentheses. The specification uses population weights at origin in 2000. The dependent variable is the relative value of emigration. The instrument interacts cropping patterns in 2000 with the HP-filtered prices of agricultural commodities in 2000 (as in [Imbert et al. 2022](#)).

These international commodity prices affect agricultural hinterlands differently, depending on local cropping patterns. We exploit this intuition and combine international prices with the revenue share of crop c at origin r in a shift-share design. More specifically, we need the following ingredients to construct a revenue share for each crop: (i) a measure of output (e.g., as measured in tonnes) across locations; and (ii) a price per tonne. We construct a measure of output by multiplying local harvested areas in 2000 (a measure “in acres”) with a local predicted yield (a measure “in quantity per acre”). The harvested areas are provided by the World Census of Agriculture 2000, and the predicted yield is constructed within the Global Agro-Ecological Zones project. Nesting these measures within Chinese prefectures requires some geographic approximation that is best described in [Imbert et al. \(2022\)](#). We weigh this predicted output in 2000 by the baseline commodity price in 1980 to construct a revenue share for each crop, α_{cr} , orthogonal to later deviations in international prices. Letting d_c denote the previous price residual at a period of interest, our agricultural productivity shock, ω_r , will be defined as:

$$\omega_r = \sum_c \alpha_{cr} \times d_c.$$

The estimates reported in [Table 4](#) rely on a two-stage specification where we instrument the relative value of migration with ω_r . We provide the first-stage specification

⁴⁷We focus on the following 21 crops (commodities): banana, cassava, coffee, cotton, fodder crops (barley), groundnut, maize, millet, other cereals (oats), potato, pulses (lentil), rapeseed, rice, sorghum, soybean, sugar beet, sugar cane, sunflower, vegetables (cabbage), tea, and wheat. The international price of these commodities is disciplined by World demand and World supply, and China is a large World supplier for a few crops. The most obvious one is tobacco, where China is the leading producer and one company enjoys a local monopoly; we thus exclude tobacco from our agricultural productivity measures.

underlying the estimations of the upper nest in Table D5. The agricultural shock from Imbert et al. (2022) is a strong predictor of the relative value of migrating across origins ($\ln V_r^m / V_r^s$).

Table D6. The migration model for the *other* migrants—single-individual households.

Parameters	λ_O	γ_O	a_O^s	a_O^c
<i>Other</i> migrants	11.143 (2.168) [17.98] 74,031	2.371 (0.596) [15.03] 257	0.966 (0.587) [6.25] 6,360	-3.815 (2.217) [7.95] 8,585

Notes: This table reports estimates for the location choice model. The first column reports estimates from the lower nest λ , as in Table 2. The second column reports the estimate from the upper nest (γ). The third and fourth columns report the relationship between migration costs and migration policies, as in Panels A and B of Table 5. Standard errors are reported between parentheses, first-stage F-statistics are reported between square brackets, and the number of observations is reported in the last rows. See Section 4.2 for a description of the empirical specifications.

The migration model for the *other* migrants While the novelty of our approach is to consider family households and their complex (choice of) living arrangements, a fraction of rural-urban migrants are single-individual households. In this section, we briefly discuss the estimation of their nested model of location choice.

In a first step, we follow the exact same strategy as in the estimation of the lower nest (λ_j) for family households—see Equation (5) and Table 2—comparing the value of moving across destinations u and instrumenting these values by the trade-induced labor demand shock (Facchini et al. 2019). The estimate is reported in column (1) of Table D6. As expected, single-individual households are very sensitive to real wages at destination, and their allocation is not strongly driven by individual idiosyncrasies.

In a second step, we aggregate these values of migrating to different cities into an inclusive value of migration from origin r —which we divide by the value of staying—and we exploit agricultural shocks across origins for identification in a similar manner as in the estimation of the upper nest for family migrants (Imbert et al. 2022). The estimate, reported in column (2) of Table D6, is very comparable to that for family households (see Table 4).

D.3 Labor demand and housing supply at destination

The identification of the production block of the model requires exogenous variation in migrant inflows to estimate their effect at destination. In the spirit of the shift-share procedure developed in Imbert et al. (2022), we combine the exogenous shocks to rural

Table D7. Labor demand and housing supply—first stage.

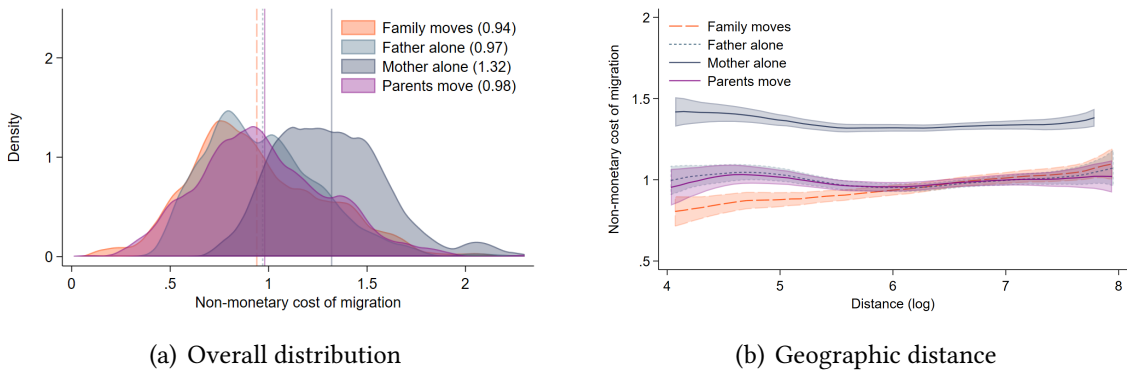
	Immigration rate (1)	Rented units (2)
Shift-share	-0.745 (0.155)	-1.662 (0.361)
Observations	289	269

Notes: A unit of observation is a prefecture. Robust standard errors are reported between parentheses. The dependent variables are the relative immigration rate between 2000 and 2005 in column (1) and the change in rented housing units between 2000 and 2005 in column (2). The explanatory variable is a shift share instrument that combines agricultural shocks between 2000 and 2005 in all rural origins and the share of migrants from each origin who lived in that destination in 2000 (as in [Imbert et al. 2022](#)).

incomes in each prefecture of origin that we used for the estimation of the upper nest of the location choice model (shifts) with the share of migrants coming from each origin in 2000 (shares):

$$z_u = \sum_{r \in R} \zeta_{ru} \omega_r,$$

While we report the second-stage estimates in the main text (Table 6), this Appendix reports the first-stage estimates (see Table D7, with column 1 focusing on the wage elasticity and column 2 focusing on the housing-supply elasticity).

Figure D2. Bilateral migration costs and distance.

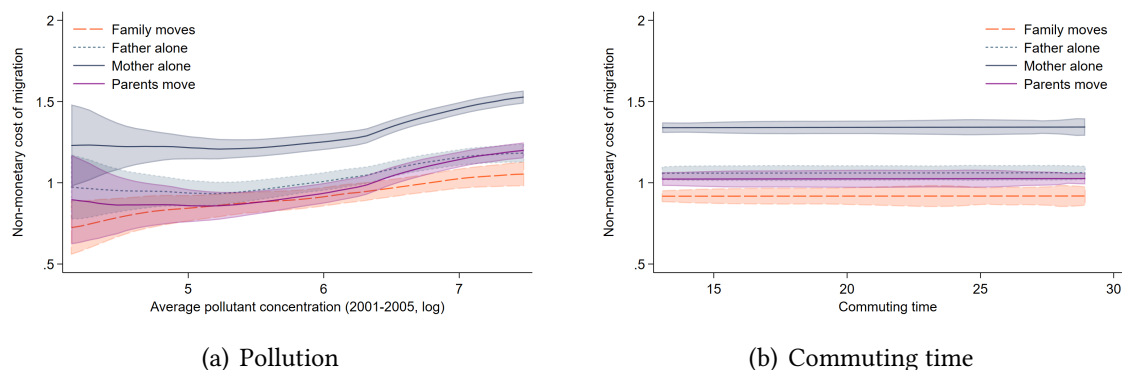
Notes: Panel (a) plots the distributions of bilateral migration costs for all the living arrangements of family migrant households. Panel (b) plots these bilateral migration costs against the (log) distance between origin r and destination u . The lines are local polynomial fits.

D.4 An interpretation of migration costs

The distribution of migration costs We first shed light on the distribution of bilateral migration costs across the different living arrangements for family migrant households in panel (a) of Figure D2 and their relationship with (log) distance between origins

and destinations in panel (b). We find that migration tends to induce high costs, and such costs are slightly higher when destinations are distant from origins (panel b). These findings reflect the relatively low incidence of migration given the rural-urban wage gap and the observed geographic gravity in movements across Chinese prefectures.⁴⁸

Figure D3. Relative cost of family migration and observable amenities.



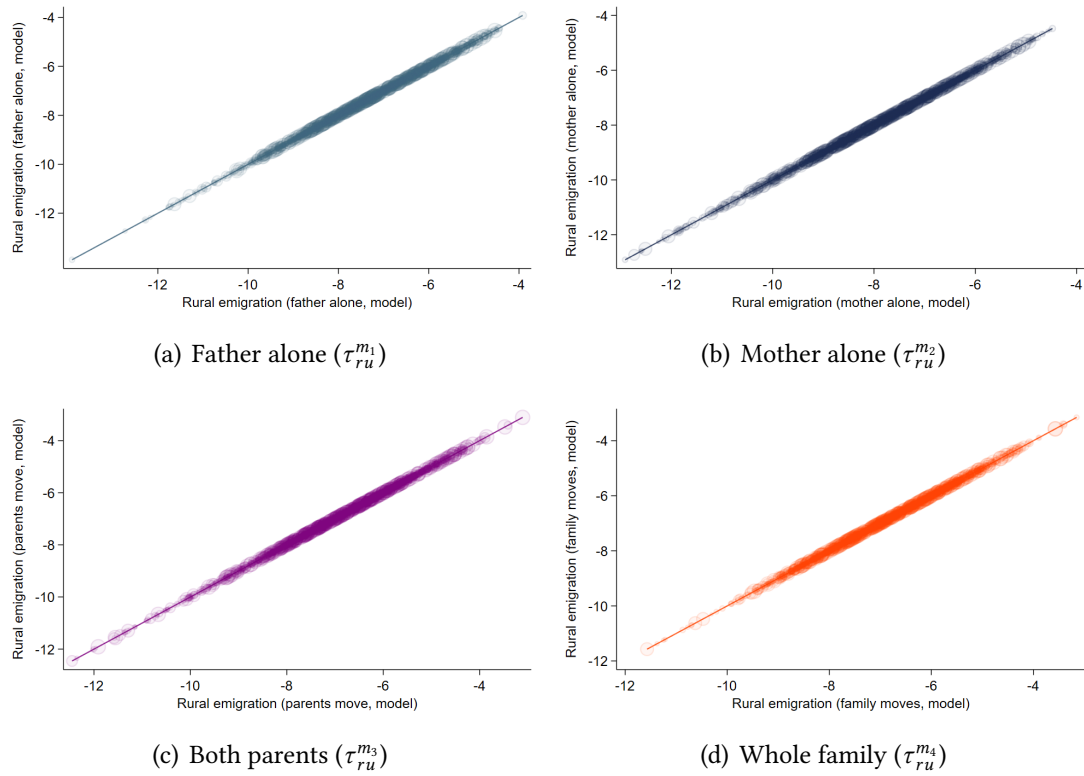
Notes: This figure plots the correlation between the average cost, τ_{ju} , for all living arrangements and destinations, against (log) pollution (2001–2005) in panel (a) and commuting time (from the “2015 Mini-Census”) in panel (b).

The role of other factors and a sanity check In Figure D3, we complement the previous evidence by plotting the relationship between the average migration costs and measures of pollution (panel a) and commuting time (panel b). We find that higher levels of pollution are associated with higher perceived barriers at destination for all migration modes.

Finally, we display in Figure D4 a validation of our bilateral migration estimates. By construction, these estimates—combined with the other estimates of the nested location model—should allow us to match migration flows between all origins and destinations of the largest connected set (Abowd et al. 1999, Card et al. 2013, Buggle et al. 2023). Figure D4 shows that our inferred migration barriers indeed allow us to match exactly migration incidences from all origins to all destinations.

⁴⁸Our estimates of migration costs can be compared with those of Bryan and Morten (2019) and Tombe and Zhu (2019). Two points are worth noting. First, Tombe and Zhu (2019) do not take into account the zeros in the migration matrix when estimating the elasticity of substitution across destinations. This attenuates the estimates of $1/\lambda$, which mechanically leads to higher estimates of migration costs. Second, when reporting average migration costs (as these two papers do), it is also important to think about the role of zeros. The model interprets zero migration flows between an origin and a destination as infinite migration costs (or as costs equal to 100 percent of the wage at destination). Hence, whether the data set has more or fewer zeros, which may be related to data collection rather than true migration costs, has a strong influence on reported average migration costs. This explains why we do not emphasize the level of our estimates as much as their heterogeneity.

Figure D4. Bilateral migration costs—matching migration flows.



Notes: Panel (a) to (d) plot the predicted non-family migration induced by bilateral migration costs ($\tau_{ru}^{m_j}$) against the actual migration rates—for each migration arrangement $j \in \{1, 2, 3, 4\}$.

E The role of displaced consumption and frictions in shaping migration

This Appendix provides complements to our counterfactual exercises in Section 5. In a first step, we focus on the main experiments discussed in Sections 5.1, 5.3 and 5.4, and we explore the normative implications of displaced consumption and migration frictions and their redistributive effects. In a second step, we consider simple extensions of our baseline quantitative model to allow for agglomeration and congestion externalities. In a third step, we illustrate the quantitative and qualitative insights induced by our precise modeling of migration and consumption choices. In settings with large differences in living standards across space, the incentives for households to migrate with or without family and split their consumption between origin and destination are instrumental in explaining migration flows. Ignoring them leads to a misspecification of bilateral migration frictions and policy evaluations.

E.1 Displaced consumption, migration frictions, and normative implications

Migration patterns, prices, and remittances In Sections 5.1, 5.3, and 5.4, we present the effects of our counterfactual experiments on migration patterns in China. In Table E1, we further report their effect on wages and rents at destination and on the amount that is remitted from urban locations to rural origins. One can see that wages and rents respond to immigration flows in similar, yet opposite, fashion. Note, however, that the wage effect is the one explaining most of the welfare response of urban-born households to migration because local prices represent a fraction of expenditures (about 0.65) such that a decrease of 1% of nominal wage has a larger impact on real wage than an increase of 1% in local prices. Table E1 also sheds light on the compositional effect of migration flows on remittances. For instance, counterfactual experiments (2a), (2b), (3a), or (3b) induce a slight increase in migration flows but a significant, yet small, *decrease* in remittances: The migration increase is indeed disproportionately explained by family migration, which typically generates smaller shares of remittances from each migrant household.

Distributional effects across space We now highlight the distributional effects of our experiments.

We first provide additional evidence on the distributional effects of our counterfactual experiments across more or less expensive cities in Figures E1 and E2. More specifically, we display the difference in the concentration of migrants across cities between the counterfactual and baseline scenarios (baseline migrant concentration can be found in Figure 2), the incidence of migration as a function of the actual level of rents in 2005 and relative to the baseline, and the differences in the welfare of urban-born households rela-

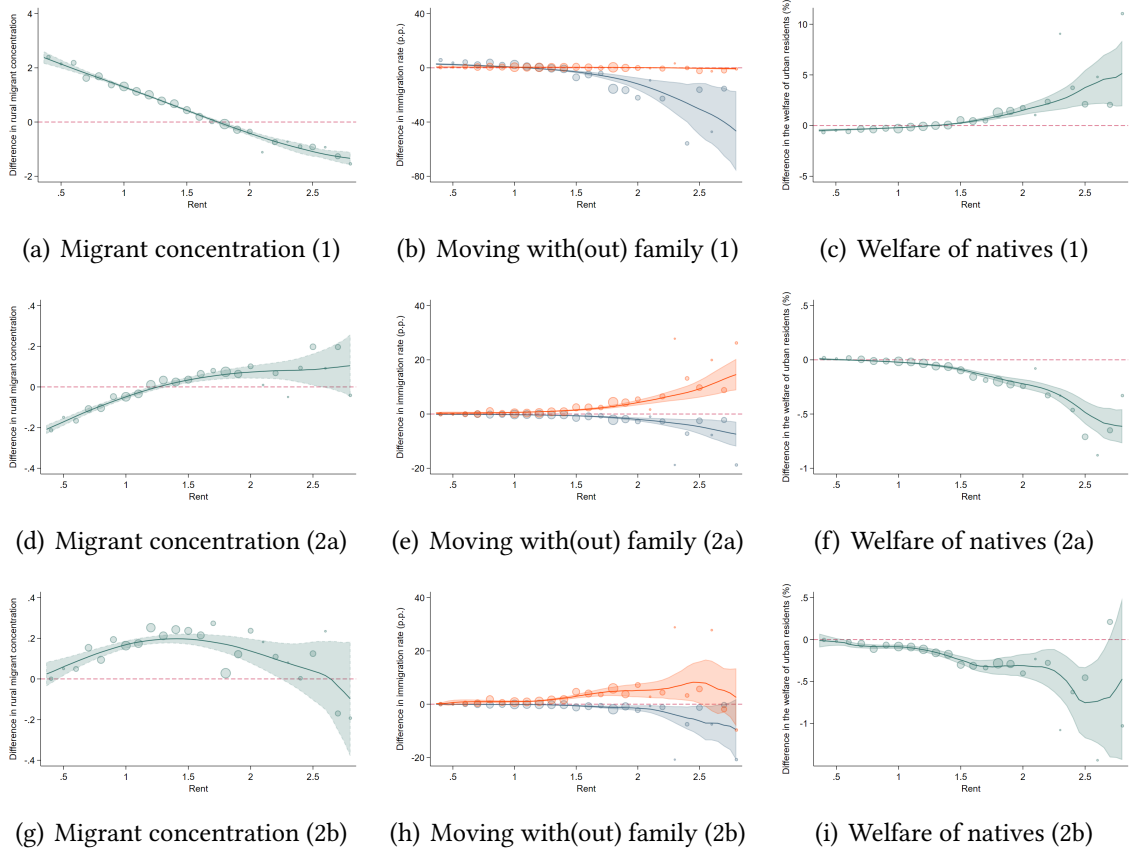
Table E1. The role of consumption patterns and migration frictions—complements.

	Wage (% rel. base.)	Rent (% rel. base.)	Remit. (% rel. base.)
<i>1. Consumption imbalances</i>			
No remittances (1)	0.298	-0.393	-100.00
<i>2. Reducing the cost of education for migrants</i>			
Blanket subsidy (2a)	-0.072	0.085	-1.74
Targeted subsidy (2b)	-0.063	0.080	-3.45
<i>3. Removing family-specific migration frictions</i>			
No family penalty (3a)	-0.089	0.117	-4.57
No policy-ind. family penalty (3b)	-0.027	0.032	-0.94
<i>4. Reducing overall migration frictions</i>			
Relaxing <i>hukou</i> policy (4)	-0.407	0.452	55.33

Notes: This Table reports statistics on the extent, nature, and consequences of migration flows in the baseline and in counterfactual experiments (1: taxing remittances), (2a: subsidizing education costs), (2b: subsidizing education costs in non-mega-cities), (3a: equalizing residual migration costs), (3b: family-specific removal of policy-induced frictions), and (4: relaxing *hukou* policy). Across all experiments, we report the differences implied by the experiment—relatively to the baseline, and in percentage points—on: urban wage in column (1), the urban rent in column (2), and the level of remittances by migrants across modes and from all urban destinations to all origins (column 3).

tive to the baseline across cities. The difference in rural migrant concentration, displayed in the left panels, allows us to visualize the change in the spatial distribution of migrants that the counterfactual would generate—a difference that can also be interpreted as the change in the (log) share of total rural migrants that a city hosts. Figure E1 illustrates that shutting down remittances very significantly reduces migrant concentration in larger, expensive agglomerations (panel a). Removing the possibility for migrants to displace their consumption leads to fewer of them moving to expensive cities without their family and slightly more of them moving with family across all cities (panel b). This experiment leads to moderate welfare losses for urban-born households in the least expensive cities and to larger welfare gains for urban-born households in the most attractive (and expensive) ones—see panel (c). Subsidizing education at destination, counterfactual (2a), leads to a massive substitution across migration modes (apparent in panel e) and ambiguous

Figure E1. Distributional effects across cities—consumption imbalances and education policy.

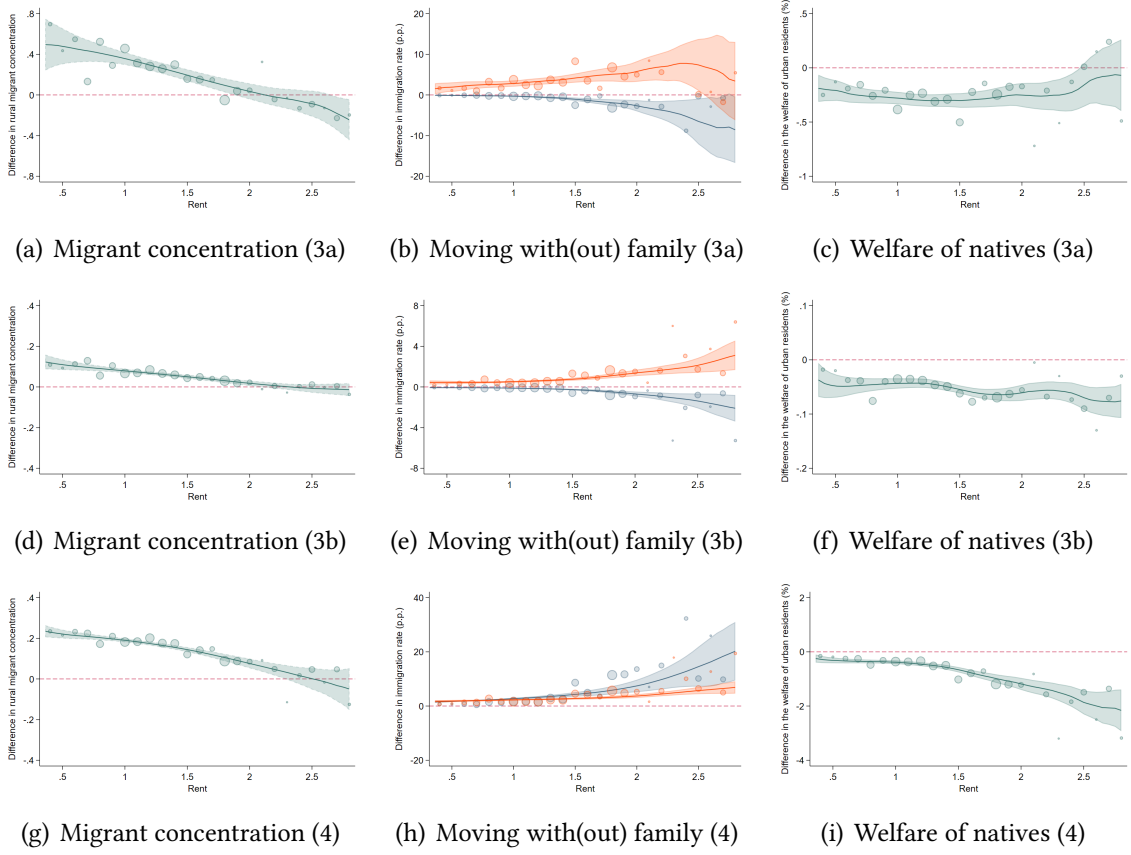


Notes: This Figure reports statistics on the extent, nature, and consequences of migration flows in counterfactual experiments (1: taxing remittances), (2a: subsidizing education costs), and (2b: subsidizing education costs in non-mega-cities). More specifically, we display: the difference in the concentration of migrants across cities between the counterfactual and baseline scenarios in panels (a/d/g); the incidence of migration in percentage points of the baseline city population (with family in orange, without family in blue) in panels (b/e/h); and the differences in the welfare of urban-born households relative to the baseline in panels (c/f/i). We group cities by bins of similar rents to limit the number of points; and their size is weighted by the total population at baseline in those cities.

effects on migrant concentration: On the one hand, more expensive cities become more affordable; on the other hand, there are more family migrants, typically more impacted by local prices. We find that the first effect dominates slightly, and migrants concentrate slightly more in more expensive cities, entirely driven by a surge in family immigration. The best way to visualize these effects is to contrast counterfactual (2a) with counterfactual (2b), shutting down the first effect: Migrants then concentrate in mid-size cities, driven by a surge in family migration.

Finally, equalizing the residual component of utilities between the different migration modes, i.e., $\tau_{ru}^{m_2} := \tau_{ru}^{m_1}$ in counterfactual (3a) or the equalization through the effect of policy as in counterfactual (3b), leads to some reduction in migrant concentration, with migrants favoring *medium-sized cities*, and a substitution away from non-family migration. However, these adjustments are one order of magnitude smaller than the

Figure E2. Distributional effects across cities—removing migration penalty.

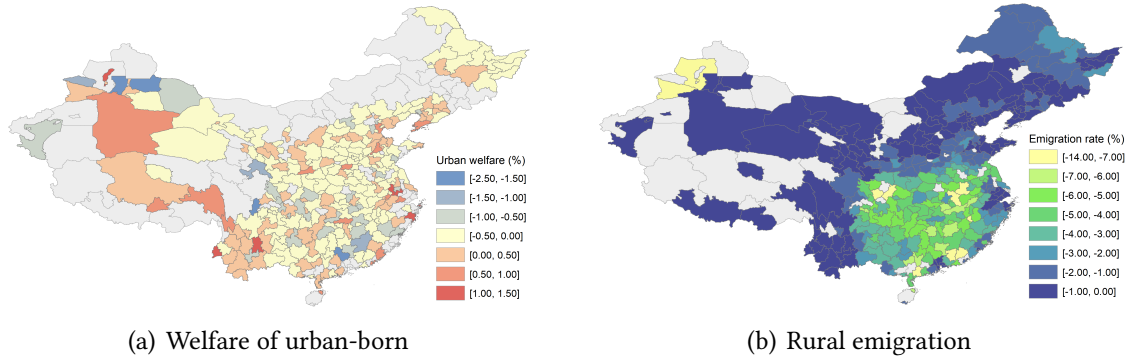


Notes: This Figure reports statistics on the extent, nature, and consequences of migration flows in counterfactual experiments (3a: equalizing residual migration costs), (3b: family-specific removal of policy-induced frictions), and (4: relaxing *hukou* policy). More specifically, we display: the difference in the concentration of migrants across cities between the counterfactual and baseline scenarios in panels (a/d/g); the incidence of migration in percentage points of the baseline city population (with family in orange, without family in blue) in panels (b/e/h); and the differences in the welfare of urban-born households relative to the baseline in panels (c/f/i). We group cities by bins of similar rents to limit the number of points; and their size is weighted by the total population at baseline in those cities.

ones induced by equalizing the *consumption* component of utilities (counterfactual 1).

We shed further light on the geography of counterfactual experiment (1) in Figure E3. In panel (a), we nest those effects across destinations and report the change due to the experiment in the welfare of urban-born households. We find that the large, aggregate drop in migration mostly concerns a few urban centers: the large cities (Beijing, Shanghai) and the new exporting regions (Shenzhen/Guangzhou, Fujian, Zhejiang). In panel (b), we nest those effects across origins and report the change in the emigrant share. We find that fewer rural migrants of the “hinterlands” of the highly productive coastal prefectures would leave. In summary, the possibility for migrants to remit explains the very large outflows from the Southern, Central regions of China to productive coastal prefectures with less amenable conditions to family migrants.

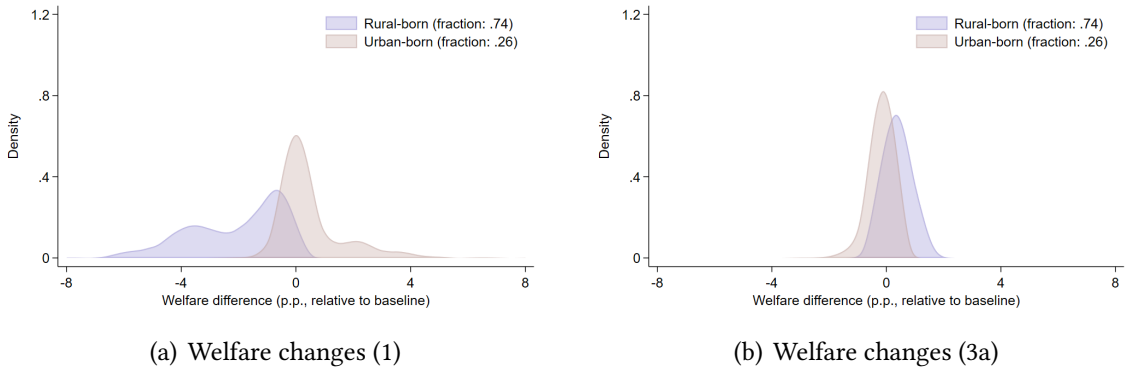
Figure E3. Displaced consumption—implication across destinations and origins.



Notes: Panel (a) displays the change in the welfare of urban-born households (in percentage points) at destination, as induced by the counterfactual (1) that prevents consumption displacement. Panel (b) displays the change in the emigration rate across origins.

Welfare and inequalities The previous section sheds some light on the distributional effects of our counterfactual experiments across space. We now discuss the redistributive effects of such experiments between rural- and urban-born households.

Figure E4. Consumption patterns and migration frictions—welfare changes.



Notes: This Figure displays the welfare effects of our main experiments: counterfactual (1) tilting consumption patterns in panel (a); and counterfactual (3a) equalizing migration restrictions in panel (b). The panels report the differences in the welfare of urban-born and rural-born households in percentage points relative to the baseline. Given our assumptions, those differences can be interpreted as (log) units of equivalent real wage. In other words, a one percentage point difference is equivalent to a change in real wage of 1%.

We provide some evidence about the normative implications of our experiments—counterfactual (1) tilting consumption patterns and counterfactual (3a) equalizing migration restrictions as discussed in Section 5—in Figure E4. Panel (a) shows that the possibility for potential migrants to remit favors rural-born households. A counterfactual economy with consumption less closely tied to origins would induce welfare gains for urban-born households and would harm most rural-born households. We see that the welfare gains for urban dwellers are dispersed, reflecting the wide heterogeneity

in attractiveness across possible destinations. The main winners would be urban-born households in booming, expensive cities. The welfare losses for rural households are also dispersed—reflecting the gravity structure of migration flows and the heterogeneous attractiveness of nearby destinations. More specifically, households living in the proximity of expensive cities are most affected by such an experiment, while households living far from any attractive cities are far less impacted. However, even the most affected households might still be able to mitigate the effect of the policy by switching between migration modes and/or destinations. In summary, counterfactual (1): increases inequality between urban-born and rural-born households; increases the welfare differences within urban-born households; and slightly decreases the welfare differences across rural-born households. Indeed, the lucky urban households born in attractive cities are better off than before, when the relatively lucky rural households born in the hinterlands of such cities are worse off.

Equalizing migration frictions across living arrangements induces much less significant redistributive effects, as illustrated in panel (b) of Figure E4. The equalization of migration barriers across living arrangements would generate moderate gains for a very large number of rural-born households, the extent of which would depend on the location of rural areas compared to the most attractive destinations. The large mass of rural households in China’s interior provinces would gain between 0 and 1% in equivalent real wage from this experiment. In conclusion, this relaxation would be a progressive policy, in the sense that it would reduce the gap between rural- and urban-born households, but with very limited overall impact.

E.2 Introducing externalities

Our quantitative model does not feature any agglomeration externalities or congestion forces other than the ones operating through the adjustment of labor and housing markets across locations. In this section, we show how agglomeration spillovers and congestion externalities at destination would affect (i) the allocation of rural-urban migrants across space and (ii) the normative implications of a relaxation of migration policies.

We consider our baseline model, as estimated in Section 4, and add the following features across four alternative models and three sources of externalities: (i) constant and size-varying production externalities in cities, e.g., the total factor productivity is $\mathcal{A}_u L_u^{0.05}$, where L_u is labor and $\mathcal{A}_u$ is an exogenous productivity shifter in urban location u , as standard in quantitative models of urban economics (see, e.g., Ahlfeldt et al. 2015); (ii) negative congestion externalities arising from urban sprawl or pollution (see, e.g., Khanna et al. 2021), i.e., $Z_u = \mathcal{Z}_u L_u^{-0.025}$, where $\mathcal{Z}_u$ is an exogenous amenity shifter; and (iii) positive externalities from remittances at origin, i.e., the total factor productivity

Table E2. The role of consumption patterns and migration frictions—externalities.

	Migrant hou. (millions)			Fam. sh.	Welfare (%)		In mega-cities	
	All	No fam.	Fam.		Rural	Urban	All	Fam. sh.
Baseline	28,49	23,01	5.48	0.192	-	-	14,53	0.141
<i>0. Baseline model</i>								
Relaxing <i>hukou</i> policy (4)	47,48	35,76	11,72	0.247	2.650	-0.705	23,45	0.186
<i>1. Agglomeration spillovers</i>								
Productive spillovers (cons.)	47,85	36,08	11,77	0.246	2.700	-0.461	23,77	0.185
Productive spillovers (var.)	47,77	36,00	11,77	0.246	2.690	-0.481	23,23	0.187
<i>2. Negative spillovers on amenities</i>								
With negative externalities	47,38	35,67	11,71	0.247	2.640	-0.766	23,35	0.186
<i>3. Spillovers from remittances at origin</i>								
With remittance spillovers	45.48	34,23	11,24	0.247	5.080	-0.638	22,48	0.186

Notes: This Table reports statistics on the extent, nature, and consequences of migration flows in the baseline and in counterfactual experiment (4) across five alternative models: the baseline model, a model with agglomeration spillovers, a model with size-dependent agglomeration spillovers, a model with negative externalities on amenities at destination, and a model with positive productive spillovers at origin as induced by the level of remittances. Across all experiments, we report: the number of migrant households (overall in column 1, without family in column 2, and with family in column 3, all reported in millions of migrant households between 2000 and 2005); the share of migrants living with family in column (4); the welfare of rural-born households in column (5), in % relative to the baseline; the welfare of urban-born households in column (6), in % relative to the baseline; the number of migrant households who are moving toward mega-cities above 5,000,000 residents (Shanghai, Beijing, Chongqing, Tianjin, Guangzhou, Shenzhen, Chengdu, Nanjing, Wuhan, Xi'an, Dongguan, Hangzhou, Foshan, Shenyang, Suzhou, Haerbin, Qingdao, Dalian, Ji'nan) in column (7); and family migration incidence among migrants who are moving toward mega-cities above 5,000,000 residents in column (8).

is $\mathcal{A}_r R_r^{0.05}$, where R_r is the level of remittances received in rural location r (conveying the idea that remittances can be used as productive investment, see [Pan and Sun 2024](#), [Khanna et al. 2022](#)).

Table E2 shows that the addition of productive spillovers (sometimes called agglomeration economies) further boosts the effect of relaxing migration restrictions (policy 4, described in Section 5.4) with a larger number of migrants moving to cities with or without family than in the baseline model. The effect remains, however, limited: The first panel of Table E2 predicts an additional inflow of about 370,000 rural-born households. Adding positive agglomeration externalities implies that rural-born households are left better off from the relaxation of restrictions than estimated through our externality-free

model and urban-born households are less worse off—both effects being driven by a muted response of wages to migration flows. Negative congestion externalities have the exact opposite effect: The migration response to the policy is lower, and its normative implications are less positive. More specifically, urban-born households lose more from the relaxation of restrictions when rural-born households gain slightly less. Finally, assuming that remittances boost production at origin changes our predictions in the most significant manner: While this spillover increases the social returns to migration, these returns are not internalized by migrants such that the increase in local wages mitigates the desire to move to urban destinations. In such a model, migrants would respond *less* positively to a relaxation policy, even though the policy would have much larger welfare effects. Their muted response implies that the level of remittances would be lower than that predicted by the externality-free model. In the presence of such externalities, a social planner would be tempted to subsidize migration rather than penalize it.

E.3 Sensitivity analysis and alternative migration models

Alternative migration models Our quantitative model of location choice is designed to best capture the choice of rural residents in transforming economies with large productivity and price differentials across urban areas and an even wider rural-urban gap. In those settings, rural migrants often consume at origin to mitigate the living costs at destination, and an important adjustment margin is whether to leave relatives behind or not (as we document in Section 2). For these reasons, we add the following ingredients to the standard migration models (see, for instance, [Bryan and Morten 2019](#), [Tombe and Zhu 2019](#), [Monras 2020](#)): (i) a three-nest structure for the location choice model allowing potential migrants to trade off whether to migrate or not, how to do so (with or without family), and where to go; and (ii) a technology to displace part of the consumption of non-tradable goods across household members and across space, depending on the migration mode (with or without family).

In this section, we illustrate the quantitative and qualitative insights gained through the adoption of those two novel features. To do so, we estimate three alternative models: (1) a simple model of location choice among numerous alternatives, and where the birth location is one of those alternatives ([Bryan and Morten 2019](#), [Tombe and Zhu 2019](#)); (2) a two-nest structure with the upper nest capturing the decision to migrate or not, and the lower nest modeling the choice of destinations ([Monras 2020](#)); and (3) a two-nest structure adding the possibility for migrants to displace part of their consumption ([Albert and Monras 2022](#)). In all these models, we treat family and non-family migrants as two separate populations, thus estimating separate elasticities. We estimate these models using a similar approach as in Section 4. We thus estimate Model 1 by assuming

a standard formulation for real wages, i.e., $\ln(w_u/p_u^\alpha)$, and estimating the parameters λ_j in a similar manner as in Table 2 (but with a slightly different explanatory variable). We estimate Model 2 by assuming the same standard formulation for real wages, i.e., $\ln(w_u/p_u^\alpha)$, estimating the parameters λ_j and γ_j in a similar manner as in Tables 2 and 4. Model 3 follows the same estimation as Model 2, except for the computation of price indices, which are calculated—across living arrangements—as in the baseline.

Table E3. The role of migration policies—alternative migration models.

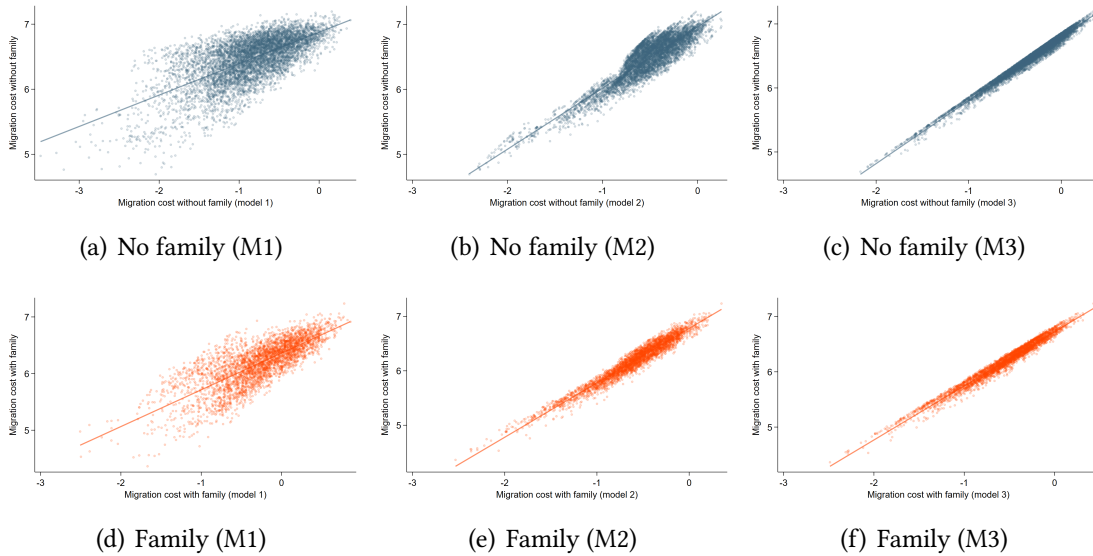
	Migrant hou. (millions)			Fam. sh.	In mega-cities	
	All	No fam.	Fam.		All	Fam. sh.
Our model	47,48	35,76	11,72	0.247	23,45	0.186
<i>rel. baseline (%)</i>	<i>+66.7</i>	<i>+55.4</i>	<i>+113.9</i>	<i>+28.6</i>	<i>+61.4</i>	<i>+31.9</i>
Model 1 (λ_i)	532,00	443,70	88,33	0.166	239,80	0.128
<i>rel. baseline (%)</i>	<i>+1767.3</i>	<i>+1828.3</i>	<i>+1511.9</i>	<i>-13.5</i>	<i>+1550.4</i>	<i>-9.2</i>
Model 2 (λ_i, γ_i)	46,51	39,07	7,44	0.160	23,52	0.118
<i>rel. baseline (%)</i>	<i>+63.2</i>	<i>+69.8</i>	<i>+35.8</i>	<i>-16.7</i>	<i>+61.9</i>	<i>-16.3</i>
Model 3 (λ_i, γ_i, P_i)	47,62	40,13	7,49	0.157	24,96	0.116
<i>rel. baseline (%)</i>	<i>+67.1</i>	<i>+74.4</i>	<i>+36.7</i>	<i>-18.2</i>	<i>+71.8</i>	<i>-17.7</i>

Notes: This Table reports statistics on the extent, nature, and consequences of migration flows in counterfactual experiment (4: relaxing *hukou* policy). We do so for the baseline model and for three alternative models: model 1, model 2, and model 3. We report: the number of migrant households (overall in column 1, without family in column 2, and with family in column 3, all reported in millions of migrant households between 2000 and 2005); the share of migrants living with family in column (4); the number of migrant households who are moving toward mega-cities above 5,000,000 residents (Shanghai, Beijing, Chongqing, Tianjin, Guangzhou, Shenzhen, Chengdu, Nanjing, Wuhan, Xi'an, Dongguan, Hangzhou, Foshan, Shenyang, Suzhou, Haerbin, Qingdao, Dalian, Ji'nan) in column (5); and family migration incidence among migrants who are moving toward mega-cities above 5,000,000 residents in column (6). For each variable and each model, the percentage difference between the counterfactual prediction and its initial level is reported in *italics*.

To show the quantitative differences implied by the different models, we evaluate counterfactual experiment (4: relaxing *hukou* policy). Model 1 provides nonsensical predictions, because it only features *one* elasticity across all locations (including the birth location) for each migrant mode, and this elasticity is very high. A significant improvement in conditions across possible destinations—as in counterfactual experiment (4)—would generate massive outflows. Model 2, by introducing an upper nest and lower estimates for the elasticity of outmigration, corrects for this issue. Overall migration incidence is quite close to our baseline estimates, as are additional migration inflows to large cities. It however misses the substitutability across living arrangements and significantly under-estimates the incidence of family migration: migration without family is predicted to increase relatively more than family migration, leading to a decrease in

the incidence of family migration in the aggregate and in large cities. This prediction starkly contrasts with our model insight. Adding realistic measures of local prices across living arrangements—as in Model 3—does not help in that respect: the overall number of additional migrant households remains closely aligned with our model prediction(s), but not their composition; altering the sensitivity to local prices has different implications in a model where households *can* switch from one migration mode to the other. In summary, our quantitative model of location choice does not only provide qualitative insights about the nature of migration in transforming economies; it also has quantitative implications for the effect of migration frictions/policies on the spatial allocation of migrant households and their composition across space.

Figure E5. Correlation between migration frictions across models.



Notes: Model 1 assumes a standard formulation for real wages, i.e., $\ln(w_u/p_u^\alpha)$, and a one-nest migration model for each migrant mode j (we estimate λ_j in a similar manner as in Table 2). Model 2 assumes the same standard formulation for real wages, i.e., $\ln(w_u/p_u^\alpha)$, but considers a two-nest migration model for each migrant mode j (λ_j and γ_j are estimated in a similar manner as in Tables 2 and 4). Model 3 follows the same estimation as Model 2, except for the computation of price indices which are calculated as in the baseline.

One crucial element of such migration models—including our baseline model—is to nest all residual, unexplained variation in migration flows in bilateral migration costs. This set of inferred parameters, labeled $\{\tau_{ru}^{mj}\}_{j,r,u}$ in our framework, are capturing actual pull or push factors, and gravity or network effects, but also residual errors and biases when the model is misspecified. In Figure E5, we show the correlation between our baseline bilateral costs and the estimated costs in alternative models. Overall, we find that our estimated bilateral costs for family migration become more closely matched when we introduce the various elements of our theory: a multi-nest location choice model, and realistic price indices. However, we argue that the variation underlying these residual

terms remains better behaved in our baseline model than in these simpler alternatives.

Table E4. Explaining bilateral costs $\{\tau_{ru}^{m_j}\}_{j,r,u}$.

Bilateral migration costs ($\exp(-\tau_{ru}^{m_j})$)	Model 1	Model 2	Model 3	Baseline
Wage $\times \mathbb{1}_{j=2}$	-0.166 (0.081)	-0.142 (0.080)	-0.171 (0.079)	-0.032 (0.086)
Local price $\times \mathbb{1}_{j=2}$	-0.085 (0.064)	-0.089 (0.063)	0.043 (0.063)	0.009 (0.065)
Bilateral distance $\times \mathbb{1}_{j=2}$	-0.202 (0.029)	-0.249 (0.029)	-0.232 (0.029)	-0.177 (0.031)
Observations	175,154	175,154	175,154	174,576

Notes: A unit of observation is a triplet of mode/origin/destination (j, r, u) . The estimation is a Poisson regression, and standard errors are reported between parentheses. The dependent variable is a transformation of the bilateral migration costs, $\exp(-\tau_{ru}^{m_j})$, and the explaining variables are the wage, local prices at destination, and bilateral distance interacted with the mode, $\mathbb{1}_{j=2}$. The reported estimates are relative differences in bilateral migration costs across living arrangements, as induced by observable living standards at destination (and distance). All specifications control for the wage, local prices at destination, a mode fixed effect, origin-fixed effects and the (log) distance between the origin and the destination. Model 1 assumes a standard formulation for real wages, i.e., $\ln(w_u/p_u^\alpha)$, and a one-nest migration model for each migrant mode j (we estimate λ_j in a similar manner as in Table 2). Model 2 assumes the same standard formulation for real wages, i.e., $\ln(w_u/p_u^\alpha)$, but considers a two-nest migration model for each migrant mode j (λ_j and γ_j are estimated in a similar manner as in Tables 2 and 4). Model 3 follows the same estimation as Model 2, except for the computation of price indices which are calculated as in the baseline.

Table E4 regresses (a transformation of) the bilateral migration costs obtained across the different models, $\exp(-\tau_{ru}^{m_j})$, on a set of observable predictors for these migration frictions (interacted with the migration mode, $j = 2$): (i) the wage of migrants at destination and (ii) the local price at destination, as placebo variables that should not be predictive outside of its indirect effect through real wages, and (iii) distance between origins and destinations to capture the iceberg costs of migration. Models 1 and 2 fail to restrict the effect of wages/local prices to their impact through real wages (see columns 1 and 2)—a feature that we attribute to their failure to account for displaced consumption and the fact that migrants allocate different shares of their income to local consumption when they migrate with or without family. Model 3 better accounts for the latter effect, as apparent from the predictive power of the price of the non-tradable good, but does not fully neutralize the correlation between bilateral migration costs and living standards at destination. Because it ignores substitutability across migration modes, any factor favoring one migration mode will appear in both bilateral costs, positively in one and negatively in the other. The impact of such misspecification is made salient through the observed gaps across migration modes in the estimates for wages—a gap that we do not observe

in our baseline model (column 4). We interpret these findings as supportive evidence for our baseline model.